

Five-Year Review Report

Third Five-Year Review Report
Kysor Industrial Corp., Superfund Site
Cadillac, Wexford County, Michigan

August 2010

EPA Region 5 Records Ctr.



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Table of Contents

<u>Section</u>	<u>Page</u>
List of Acronyms.....	5
Executive Summary.....	7
Five-Year Review Summary Form.....	9
 I. Introduction.....	 11
II. Site Chronology.....	12
III. Background.....	13
Physical Characteristics.....	13
Land and Resource Use.....	15
History of Contamination.....	16
Initial Response.....	17
Basis for Taking Action.....	18
 IV. Remedial Actions.....	 19
Remedy Selection.....	19
Remedy Implementation.....	21
Institutional Controls and Access Restrictions.....	24
Operation and Maintenance (O&M).....	30
Costs and Operation	31
 V. Progress Since the Last Five-Year Review.....	 32
VI. Five-Year Review Process.....	33
Administrative Components.....	33
Community Involvement.....	33
Document Review.....	34
Data Review.....	34
Site Inspection.....	40
 VII. Technical Assessment.....	 41
Question A: Is the remedy functioning as intended by the decision documents?.....	41
Question B: Are the exposure assumptions, toxicity data cleanup levels, and remedial action objectives (RAOs) used at the time of the remedy still valid?.....	42
Question C: Has any other information come to light that could call into question the protectiveness of the remedy?.....	42
Technical Assessment Summary.....	43
 VIII. Issues.....	 43
 IX. Recommendations and Follow-up Actions.....	 44
 X. Protectiveness Statement.....	 45
 XI. Next Review.....	 45

Tables

Table 1 – Chronology of Site Events.....	12
Table 2 – Institutional Controls Summary.....	29
Table 3 – Annual System Operations/O&M Costs.....	31
Table 4 – Actions Taken Since the Last Five-Year Review.....	32
Table 5 – Contaminant Trend Summary.....	40
Table 6 – Issues.....	43
Table 7 – Recommendations and Follow-Up Actions.....	44
Table 9 – LDFA Extraction Well Construction Summary	Attachment 9
Table 10 – LDFA Monitoring Plan	Attachment 9
Table 11 – Boring Well Placement for MDNRE Plume Investigation	Attachment 9

Figures

Figure 1 – Site Location Map
Figure 2 – Plume Map
Figure 3 – Cadillac Industrial Park Area
Figure 4 – Location of Cadillac City Well Field
Figure 5 – SVE System Layout and Former Excavation Area
Figure 6 – Diagram of Extraction Well Discharge and Piping
Figure 7 – North Park Subdivision Residential Well Users
Figure 8 – Residential Well Reconnaissance of November 2004
Figure 9 – Identified Areas of Principal Uncertainty for Contamination
Figure 10 – MDNRE Plume Investigation in Deep Well Aquifer
Figure 11 – MDNRE Plume Investigation in Shallow/Intermediate Well Aquifer
Figure 12 – City of Cadillac New Well Field Site Map and Water System Improvements

Attachments

Attachment 1 – LDFA Operating Budget and O&M Statistics
Attachment 2 – Cadillac Ordinance and Recertification
Attachment 3 – Haring Township Ordinance
Attachment 4 – Advertisement of Five-Year Review Start
Attachment 5 – List of Documents Reviewed
Attachment 6 – Graphs
Attachment 7 – Site Inspection Checklist
Attachment 8 – Inspection Documentation Photos
Attachment 9 – Additional Tables

List of Acronyms

ARAR	Applicable or Relevant and Appropriate Requirement
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act
CFR	Code of Federal Regulations
ESD	Explanation of Significant Difference
GAC	Granular Activated Carbon
ICs	Institutional Controls
LDFA	Local Development Finance Authority
MCL	Maximum Contaminant Level
MDEQ	Michigan Department of Environmental Quality
MDNR	Michigan Department of Natural Resources
NCP	National Contingency Plan
NPL	National Priorities List
O&M	Operation and Maintenance
OU	Operable Unit
PRP	Potentially Responsible Party
RA	Remedial Action
RCRA	Resource Conservation and Recovery Act
RD	Remedial Design
RI/FS	Remedial Investigation/Feasibility Study
ROD	Record of Decision
RPM	Remedial Project Manager
SVE	Soil Vapor Extraction
TCE	Trichloroethylene
UAO	Unilateral Administrative Order
USEPA	United States Environmental Protection Agency
VOC	Volatile Organic Compound

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EXECUTIVE SUMMARY

The Kysor Industrial Corp., Superfund Site is located in Cadillac, Michigan, in an industrial park that is approximately one square mile in area. The Kysor Industrial Corp. Superfund Site ("Kysor Industrial Site" or "Kysor Site") and another Superfund site, the Northern Plating Site, are among several manufacturing plants located in the industrial park. The primary concern at the Kysor Site is groundwater contaminated with volatile organic chemicals (VOCs). The intent of the remedy was to address six commingled groundwater contaminant plumes resulting from releases of waste from Kysor Industrial Corporation and three other facilities.

The city of Cadillac municipal well field is located approximately 2,500 feet northeast of the Kysor Site. Since 1993, the VOC trichloroethylene (TCE) has been detected in one of the seven wells in the municipal well field. It has not been definitively concluded whether the VOCs in the municipal well are due to the Kysor Site, to a facility east of the Kysor Site, or to a commingling of the two groundwater plumes.

The cleanup of the Kysor Industrial Site included installing a soil vapor extraction system to address contaminated soils in an area behind the Kysor plant, extracting and treating groundwater, and implementing access and use restrictions related to well installation and groundwater use. The Unilateral Administrative Order for Kysor Industrial Corporation also specified that soils and drummed waste at a nearby property that received waste from Kysor be cleaned up.

A partial excavation of soils, along with the soil vapor extraction system, addressed the risks associated with direct contact and ingestion of contaminated soil. These actions also eliminated areas of significant soil contamination that would otherwise act as a continuing source of groundwater contamination and would prolong the operation of the groundwater extraction and treatment system.

The groundwater treatment system is achieving all surface water discharge limits and performance standards. Data indicate that the Site groundwater plume may be in the vicinity of a residential area (North Park and Pine View subdivisions) where private wells exist. However, to date, residential well samples collected have not shown evidence of contamination. Although attaining the groundwater target cleanup levels established in the ROD is not expected to occur for a number of years, significant clean-up progress has been made. The VOC treatment system operates at a very high rate of efficiency, and groundwater exiting the system has not shown detections of total VOCs since the system began operating in 1996.

The remedy for the Kysor Industrial Site continues to be protective of human health and the environment in the short term. Long-term protectiveness will be achieved when groundwater cleanup goals are met and the implementation of effective ICs is monitored, maintained and enforced. In the meantime, ICs are in place in the form of two local ordinances in Haring Township and the city of Cadillac to regulate groundwater use. Initial IC evaluations have revealed that these ordinances should be strengthened by requiring, all current and future residents to connect to their respective municipal systems¹; requiring all current and future residents in Haring Township and the city of Cadillac to abandon private wells; and, developing methods to achieve better compliance with current and future ICs. Long-term protectiveness requires compliance with effective groundwater use restrictions. This will be accomplished by ensuring effective ICs are in place and planning for long-term stewardship that includes maintaining, monitoring and enforcing effective ICs. In addition, the existing private wells should be periodically tested.

¹ Haring Township and the city of Cadillac Ordinances currently require connection to the municipal supply under certain circumstances.

This is the third Five-Year Review report for the Kysor Industrial Site. The report covers the Kysor Site only. A separate Five-Year Review report was prepared for the Northernair Plating Superfund Site, located about 750 feet to the northeast of the Kysor Site.

Five-Year Review Summary Form

SITE IDENTIFICATION		
Site name (from WasteLAN): Kysor Industrial Corporation		
EPA ID (from WasteLAN): MID043681840		
Region: 5	State: MI	City/County: City of Cadillac, Wexford County
SITE STATUS		
NPL status: ☒ Final ☐ Deleted ☐ Other (specify)		
Remediation status (choose all that apply): ☐ Under Construction ☒ Operating ☐ Complete		
Multiple OUs?* ☐ YES ☒ NO		Construction completion date: 09/23/1996
Has site been put into reuse? ☒ YES ☐ NO Although it is under new ownership, the Kysor Corporation facility is still operating		
REVIEW STATUS		
Lead agency: ☒ EPA ☐ State ☐ Tribe ☐ Other Federal Agency		
Author name: Sheila A. Sullivan		
Author title: RPM	Author affiliation: U.S. EPA, Region 5	
Review period:** 09/30/2005 to 07/15/2010		
Date(s) of site inspection: May 21, 2010		
Type of review: ☒ Post-SARA ☐ Pre-SARA ☐ NPL-Removal only ☐ Non-NPL Remedial Action Site ☐ NPL State/Tribe-lead ☐ Regional Discretion		
Review number: ☐ 1 (first) ☐ 2 (second) ☒ 3 (third) ☐ Other (specify)		
Triggering action: ☐ Actual RA Onsite Construction at OU #____ ☐ Actual RA Start at OU#____ ☐ Construction Completion ☒ Previous Five-Year Review Report ☐ Other (specify)		
Triggering action date (from WasteLAN): 09/30/2005		
Due date (five years after triggering action date): 09/30/2010		

* ["OU" refers to operable unit.]

** [Review period should correspond to the actual start and end dates of the Five-Year Review in WasteLAN.]

Five-Year Review Summary Form (cont'd.)

Issues:

1. Haring Township and the city of Cadillac ordinances that serve to prevent drinking water exposures to contaminated groundwater should be strengthened to ensure effectiveness by: 1) requiring all current and future residents to connect to their respective municipal water systems; 2) requiring current and future residents in Haring Township and the city of Cadillac to abandon their private wells, and 3) developing methods to achieve better compliance with current and future ICs.
2. Initial IC evaluations revealed that additional steps must be taken to assess the effectiveness of ICs and to ensure long-term protectiveness. The evaluation supports follow-up actions to assure that the remedy remains protective. LTS procedures should be reviewed to ensure long-term protectiveness.

Recommendations and Follow-up Actions:

1. EPA should encourage Haring Township and the city of Cadillac governments to tighten the existing ordinances and report the status and/or changes to the ordinances to EPA and MDNRE in order to evaluate future protectiveness.
2. An IC Plan will be developed by EPA with input from MDNRE. The Plan will include the results of the IC evaluation, plan for additional IC activities as needed (such as mapping), and include a plan to develop more effective ordinances and LTS procedures to ensure that effective ICs are maintained, monitored and enforced. The IC Plan will also include a plan to periodically sample the potentially affected private wells in order to ensure the drinking water is safe until local ICs are improved.

Protectiveness Statement:

The remedy for the Kysor Industrial Site continues to be protective of human health and the environment in the short term. Long-term protectiveness will be achieved when groundwater cleanup goals are met and the implementation of effective ICs is monitored, maintained and enforced. In the meantime, ICs are in place in the form of two local ordinances in Haring Township and city of Cadillac to regulate groundwater use. Initial IC evaluations have revealed that these ordinances should be strengthened by: 1) requiring all current and future residents to connect to their respective municipal systems; 2) requiring all current and future residents in Haring Township and the city of Cadillac to abandon private wells; and 3) developing methods to achieve better compliance with current and future ICs. Long-term protectiveness requires compliance with effective groundwater use restrictions. This will be accomplished by ensuring effective ICs are in place and planning for long-term stewardship that includes maintaining, monitoring and enforcing effective ICs. In addition, the existing private wells should be periodically tested.

Other Comments:

None

Environmental Indicator Data:

Date of last Regional Review of Human Exposure Indicator (from CERCLIS): **6/15/2009**

Human Exposure Survey Status (from CERCLIS): **Current Human Exposure Controlled**

Date of last Regional review of Groundwater Migration Indicator (from CERCLIS): **6/15/2009**

Groundwater Migration Survey Status Form (from CERCLIS): **Contaminated Groundwater Migration under Control**

Ready for Reuse Determination Status (from CERCLIS): **Not Ready for Reuse**

**KYSOR INDUSTRIAL CORP. SUPERFUND SITE
WEXFORD COUNTY, MICHIGAN
FIVE-YEAR REVIEW REPORT**

I. INTRODUCTION

Authority and Purpose

The purpose of a Five-Year Review is to determine whether the remedy at a site is protective of human health and the environment. The methods, findings, and conclusions of reviews are documented in Five-Year Review reports. In addition, Five-Year Review reports identify issues found during the review, if any, and identify recommendations to address those issues.

The United States Environmental Protection Agency (EPA) is preparing this Five-Year Review report pursuant to CERCLA §121 and the National Contingency Plan (NCP). CERCLA §121 states:

[i]f the President selects a remedial action that results in any hazardous substances, pollutants, or contaminants remaining at the site, the President shall review such remedial action no less often than each five years after the initiation of such remedial action to assure that human health and the environment are being protected by the remedial action being implemented. In addition, if upon such review it is the judgment of the President that action is appropriate at such site in accordance with section [104] or [106], the President shall take or require such action. The President shall report to the Congress a list of facilities for which such review is required, the results of all such reviews, and any actions taken as a result of such reviews.

EPA interpreted this requirement further in the NCP. 40 CFR §300.430(f)(4)(ii) states:

[i]f remedial action is selected that results in hazardous substances, pollutants, or contaminants remaining at the site above levels that allow for unlimited use and unrestricted exposure, the lead agency shall review such action no less often than every five years after the initiation of the selected remedial action.

EPA Region 5 conducted this Five-Year Review of the remedy being implemented at the Kysor Industrial Superfund Site (Kysor Industrial Site, Kysor Site, or the Site) in the city of Cadillac, Wexford County, Michigan. This review was conducted by EPA Remedial Project Manager, Sheila Sullivan, with assistance from Scott Cornelius, MDNRE, Nick Swiger and Steve Cunningham, MDNRE, Cadillac District Office, and Jeffrey Dietlin, city of Cadillac Utilities. The review was conducted from September 30, 2005 through July 15, 2010. This report documents the results of the review. The final review report will be placed in the EPA site files and at the local repositories for the Kysor Industrial Site at the Cadillac-Wexford County Public Library, 411 South Lake Street, Cadillac, Michigan, and the Cadillac City-Municipal Complex, 200 North Lake Street, Cadillac, Michigan. This is the third Five-Year Review for the Kysor Industrial Site.

The triggering action for this statutory review is the last Five-Year Review completed on September 30, 2005. This Five-Year Review is required due to the fact that hazardous substances, pollutants, or contaminants remain at the site above levels that allow for unlimited use and unrestricted exposure.

II. SITE CHRONOLOGY

Table 1 - Chronology of Site Events

EVENT	DATE
Kysor Industrial Corporation begins operating	1955
Disposal practices at Kysor include dumping barrels of spent solvent directly on ground behind plant	As early as 1955 (exact date unknown) to 1980s
Four Winns (Frisbie Street) begins operating	1975
A.H. Joynt begins operating	1977
Voluntary excavation of some soils from waste pits behind Kysor plant and from nearby property	July 22 and 23, 1981
Hydrogeological Study, Kysor of Cadillac, Inc.	March 1982 (Phase I) and August 1983 (Phase II)
ROD (OU1) for Northernair Plating Site signed	September 11, 1985
MDNR inspection of Four Winns (Frisbie Street)	1986
Proposal of Kysor Industrial Site to NPL	June 24, 1988 (initially proposed in September 1985)
OU1 remediation (Northernair Plating Site)	October/November 1988
General notice letters and 104(e) requests sent	May 20, 1988
FS for Cadillac area groundwater completed	August 1988
ROD (OU2) for groundwater cleanup signed	September 29, 1989
Final NPL listing	October 4, 1989
UAO issued for remedial design of OU2	May 16, 1990
Remedial Design Additional Studies report	October 1992
TCE detected above MCL in city drinking well	1993
Explanation of Significant Difference (ESD)	March 3, 1994
Two separate UAOs for RA issued (Kysor Industrial and Four Winns/A.H. Joynt)	January 30, 1995
Third UAO for RA issued (Northernair Plating)	April 11, 1995
Onsite construction for OU2 begins	June 29, 1995

EVENT	DATE
Groundwater extraction and treatment begins	August 1996
Final inspection of RA construction by USEPA	September 19, 1996
Construction Completion	September 23, 1996
First Five-Year Review completed	July 26, 2000
Second Five-Year Review completed	September 30, 2005
Approval to city of Cadillac to shut down SVE system	April 16, 2010
Third Five-Year Review inspection conducted	May 21, 2010

III. BACKGROUND

Physical Characteristics

The Kysor Industrial Site is located about one-half mile northwest of and within the limits of the city of Cadillac, Michigan (see Figure 1). Cadillac has a population of about 10,000 and is approximately 100 miles north of Grand Rapids, Michigan. The dominant physical feature of the city is Lake Cadillac, which is about three square miles in area and is located about 2,000 feet southeast of the Kysor Site.

The Kysor Industrial Site is located in one of Cadillac's industrial parks. The park is about one square-mile in area and includes about 40 manufacturing facilities, a number of residences, municipal buildings, a trailer park, and a few light industrial/commercial facilities. The city of Cadillac municipal well field is situated in the center of the industrial park. A subdivision borders the Site to the north. The Kysor Site is defined by the plume within the one square-mile park area and the areas in Haring Township to which the plume has migrated. There are no environmentally sensitive ecosystems in the immediate area of the Site.

The former Kysor Industrial Corporation² was located at 1100 Wright Street, in Cadillac, Michigan. The Kysor property is about a 30-acre area, and is now owned and operated by BorgWarner. The Kysor Industrial Site, however, refers not only to the groundwater contamination originating from the Kysor facility on Wright Street, but also the contamination from another location where the Kysor Corporation deposited waste and to three other groundwater contamination plumes from other facilities. Some reports maintain that the two Kysor Corporation plumes may be responsible for the majority of the groundwater contamination being addressed by the cleanup of the Kysor Site.

² Throughout this report, references to the "Kysor Industrial Site," the "Kysor Site," or the "Site" all refer to the Superfund site, which includes a number of different facilities. When referring to the Kysor Corporation itself, the terms "Kysor Corporation" or "Kysor" are used. In some site file documents, the Kysor Corporation is also referred as "Kysor of Cadillac, Inc."

The first Kysor Corporation plume is due to the disposal of wastes in two unlined lagoons behind its plant on Wright Street. The second Kysor Corporation plume is due to waste disposal on the Leo Ingraham property, a private residential property located 1,500 feet northwest of the Kysor plant.

In addition to those plumes attributable to Kysor Corporation, there are additional plumes included as part of the Kysor Site from releases at three other facilities: Four Star Service Corporation (Four Star), Four Winns (Frisbie Street), and Four Winns/A.H. Joynt. All are hydraulically downgradient from the Kysor Corporation location. Four Star and Four Winns/A.H. Joynt are northeast of Kysor and Four Winns (Frisbie Street) is directly north of the Kysor plant.

Groundwater in the commingled plumes is contaminated by a number of volatile organic chemicals (VOCs), which include the main contaminants trichloroethylene (TCE), 1,1,1-trichloroethane (1,1,1-TCA), and tetrachloroethylene (PCE). The approximate extent of the plume³ being addressed by the remedial action for the Site is 4,000 feet long (north to south and beginning at the Kysor property) and, at its widest (east to west), 2,500 feet wide. At its widest point, the eastern boundary of the plume extends to the west edge of the Cadillac municipal well field; the western boundary extends approximately to Leeson Avenue [although some trace amounts have been found west of Leeson (see Figure 2)]. The municipal wells draw water from about 340 feet below ground level (bgl), while the greatest depth of the Kysor Site plume is estimated at slightly over 200 feet bgl. In addition, a clay confining layer is present over part of the plume area between the intermediate aquifer and the aquifer from which the municipal wells draw.

Although a number of other facilities in the area are known or possible sources of groundwater contamination these facilities are being addressed separately from the Kysor Industrial Site (see Figure 3). A partial list of other documented releases and sources of groundwater contamination are as follows.

Paulstra/CRC Corporation, formerly known as Cadillac Molded Rubber, Inc. (Paulstra)

- Located at 600 Seventh Street and northeast of the Kysor plant
- Rupture of above-ground storage tank containing TCE in 1984
- Installed recovery wells in 1984; operated until November 1986
- Involved in third-party lawsuit related to the Kysor Site: dismissed from suit by Federal District Court
- Installed additional extraction well in 2000 that is still operating

Mitchell-Bentley Corporation

- 1,400 feet east of and side-gradient to Kysor plant
- Began operating in late 1960s
- Investigation in 1991 indicates spill of PCE occurred in area of former underground storage tank; date of release unknown

³ In this report, the commingled plumes from the facilities that are part of the Kysor Site are referred to as a single plume. If a groundwater plume from an individual plume is being referenced, the name of the specific facility is used, e.g., the Four Winns (Frisbie Street) plume.

- Extraction well system installed to capture PCE plume
- Company declared bankruptcy in 2006
- Since 2006, the MDNRE has been operating the extraction system

Rexair, Inc.

- East-northeast of and side-gradient to Kysor plant
- MDNR⁴ files indicate that approximately twenty barrels containing unknown substances were stored outside on the east side of the facility; barrels appeared to be leaking
- In late 1980s, investigation of area groundwater revealed a plume consisting largely of TCE (nearly 100 percent) from the facility
- State of Michigan Consent Judgment in 1991
- *Installed a number of extraction and monitoring wells*
- Michigan Department of Environmental Quality (MDEQ) report reviewed by three independent experts maintains that Rexair extraction well system is not adequately capturing its plume, leading to commingling with the Kysor Site plume and resulting in the Kysor remediation system capturing and treating part of the contamination arising from Rexair since 1997

Because of the number of groundwater plumes in the vicinity of the Kysor Site, over a dozen different parties have installed monitoring and extraction wells. A total of 340 wells are present in the area.

Land and Resource Use

The industrial park in which the Kysor Industrial Site is located has historically been referred to as the “Cadillac Industrial Park” or the “Cadillac Industrial Park Area.” The industrial park is approximately one square-mile in area and is generally bounded by Thirteenth Street to the north, Mitchell Street (Route 131) to the east, Wright Street to the south, and Leeson Avenue to the west (see Figure 3). Kysor Corporation, the Ingraham property, Four Winns (Frisbie Street), Four Winns/A.H. Joynt, and Four Star are located in the western half of the industrial park.

Although the area is predominantly industrial, in addition to the more than 40 manufacturing facilities, the industrial park also includes garages and storage areas for the city of Cadillac, the city of Cadillac municipal well field, a number of residences, a trailer park, and a baseball diamond. The city of Cadillac municipal well field is approximately 1,000 feet south of Thirteenth Street and mid-way between Mitchell Street and Leeson Avenue.

Currently, the Four Star plant is inactive. The residence on the Ingraham property was demolished and the lot is vacant. Four Winns intends to use it as a storage facility and parking lot. The Four Winns (Frisbie Street) facility is still operating and the Four Winns/A.H. Joynt

⁴ The Michigan Department of Natural Resources (MDNR) was created in 1968 as the successor of the Michigan Department of Conservation. In 1995, the environmental regulatory programs of the MDNR were transferred to the Michigan Department of Environmental Quality (MDEQ). In 2010, the MDEQ was merged back with the MDNR and became the Michigan Department of Natural Resources and Environment (MDNRE).

facility is also still operating as a plant that manufactures fiberglass boats and is owned by Four Winns. The Kysor Corporation facility is an operating plant owned by BorgWarner. Future land use for each of these properties is projected to be industrial.

The land and resource use of primary concern near the Site is groundwater that is used for drinking water. The city of Cadillac well field includes seven municipal wells and is the sole community drinking water source for the city's 10,000 residents. One of the seven city wells (CW-7) has been shut down since 1994 because TCE was detected in 1993 (see Figure 4). Also, in a residential area to the north of the industrial park, commonly referred to as the North Park⁵ subdivision, located in Haring Township, a charter township adjacent to Cadillac, a number of the residences still use private wells for drinking water, although the exact number of these wells is unknown.

Studies of the area have shown that there are three distinct aquifers at, or near, the Kysor Site. The three aquifers are referred to as shallow, intermediate, and well field (or deep). The shallow and intermediate aquifers consist of sand with some silty clay and gravel. Clay layers separating the three aquifers from one another are predominantly seen in the southwest part of the industrial park. It appears, however, that the shallow and intermediate aquifers become hydraulically connected northeast of the Kysor Corporation, near the North Park subdivision.

The clay layer also appears to be absent in the vicinity of the city of Cadillac well field. The city wells are screened at 240-340 feet and draw from a deep, predominantly sand aquifer. Flow in the shallow aquifer is generally to the north, while flow in the intermediate is more to the northwest.

The closest surface water bodies to the Site are the Clam River and Lake Cadillac. Both are approximately three-quarter miles to the southeast. Some of the surface water drainage from the industrial park ends up in a water retention pond located near Leeson Avenue, but the majority of it flows south and east towards the Clam River.

The effluent from the groundwater treatment facility discharges via an outfall to the Clam River. According to the application submitted to MDNR for National Pollutant Discharge and Elimination System (NPDES) permitted discharge limitations, the river is protected for agricultural use, navigation, industrial water supply, cold-water fish, partial body contact recreation, and total body contact recreation. The main uses are recreation and sport fishing.

History of Contamination

Kysor Corporation began operating in 1955 at its location on Wright Street. The company manufactured truck parts and temperature control systems for the automotive industry. Processes at the plant included stamping, machining metal parts, and painting. A variety of solvents, degreasers and paints were used at the facility.

⁵ Throughout this document, references to the "North Park" subdivision also include the Pine View subdivision, which is adjacent to North Park, but on a separate plat of land. Both the North Park and Pine View subdivisions are in Haring Township and subject to the same contamination issues and private well use concerns.

Two adjacent areas behind (north of) the plant were used as waste disposal areas, sometimes referred to as lagoons or waste pits. Spent solvents were disposed of in these areas and directly to the ground upgradient of the Ingraham property northwest of the plant. The contaminants from the Kysor plant included toluene, xylenes, ethylbenzene, TCE, 1,1,1-TCA, and PCE. In the late 1970s it was discovered that Kysor Industries had dumped drums in a depression on the Ingraham property. Drum and soil samples collected from the Ingraham property in 1981 indicated VOC contamination. Kysor subsequently removed the drums and yards of contaminated soil from the property. Several years later, residual VOC contamination was confirmed in the soil and groundwater beneath the property. The Ingraham site was listed on the National Priorities List in 1985. The contaminated soil was subsequently excavated and replaced with clean soil.

The Four Star plant operated from 1967 to 1979. Between 1979 and 1981, the facility was used for storage. During its operation, the plant manufactured polished aluminum and stainless steel trim for the automotive industry. Steel was used as the primary raw material, and parts were stamped, welded, and assembled into roll-formed or stamped products. TCE was used to clean the finished parts. Soils collected at Four Star contained elevated levels of 1,1,1-TCA, TCE, toluene, and methylene chloride. Water samples from the Four Star site indicated that PCE and TCE were the principal contaminants released into groundwater.

The Four Winns plant on Frisbie Street began operating in the mid-1970s. Four Winns manufactures fiberglass boats. Solvents and degreasers used in the manufacturing process included acetone, methylene chloride, toluene, methanol, PCE, TCE, and 1,1,1-TCA. In 1936, the MDNR found soils contaminated with these contaminants as well as styrene, ethylbenzene and xylenes. 1,1,1-TCA is a compound used in fiberglass hardening.

A.H. Joynt operated its facility on Thirteenth Street from 1977 to 1982. A.H. Joynt was an automotive parts wholesaler and machine shop that also used degreasers in its operations. Four Winns bought the facility in 1982 and has operated the Four Winns' Cruiser Division at the location since that time. In 1992, the State of Michigan found hazardous substances, including TCE, in a concrete sump located at the Four Winns/A.H. Joynt property.

Initial Response

In 1980 and 1983, Kysor Corporation installed a series of borings and monitoring wells (designated by "K") for hydrogeological studies of groundwater contamination. On July 22 and 23, 1981, Kysor voluntarily excavated and disposed of approximately 700 cubic yards of contaminated soil from the two disposal areas behind the facility (see Figure 5). The average depth of the excavation was six feet, and the average depth to the water table is 25 feet. Drums, buckets, rags, and paint sludge were removed along with the soil. MDNR observers noted strong odors remaining in the completed excavation pit. Samples taken by MDNR indicated greater than 100 parts per million (ppm) of TCE, TCA, PCE, ethylbenzene, and xylenes remaining in the soils. The excavation was backfilled with clean material. The Kysor Industrial Corporation facility is where the most significant actions were taken prior to the remedial action (RA) specified in the Record of Decision (ROD) for the second operable unit (OU2).

The Statement of Work attached to the UAO issued to Kysor Industrial in January 1995 established one additional remedial component for the Kysor site. The additional requirement was to address soil contamination, via removal or otherwise, at the Ingraham property, where Kysor had disposed of industrial waste from the Wright Street plant.

Four Winns conducted limited investigations in 1986 and 1987, but no remediation is known to have been done. Under a later agreement with the State of Michigan, Four Winns installed a number of monitoring wells (designated by "FWMW"). Four Star completed a two-phase hydrogeological study in 1983. The 26 monitoring wells (designated by "FS") installed did not extend more than 20 feet below the water table (to approximately 52 feet bgl), so the plume was not considered adequately characterized by MDNR. Additional wells installed by MDNR at this time were designated with the prefix "DNR." At the Four Star site, the MDNR conducted soil vapor extraction (SVE) to remove contaminants, and the plant owner conducted a soil removal.

When 30 soil samples from the former lagoon areas were analyzed in 1987 six years after the excavation, total VOC concentrations ranged from a low of 29 micrograms-per-kilogram (ug/kg), also referred to as parts-per-billion (ppb), to as high as 752,000 ug/kg. The depths of the borings ranged from six to 25 feet bgl. On average, the total concentration of VOCs in the 30 samples was nearly 88 mg/kg. The contamination was made-up primarily of toluene, xylenes, ethylbenzene, and TCE. High levels of 1,1,1-TCA were also detected in some of the borings. PCE ranged from non-detect to 0.05 mg/kg in most samples; however, one sample showed a concentration of 2.2 mg/kg. The highest concentrations of the five predominant contaminants are listed below:

Toluene	95 mg/kg
Xylenes (total)	520 mg/kg
Ethylbenzene	68 mg/kg
TCE	74 mg/kg
1,1,1-TCA	24 mg/kg

Inorganic chemicals included:

Zinc	17 mg/kg
Lead	2.4 mg/kg
Hexavalent chromium	0.028 mg/kg

At the Ingraham property, inorganic constituents were found at comparatively higher levels: zinc (1,900 mg/kg), lead (500 mg/kg) and total chromium (360 mg/kg), while the VOCs were measured at several orders of magnitude lower.

Basis for Taking Action

Hazardous substances that were released into groundwater and soil from the Kysor Industrial Site included acetone, methylene chloride, toluene, methanol, styrene, ethylbenzene, xylenes, PCE, TCE, 1,1,1-TCA, 1,2-dichloroethylene (1,2-DCE), 1,1-DCE, and 1,2-dichloroethane (1,2-DCA).

The primary exposure pathway of concern at the Site is the ingestion of groundwater. The eastern extent of the contaminated groundwater plume from the Kysor Industrial Site is close to the western boundary of the city of Cadillac municipal well field; TCE, a primary compound of concern at the Kysor Site, has been detected in CW-7 above the maximum contaminant level (MCL) of five ug/l on a number of occasions since 1993; and, a number of residents in the North Park subdivision, where the northern extent of the plume has migrated, still use their private wells for drinking water or for other domestic purposes. Evaluation of the potential exposure and risk showed that a resident drinking groundwater from either the shallow or the intermediate aquifer would be exposed to excess lifetime cancer risks ranging from a low of 10^{-5} to a high of 10^{-3} .⁶

IV. REMEDIAL ACTIONS

A ROD for Operable Unit #1 (OU1) was signed by EPA in September 1985. This ROD related to the Northernair Plating Site and was a source control remedy that focused on removal of contaminated soils and sediments. The OU1 remedial action took place in October and November 1988.

The second part of the investigation involved a broader study of the groundwater contamination in the Cadillac Industrial Park area. A State-led, federally-funded study was completed in August 1988. Based on these results, the ROD for OU2 for both the Kysor and Northernair Plating Sites was signed on September 29, 1989. In 1990, MDNR completed Remedial Action for OU1, including removal of contaminated soil and sewer line sediments, limited modification of the Northernair building and restoration of the sewer line.

Remedy Selection

The September 1989 ROD for OU2 addressed the groundwater contamination at both the Kysor Industrial and the Northernair Plating Sites. The ROD served as the second and final action for the Northernair Plating Site and the first and final action for the Kysor Industrial Site. The predominant contaminant of concern (COC) at the Northernair Site is hexavalent chromium, while COCs in groundwater arising from the Kysor Site include a number of VOCs.

The OU2 ROD basically served as a ROD for both the Northernair Plating and the Kysor Industrial Sites. After this ROD was signed, the Sites were more often referred to as the Northernair/Kysor Site rather than by their individual names. The Sites were treated as one in the OU2 ROD because the designs for the treatment systems were developed jointly. In addition, the plume from the Northernair Plating Site was contained within the Kysor Site plume, so designing separated groundwater extraction systems would not have been feasible.

⁶ Risks are probabilities that are expressed in scientific notation (e.g., 1×10^{-6}). An excess lifetime cancer risk of 1×10^{-6} indicates that an individual experiencing the reasonable maximum exposure estimate has a one-in-one-million chance of developing cancer solely as a result of site-related exposure. This is referred to as an "excess lifetime cancer risk" because it would be in addition to the cancer risk individuals face from other activities. The chance of an individual developing cancer from all other causes has been estimated to be as high as one in three. EPA's generally acceptable risk range for site-related exposures is one-in-ten-thousand (1×10^{-4}) to 1×10^{-6} .

The chromium and VOC treatment systems are housed in one building and are operated and maintained by the same party.

The remedial action objectives for the OU2 ROD were to: 1) eliminate any human exposure to residual hazardous waste or contaminated materials disposed of at the Site; and 2) address all potential risks to human health and/or impacts to the environment. The components of the groundwater remedies required for each Site were identical, except that the remedy for the Kysor Site also called for the installation of a soil vapor extraction (SVE) system. For the Kysor Site, the OU2 ROD called for the following:

- Install a groundwater extraction and treatment system to remove groundwater contamination from the area surrounding the Site. Discharge treated water to the Clam River.
- Construct a fence around the site to prevent trespassing
- Conduct groundwater monitoring to assess the effectiveness of the groundwater remediation
- Install a vapor extraction system to remove VOC contamination from soils
- Restrict groundwater use for drinking purposes in the shallow and intermediate aquifer. Ensure that residences with contaminated wells are not using these wells for consumption purpose. Take appropriate actions to ensure that current or future land owners do not use the contaminated groundwater as drinking water.

A Unilateral Administrative Order (UAO) issued to Kysor in January 1995 added one additional remedial component--addressing soil contamination at the Ingraham property where Kysor had disposed of a number of drums of industrial waste from its Wright Street plant.

Groundwater cleanup levels [i.e., target cleanup levels (TCLs)] for VOC contamination in groundwater are:

<u>Compound</u>	<u>Target Cleanup Level (ug/l)</u>
1,1,1-TCA	200
trans-1,2-DCE	70
1,1-DCE	5
1,2-DCA	5
PCE	1
TCE	5
xylene	440
toluene	40
Methylene chloride	5
Hexavalent Chromium	50

Soil cleanup levels in the ROD for VOC contamination being treated by the SVE system are:

<u>Compound</u>	<u>Target Cleanup Level (mg/kg)</u>
1,1,1-TCA	7.6
TCE	0.07

xylene
toluene

141
724

Remedy Implementation

EPA issued a UAO for the remedial design (RD) of OU2 in May 1990. Nine parties, including the Northern Plating Company, were named by the Order. The other parties were: Top Locker Enterprises, Inc.; R.W. Meyer, Inc./Meyer Construction, Co.; Willard S. Garyood; Kysor of Cadillac; Four Winns Company; Four Star Corp.; Jomar Company; and, Leo Ingraham, Sr. The RD, finalized in March 1995, addressed the work required in the OU2 ROD for both the Northern Plating and Kysor Industrial Sites.

In 1994, EPA signed an Explanation of Significant Difference (ESD) to document that the selected remedy would also address the contaminated groundwater from the Northern Plating/Kysor Site that had migrated into the North Park subdivision located north of the Cadillac Industrial Park. None of the other required actions in the OU2 ROD were changed. The cleanup established in the OU2 ROD and ESD called for the remediation of the releases from the nine facilities to which EPA issued a UAO (see below). Through a mechanism put in place by the city of Cadillac, other facilities in the Cadillac Industrial Park are also contributing to the cleanup costs.

In 1995, three separate UAOs compelling the PRPs to conduct RAs were issued. Each UAO involved a separate group of PRPs. By 1996, the groundwater extraction and treatment system, as well as the SVE system at the Kysor Site, were constructed and operating.

Based on the findings of the remedial investigation, the areas of contamination to be addressed by this RA were:

- A VOC plume in the shallow aquifer containing up to 115 milligrams-per-liter (mg/l) of VOCs
- A less concentrated VOC plume (up to 12 mg/l) in the intermediate aquifer
- A hexavalent chromium plume in both the intermediate and shallow aquifers
- An area of soil contaminated with VOCs

The various components of the systems designed to remediate these areas are:

- A groundwater extraction system consisting of 17 wells [originally 18 wells; one (S-7) is no longer used due to low yield (see Figure 6)].
- Associated pipelines to convey VOC and VOC/chromium-contaminated groundwater to the treatment system
- A discharge pipe to convey treated groundwater to the Clam River
- A packed tower air stripping (PTAS) system to remove VOCs from groundwater
- A carbon adsorption system to remove hexavalent chromium from groundwater
- A treatment building to house the treatment system components
- Expansion of a pilot SVE and treatment system to remediate soils contaminated with VOCs at the Kysor property

Air Stripping System for Removal of VOCs from Groundwater

For groundwater treatment, water entering the treatment plant from the 17 groundwater extraction wells is sent directly to the air stripping towers. The treatment system for removing VOCs from groundwater consists of two PTASs connected in series. The design of the air stripping system was based on the TCE concentrations detected during pilot tests in 1991. The TCE concentrations in the influent groundwater were within the range of 970 to 1,430 ug/L. TCE was roughly 30 times greater than the next most prevalent contaminant, PCE. As TCE was the COC with the highest concentrations in groundwater, the treatment system was built to reduce TCE in groundwater effluent to the TCL of five ug/L.

Granular Activated Carbon System for Removal of Hexavalent Chromium from Groundwater

The design consisted of one extraction well to pump chromium-contaminated groundwater via a separate piping system to the treatment plant. The water entering the treatment plant from this well flows through the carbon adsorption treatment system and is then routed to the air stripping system. The treatment system for chromium includes two granular activated carbon (GAC) contactors, a pH control system, and associated valves and piping. The effluent from the chromium removal system is sent directly to the air stripper.

Discharge of Treated Groundwater to Clam River

The effluent from the groundwater treatment system discharges to the Clam River via gravity outfall piping. The treatment facility design includes vertical turbine pumps, which are designated as cooling water make-up pumps. These pumps may be used to pump up to 800 gpm to the Co-generation Power Plant located 4,000 feet west of the facility. Revenue from sales of cooling water to the power plant is used to help fund the costs of operating the treatment plant.

In 1994, MDNRE prepared a Substantive Requirements Document (SRD). The NPDES discharge permit, issued in 1996, specified the discharge monitoring requirements (DMRs) for the treatment plant. The purpose of the SRD and the discharge permits was to establish requirements and limits for discharging treated groundwater to the Clam River. For Superfund sites, the SRD can substitute for actual permits such as an air emissions permit or an NPDES permit. The 1994 SRD and 1996 permit provide the following limits for discharging to the Clam River:

<u>Compound</u>	<u>Daily Maximum (ug/l)</u>
1,1,1-TCA	5
1,2-DCE	5
PCE	5
TCE	5

<u>Compound</u>	<u>Monthly Average (ug/l)</u>
Hexavalent chromium	8.3
Total chromium	59

In 2002, a reissued permit increased the DMR monthly averages for hexavalent chromium from 8.3 ug/l to 12 ug/l, and for total chromium from 59 ug/l to 92 ug/l. DMRs are requirements that are expected to be met by the start of treatment and throughout the cleanup. Compliance with DMRs is based on samples collected at the Clam River outfall pipe, after the treated groundwater travels through approximately 4,000 feet of underground piping. However, the goal of the RA is to achieve the groundwater TCLs in the ROD and when the TCLs are met, the cleanup will be complete. The TCLs are compared to results of groundwater samples as the water exits the treatment systems.

SVE System for Removal of VOCs from Soil at the Kysor Plant

The January 30, 1995 UAO specified the construction and operation of a SVE to achieve cleanup standards. Residual VOC concentrations in soil indicate whether an SVE system was successful. Contaminant concentration trends in groundwater are also useful for assessing the effectiveness of a SVE system. In general, SVE is a remedy for source zones where contaminants leach to groundwater. If the SVE is successful, the ongoing contaminant source will decrease and ultimately be eliminated, thereby resulting in decreasing contaminant trends in the affected groundwater. Onsite construction of the RA for the OU2 ROD began in June 1995.

The SVE system at the Kysor plant consisted of 23 four-inch-diameter extraction/induction wells at a depth of approximately 25 feet (about five feet above the water table). A mounted blower creates a negative pressure in the system. Collected vapors are blown through two vapor-phase carbon units, each containing 2,000 pounds of vapor-phase carbon. The treated air is discharged through a four-inch stack. The system can draw from four wells at any given time, with the remaining wells alternating as induction wells (see Figure 5).

The SVE system has been operating from the spring through the fall each year since 1996. The ROD estimated that the soils would be able to meet the cleanup criteria to prevent continuing contamination of the groundwater within two years.

Construction Implementation and Operation and Maintenance

Typically, Respondents to a UAO incur the full costs and responsibility for constructing the remedy and conducting O&M. For the Kysor Industrial and Northern Plating Sites, however, a public/private sector partnership was formed to address the cleanup. Although the city of Cadillac was not identified as a potentially responsible party (PRP) and is not a Respondent to the UAO, it has taken the lead in constructing and operating the remedy. Using State legislation, the city formed a Local Development Financing Authority (LDFA) in the early 1990s to facilitate construction of the remedy and assist with the capital expenditures of building the treatment system. The development project that was the genesis of the LDFA was the construction of the Co-generation Power Plant. Revenue generated through tax increment funding (TIF) authorities, and funds from \$7.4 million in bonds issued by the city, helped to finance the remedial construction.

The city established a Special Assessment District to finance the operation and maintenance of the treatment plant. Annual costs include running the plant, conducting monitoring, and any

unexpected costs. The area covered by the Special Assessment District coincides to an area of the Cadillac Industrial Park. All non-exempt property owners, including the city of Cadillac, within the Special Assessment District pay a yearly special assessment in addition to their property taxes. An example of an exempt party would be a resident.

In addition, revenues from the sale of treated groundwater to the power plant for its cooling processes also go towards annual operating expenditures. Further description of the LDFA Remediation Project and a diagram showing the Special Assessment District is in Attachment 1.

Institutional Controls and Access Restrictions

Institutional controls (ICs) are required to ensure the protectiveness of the remedy. ICs are non-engineered instruments, such as administrative and/or legal controls, that help minimize the potential for exposure to contamination and protect the integrity of the remedy. Compliance with ICs is required to assure long-term protectiveness for any areas that do not allow for unlimited use or unrestricted exposure (UU/UE).

Decision Document:

For the Kysor Industrial and Northernair Plating Sites, the April 11, 1995 UAO Statement of Work and the September 1989 ROD for OU2 required, as a component to the remedy, that access and use restrictions be imposed. Specifically, restrictions on groundwater use for drinking water purposes in the shallow and intermediate aquifers were to be placed on the Kysor Industrial and Northernair Plating Sites where groundwater contamination is located. There are wells that were previously used for residential consumption, but are now contaminated. These residents were placed on city water. The ICs were intended to prevent interference with the performance or maintenance of the RA required under the UAO. The ICs were to be in the form of deed restrictions unless the city of Cadillac or other government entity adopts, and maintains an enforceable ordinance meeting each of the requirements above. All deed restrictions were to run with the land and remain in full force and effect until approval of the certification of completion of work.

The RA Work Plan specified the IC goals as:

- a) ensure that no drinking water or domestic use wells are constructed onsite and all existing wells are restricted from being used for drinking water and other domestic uses;
- b) ensure that no wells are installed that may interfere with the operation and maintenance of treatment and monitoring systems required by the UAO;
- c) prohibit new contaminant releases at the Kysor facility and the Ingraham property, due to building on or excavating the contaminated soils from those areas.

The city of Cadillac passed an ordinance imposing restrictions on the real estate described in the ordinance. The restrictions include prohibiting installation of drinking water wells on the Site and installation of any wells that might interfere with the operation and maintenance of the

groundwater extraction or treatment systems, except with written consent by EPA. The ordinance also prohibits tampering with or removal of the containment or monitoring systems at the Site. A copy of the ordinance, along with a certification that it is still in effect, is in Attachment 2. The ROD or ESD did not require ICs be put in place for on-site soils because previous removal actions have allowed for unlimited use and unrestricted exposure.

In addition, the PRPs were required to install and maintain a fence to limit access to portions of the Site where the RA activities were to occur. This was to reduce risks due to construction activity or exposure to hazardous chemicals during the cleanup and to protect the RA from vandalism. The fence was to enclose remedy components such as the SVE and the PTAS. Fencing was to be six-foot high perimeter chain link fence topped by minimum three-strand barbed wire. One warning sign was to be posted on each side of the SVE and groundwater treatment facilities and at all gates.

Land and Resource Use:

Although attaining the groundwater target cleanup levels established in the ROD is not expected to occur for a number of years, significant clean-up progress has been made. The majority of facilities in the industrial park, except for the Four Star plant which is inactive and the Ingraham property lot which will be used as a storage facility and parking lot, are still operating and future use of these properties is predicted to be industrial. Certain areas of the industrial park, such as along Leeson Avenue, also contain commercial establishments.

The land and resource use of primary concern near the Site is groundwater that is used as a drinking water source. The city of Cadillac well field is the sole community drinking water source for the city's 10,000 residents. The residential area to the north of the Industrial Park, commonly referred to as the North Park subdivision, contains a number of residences that still use private wells for drinking water (see Figure 7).

The effluent from the groundwater treatment facility discharges via an outfall to the Clam River. According to the application submitted to MDNR for NPDES permitted discharge limitations, the river is protected for agricultural use, navigation, industrial water supply, cold-water fish, partial body contact recreation, and total body contact recreation. The main uses are recreation and sport fishing.

Physical Area:

Maps which depict the current conditions of the site and areas which do not allow for UU/UE will be developed as part of the IC Plan. This will include the current groundwater area that exceeds cleanup standards relative to the area covered by the ordinances. This comparison, will demonstrate whether the ordinances cover the geographical area of groundwater that exceeds TCLs as well as a buffer zone.

IC Objectives:

The ROD called for the following: (1) restrict groundwater use for drinking purposes in the shallow and intermediate aquifer; (2) ensure that residences with contaminated wells are not

using these wells for consumption purpose; and (3) take appropriate actions to ensure that current or future land owners do not use the contaminated groundwater as drinking water.

Status of ICs - Implemented or Planned ICs

The Haring Township and city of Cadillac ordinances that serve to prevent drinking water exposures to contaminated groundwater should be strengthened. Initial IC evaluations have revealed that additional steps must be taken to evaluate the effectiveness of the ICs and to ensure long-term protectiveness. As mentioned below, an IC Plan will be prepared.

As background, in the late 1970s, when hexavalent chromium releases from the nearby Northern Plating Site were found in onsite groundwater, residences located in the Industrial Park were connected to the city of Cadillac municipal water system. The closest offsite residences are in the North Park subdivision across Thirteenth Street, which forms the northern border of the Industrial Park.

The city of Cadillac passed an ordinance in 1997 restricting groundwater use and well installation in the Cadillac Industrial Park. The ordinance was recertified in August 2005 and was codified in 2009. A copy of the ordinance, along with the August 2005 recertification is in Attachment 2. The city ordinance requires new construction to connect to the municipal supply. Currently, as long as a private well is still operable, the resident is permitted to use it. Should the well fail, however, the owner must hook up to the municipal supply. The ordinance does not require abandonment of private wells. The Wexford County Health Department oversees the implementation of the requirements under the ordinance.

When the ROD was written in 1989, data indicated that contaminated groundwater had not migrated beyond Thirteenth Street. After the ROD was finalized, however, site-related VOCs were detected in monitoring wells nearby and within the North Park subdivision. The North Park subdivision is in Haring Township and is not covered by the city of Cadillac ordinance. Haring Township covers approximately 36 square miles and is considered largely rural. The Haring Township water system is small and draws water from a well field located two miles northeast of the North Park and Pine View subdivisions.

In mid-July, 2006, EPA inventoried the private wells near the Kysor Industrial and Northern Plating Sites to build a database for information about well use and locations. The primary area for the inventory was in the North Park and Pine View subdivisions, located north of Thirteenth Street and south of the airport (see Figure 8).

A summary of the results of the well survey were:

Low or no potential for well existence ⁷	103
Well is not used for drinking but is not properly abandoned	49

⁷ This determination was made based on the fact that the facility was either a commercial establishment, the well was supposedly abandoned, no well had been previously identified, the residence could not be located, or the address was a vacant lot/garage.

Subsequent EPA testing has shown that, while there historically has been contamination in the deep groundwater underneath the area where the private well samples were collected, there is no contamination in water samples taken from residential wells in the Haring Township area. Most residential wells draw from shallower depths (i.e., 40-60 feet bgl) than from depths where the contamination has been observed. In May, 2008, EPA tested 16 wells of concern identified from a survey of residential wells near the Kysor Industrial Site. No contamination above health-based limits was found in the residential well samples. The results showed one well with TCE contamination well below the MCL.

At least 12 residences in the city of Cadillac and in Haring Township use private wells, which the Wexford County Department of Public works (DPW) tries to monitor. Of these, about six are connected to the township system, but are still using their private wells for drinking water. Three of these wells are within the city of Cadillac and near the Kysor Industrial Site. Although water mains and service connection piping have been extended to these three homes, the residents have refused to connect. About five private wells are on Thirteenth Street on the north end of the city well field. The wells identified to be at risk in the Cadillac and Haring Township were sampled and found to be safe.

In early 2008, EPA asked Haring Township trustees to pass an ordinance restricting the use of existing private wells in North Park and Pine View subdivisions of Haring Township. Haring Township trustees, however, refused and noted that its residents did not want to bear the costs of a prohibition on use of private residential drinking water wells. The Wexford County Department of Public Works (DPW) manages and operates the Haring Township water supply system and is the enforcement agency for any ordinance that is put in place for the water supply system. However, the Township has the ultimate authority regarding the adoption and implementation of the proposed ordinance.

There is a Haring Township Zoning Ordinance (see Attachment 3), effective April 1986 that requires township homeowners and property owners to connect to the township water system if they are located within 200 feet of a water distribution line. However, the ordinance does not require these residents to abandon their private wells. In areas where public water lines are available, all new structures or uses are required to hook up to the public system. Because abandonment of private wells is not required, the private wells may still be used even when the residence was connected to the water supply. Although fines could be imposed, the ordinance is enforceable, but compliance has been difficult to achieve.

The DPW is currently proposing a more protective Haring Township ordinance to prohibit use of private wells in the township water service area and to require abandonment of private wells. The proposed ordinance also will provide that any resident required to abandon their well must use a professional well driller. The driller must provide paper certification to the DPW of the abandonment. The DPW is crafting the ordinance in plain language that will allow for more effective enforcement by the township and will provide better records of residential private well use. The DPW plans to present the new ordinance to the Haring Township board by Fall 2010.

A well-head protection program is also being developed, which will allow DPW to track those areas where private wells are restricted, and will allow for regular inventories of private wells.

All Michigan counties have ordinances requiring that a well permit be obtained from the local health department (or that the local health department be notified) before the well is drilled. Permits contain important information about where the well must be placed on the property to avoid contamination sources. Minimum well depths, minimum capacity, and other conditions are noted on permits. Failure to comply with permit conditions can result in the need to redrill the well or in the revocation of the well drilling contractor's Certificate of Registration.

The Haring Township and City of Cadillac ordinances that serve to prevent drinking water exposures to contaminated groundwater cover the areas of groundwater contamination that do not support UU/UE. Further, these ordinances are enforceable, but should be strengthened by 1) requiring connection to the municipal system for all current and future residents in Haring Township; 2) requiring abandonment of private wells in both Haring Township and the city of Cadillac for all current and future residents; and 3) developing methods to achieve better compliance with the current and future ICs.

The initial IC evaluation indicated that additional steps must be taken to assess the effectiveness of the ICs and to ensure long-term protectiveness. An IC Plan will be developed by EPA with input from MDNRE. The Plan will incorporate the results of the IC evaluation, plan for additional IC activities as needed (such as mapping), and include a plan to develop more effective ordinances and long-term stewardship procedures to ensure that effective ICs are maintained, monitored and enforced. The IC Plan will also include a plan to have the potentially affected private wells periodically sampled in order to ensure the drinking water is safe until local ICs are improved.

Table 2 –Kysor Industrial Site Institutional Controls Summary

Media, Engineered Controls, & Areas that Do Not Support UU/UE Based on Current Conditions.	IC Objective	Title of Institutional Control Instrument Implemented (note if planned)
<u>Onsite and Offsite Groundwater:</u> The areas with groundwater contamination do not support UU/UE. The current area that exceeds groundwater cleanup standards is groundwater within the Industrial Park and areas of groundwater north of Thirteenth Street in Haring Township.	(1) Restrict groundwater use for drinking purposes in the shallow and intermediate aquifer. (2) Ensure that residences with contaminated wells are not using these wells for consumption purposes. (3) Take appropriate actions to ensure that current or future land owners do not use the contaminated groundwater as a drinking water source. (4) Ensure that no wells are installed that may interfere with the operation and maintenance of treatment and monitoring systems required by the UAO.	A mixture of governmental controls and periodic monitoring is in effect: The city of Cadillac ordinance (Ord. No. 97-10, ch.25, sect. 2.300) prohibits the installation of drinking water wells onsite; the installation of any wells that might interfere with the operation and maintenance of the groundwater pump-and-treat systems, except with written consent by EPA; and, tampering with or removing the containment or monitoring systems at the Site. (Implemented) A Wexford County ordinance is in effect that requires drillers to obtain a well permit from the local health department (or notify the health department) before the well is drilled. (Implemented) Haring Township Zoning Ordinance (Ord. No. 36-01, Sect. 2.1) requires township homeowners and property owners, located within 200 feet of a water distribution line, to connect to the system within 180 days after the line becomes available. The ordinance also restricts cross connections between the township system and any other source. The ordinance does not require abandonment of private wells after connection. In areas where public water lines are available, all new structures or uses are required to hook up to the public system. (Implemented). The Wexford County DPW is currently proposing a Haring Township ordinance to prohibit use of private wells in the township water service area and to require abandonment of private wells. Further, anyone required to connect and abandon their well must use a professional well driller to abandon the well and provide certification to DPW of this action. (Planned)

Table 2 summarizes institutional controls for these restricted areas.

Current Compliance

The Wexford County DPW indicated that the current Haring Township ordinance has not been complied with and was not enforced by the previous DPW management. The ordinance currently being developed by the DPW will protect public health from potential exposure to contaminated groundwater by prohibiting the use of existing and future private wells, and by providing stronger enforcement capability. EPA will continue to attempt to monitor the situation

and the wells. EPA will encourage the Haring Township and city of Cadillac governments to make the existing ordinances more effective and report the status and/or changes to the ordinances to EPA and MDNRE, in order to evaluate their future protectiveness. EPA will also arrange to have the potentially affected private wells periodically sampled in order to ensure the drinking water is safe until local ICs are improved.

Long-Term Stewardship

Long-term protectiveness at the site requires compliance with groundwater use restrictions which will be achieved by long-term stewardship (LTS). LTS will ensure that effective ICs are maintained and monitored and enforced and the remedy continues to function as intended with regard to ICs. An LTS plan (or O&M Plan revision) will be required to document LTS procedures. This plan will include a requirement to notify EPA and MDNRE of any changes to local ordinances or if additional ICs are implemented. In addition, annual certification should be provided to the agencies that ICs remain in place and are effective. In addition, a communications plan should be developed and use of the State's one-call system should be explored for LTS.

Operation and Maintenance (O&M)

Routine maintenance of the wells, extraction wells, SVE system, and groundwater treatment plant is performed by the employees of the city of Cadillac, as part of the LDFA system. Some of the routine maintenance includes checking and replacing oil, belts, heat tape, worn valves and packing; cleaning roof-top heating unit; pulling and cleaning extraction wells, and checking wells for freezing, damage, secured locks, and extraction well failure. Daily bench logs are kept current. Logs and data are electronically stored.

During the initial two years of operation, sampling frequency was quarterly. Annual monitoring began in October 1998. Currently, 23 shallow and 22 intermediate monitoring and extraction wells are monitored for VOCs annually. Generally, shallow wells are screened between 35 and 70 feet bgl, and intermediate wells are screened between 120 to 200 feet bgl. Several intermediate wells (F-6, CMS-5W, and SLW-34) are still monitored for VOCs on a quarterly basis. A submersible pump is used for purging the wells and disposable bailers are used for sampling.

The chromium sampling information is provided in the Northernnaire Five-Year review report also being issued in 2010. The VOC sampling is as follows:

- VOCs – 17 shallow wells; 6 shallow extraction wells; 11 intermediate wells; 11 intermediate extraction wells
- Sampled annually – September (VOCs)
- Monthly sampling at the treatment plant of the influent, midpoint and effluent flows
- Weekly sampling of effluent for compliance with DMRs
- Annual static water level measurements at about 100 wells

Air emission samples exiting the SVE system are analyzed for VOCs as part of O&M requirements. The Performance Monitoring Plan required quarterly soil sampling to evaluate the progress of the SVE cleanup.

The State of Michigan establishes maximum allowable emission rates from new sources of VOCs on a case-by-case basis. The report specified that air emissions from the air strippers were not to exceed the following:

Parameter	Rate
Hexavalent chromium	0.00007 lb/hour
TCE (Stripper 1)	38.8 mg/cubic meter and 0.487 lb/hour
TCE (Stripper 2)	2.4 mg/cubic meter and 0.101 lb/hour
Total VOCs	0.95 lb/hour
Opacity	0 percent

Costs and Operation

Approximate annual costs of O&M for both the Kysor Industrial and Northernair Plating Sites are shown in the table below. The average amount spent per year is approximately \$222,000. The estimate for annual O&M expenditures in the 1989 ROD was \$125,000. Capital and construction costs of the remedy were less than expected (about \$1.3 million) and close to the amount projected in the OU2 ROD. Approximately 60 percent of the 2010 and proposed 2011 budget respectively, is allotted to construction and utilities costs.

Table 3 - City of Cadillac Annual Operations/O&M Costs

Dates		Total Cost
From	To	
7/2000	6/2001	\$160,000
7/2001	6/2002	\$210,000
7/2002	6/2003	\$240,000
7/2003	6/2004	\$240,000
7/2004	6/2005	\$226,266
7/2005	6/2006	\$227,556
7/2006	6/2007	\$289,124
7/2007	6/2008	\$244,717
7/2008	6/2009	\$242,682
7/2009	6/2010	\$380,800
7/2010	6/2011	\$380,400 (<i>Proposed Budget</i>)

Note: The city of Cadillac fiscal year runs from July 1st through June 30th

The increase in total annual costs between the 2008-2009 cycle and the last two cycles of 2010-2011 (\$380,400) is due to the upgrading of the LDFA telemetry system in 2010. The project will be funded through TIF funds spread out over two years and will cost about \$100K. See Attachment 1 for LDFA operation statistics.

V. PROGRESS SINCE THE LAST FIVE-YEAR REVIEW

The protectiveness statement in the previous Five-Year Review for the Kysor Industrial Site indicated that the remedy was protective of human health and the environment in the short term. However, long-term protectiveness would not be achieved until: 1) it was confirmed that the extraction system was capturing all contaminated groundwater from the Kysor Industrial site; 2) groundwater cleanup levels were achieved; and 3) when ICs were implemented to restrict the installation and use of private wells in the subdivision north of the site, and the adequacy of existing restrictions was confirmed.

Table 4 - Actions Taken Since the Last Five-Year Review

Recommendations from Previous Review	Party Responsible	Action Taken
Develop an Institutional Controls Study Plan to evaluate the city ordinance currently in place and to devise and implement a plan to prevent exposure of residents in the North Park subdivision to contaminated groundwater from the Northern/Kysor Sites.	Respondents	An IC Study Plan was not completed; however residential areas were investigated to determine what ICs were appropriate for the site. The private wells in the North Park and Pine View subdivision were inventoried in July 2006. In early 2008, EPA proposed an ordinance to Haring Township to restrict the use of private wells in North Park and Pine View subdivisions and to require residents to abandon private wells in the affected area and connect to the township water system. The proposed ordinance was rejected. EPA tested 16 wells of concern identified from the survey on May 2008. No well contamination above health-based limits was identified in the residential wells. One well contained TCE below the MCL. The Wexford County DPW is currently proposing to Haring Township a new ordinance.

Recommendations from Previous Review	Party Responsible	Action Taken
Re-evaluate the LDFA plume capture zone, and review design and current operation of the extraction well system to determine if the system is effective and/or can be optimized. State has filed, and will continue to enforce, a Consent Judgment against the facility that is the source of the plume possibly being captured by the Kysor remediation system. Continue to track progress of enforcement actions and to evaluate data for indications of commingling of plumes.	USEPA/ Respondents MDNRE	EPA evaluated the capture zone and issued a report in April 2007 summarizing data collected through then. Preliminary analysis showed that the plume capture system is effective. The Michigan Consent Judgment requires Rexair to purge 8-9 wells on its property for TCE removal; however, Rexair is not required to take any actions for groundwater contamination beyond its property boundary. The MDNRE is unable to take further actions against Rexair. The Brooks and Perkins site (now AAR) is under settlement negotiations with MDNRE regarding a TCE plume.
Review standards to determine the most appropriate requirements for the Kysor Site. Verify estimated pounds per hour of VOCs in the discharge to the air from the treatment plant.	Respondents	These standards were reviewed and determined to be appropriate. The estimated rate of discharge for VOCs to the air was verified in June 2006.

VI. FIVE-YEAR REVIEW PROCESS

Administrative Components

MDNRE and the city of Cadillac LDFA were notified of the initiation of the Five-Year Review in February 9, 2010. The preparation of the Kysor Industrial Site Five-Year Review was led by Sheila Sullivan of EPA, with assistance provided by Bill Bolio, Nick Swiger and Steve Cunningham of the MDNRE. EPA was the lead agency for the review.

The components of the Five-Year Review schedule include:

- Community Notification
- Document Review
- Data Review
- Site Inspections
- Report Development and Review

Community Involvement

A Public Notice announcing that a Five-Year Review of the Kysor Industrial Site was to be conducted, was published on March 13, 2010 in the Cadillac News (see Attachment 4). The city has been very satisfied with the progress of the cleanup and has requested that MDNRE and EPA evaluate its proposal to shut down the chromium treatment system and conduct preliminary soil

sampling for the SVE system as part of the first step to develop a closure plan, if sample results support it, for the SVE system.

Document Review

This Five-Year Review consisted of a review of relevant documents including RODs, the ESD, investigatory reports and studies, correspondence, O&M records, hydrogeological studies, city of Cadillac summary description of LDFA and annual budgets, expert opinion reports, annual evaluation reports, and monitoring data (see Attachment 5).

Data Review

Descriptions of data related to the Kysor Industrial are provided below.

SVE System

Since the SVE system began operating in September 1996, over 220 million cubic feet of air have been treated, and 669 pounds of VOCs have been extracted from the waste pit areas behind the former Kysor Corporation plant. During the 1991 predesign studies, almost 500 pounds of VOCs were removed for a total of 1,168 pounds of VOCs removed. The estimated time in the OU2 ROD for achieving cleanup levels was two years.

The 669 pounds removed during full operation of the system was extracted from September 1996 to September 2000. Since September 2000, no VOCs have been detected at the influent sampling ports, meaning no VOCs from the soil have entered the system. Thus, a change-out of the activated carbon has not been required since April 1999 and the system has captured 99 percent of incoming VOCs. The lack of VOCs in the influent air to the system is one indication that soil cleanup levels have been achieved. The SVE system design was reviewed and a soil sampling plan developed and executed to provide more information about residual contamination in soils both in the area of influence and east of the area that was treated (see Figure 5).

The LDFA, in consultation with the city of Cadillac, EPA, and MDNRE, recommended shutting down the SVE remedy based on: 1) site characterization; 2) design; 3) monitoring; and 4) mass flux to and from ground water. The evaluation of these components provided evidence to support shutting down the SVE. This evidence was combined with hot spot-biased soil sampling in the vadose and saturated zones to evaluate the SVE system. The biased soil sampling was to target potentially remaining hot spots and SVE stagnation points. EPA and MDEQ approved the SVE soil sampling in November 2005. VOC concentration measurements were made at the individual extraction wells, the combined vapor before carbon treatment, the mid-point of carbon treatment, and after carbon treatment. The last detection of VOCs in an extraction well was September 6, 2000 (see Attachment 6, Graph 1).

Another indicator that residual soil contamination is low enough to be protective are results from the four groundwater monitoring wells immediately downgradient of the SVE area. All four wells nearest the Site, S-1 and SLW-25 (closest to the plant) and S-2 and SLW-9, have shown decreasing concentrations of COCs. The table below shows the maximum concentrations in ug/L observed at the wells between 1996 and 2007:

Maximum Concentration in Groundwater (ug/L)

Parameter	SLW-25	S-1	TCL
1,1-DCE	2.0	1.2	5
cis-1,2-DCE	0.5	58	70
trans-1,2-DCE	0.3	0.3	100
PCE	9.8	12	1
Toluene	2.2	36	40
TCE	49	230	5
TCA	80	26	200
Xylenes	0.7	23	440

The SVE system's progress has been monitored over the past 13 years. The COC concentration trends in groundwater were evaluated for the two closest wells downgradient of the Kysor Industrial Site, shallow extraction well S-1 and monitoring well SLW-25. Attachment 6, Graphs 1 and 2, display the results for trend evaluation for S-1 and SLW-25, respectively.

Extraction well S-1 showed a sharp increase in contaminants during the first two years of operation. By December 1999, most contaminants were below the groundwater TCLs in the ROD, and have remained so since the last annual sampling event in September 2009.

SLW-25 also showed an increase in contaminant levels within the first two years of active groundwater remediation and a gradual decrease since then. However, TCA had a second peak that was almost four times higher in concentration than the first peak four years into operation of the groundwater remediation system. At this time there was also a small spike in the TCE and PCE concentrations. Since 2003, the primary contaminants of concern have shown a decreasing trend in SLW-25. TCE has decreased from a maximum concentration of 49 µg/L to the current concentration of 13 µg/L, above the TCL (5 µg/L). PCE decreased from a maximum of 9.8 µg/L to a current concentration of 0.87 µg/L, below the TCL (1 µg/L). The decreasing concentrations observed in these wells suggested that the soils at the Kysor Site were no longer a source of contaminants to the groundwater.

A formal closure plan was prepared and closure sampling was conducted. Based on a review and concurrence of EPA and MDNRE, the system is being dismantled. The underground piping will remain in place should the system need to be reactivated for any reason. The protruding well points will be sheared to ground level to allow for reuse of the area.

Discharge to Clam River

VOC requirements in the discharge of treated groundwater to the Clam River have been met without exception since treatment began in September 1996. The NPDES permit will expire in October 2011. Concentrations of VOCs in the effluent from the air stripping system have also been non-detect since then. Since start-up of the system, TCE has always been less than 450 µg/l. The concentration of TCE entering the treatment plant from the VOC extraction wells has declined since September 1996; currently the average concentration in the air stripping influent is about 150 µg/l. This indicates that the system is more than sufficiently designed for the levels of TCE extracted so far. The decreasing concentrations in the influent also may indicate that

progress is being made toward cleaning up the Site. At the midpoint between the PTAS, only two VOCs (TCE and 1,2-DCE) are consistently detected. TCE ranges from about 4 to 6.5 ug/L and 1,2-DCE from 1-3 ug/L.

City Well #7 (CW-7)

CW-7 is one of seven wells in the city of Cadillac's municipal well field. At its widest point, the eastern edge of the Kysor Site plume is adjacent to the western border of the well field. CW-7 is the northernmost of the seven city wells. The well has been out of operation since 1994 due to TCE detections in 1993.

Because TCE and 1,2-DCE are the only two VOCs being detected in CW-7, and based on historical ratios of PCE, 1,1,1-TCA and other constituents in the various plumes in the area, an MDNRE report, reviewed by three independent experts, concluded that it was probable that the contamination in CW-7 is from the Rexair groundwater plume.

Areal Extent of Kysor Industrial Site Plume

The previous Five-Year Review report for the Kysor Industrial Site concluded that the concentration of TCE entering the treatment plant had declined from around 450 ug/l during the early years of operation, to about 180 ug/l; that only 12 percent of sampling results from monitoring wells were above TCLs; and, that the number and magnitude of exceedences had decreased since 1997. It also stated that the VOC groundwater plume extends into the North Park and Pine View residential areas where private wells still exist. Well sampling identified that groundwater contaminated by TCE and cis-1,2-DCE above their respective cleanup levels is present in the area of the North Park and Pine View subdivisions north of Thirteenth Street; however no residential wells in this area are contaminated.

Under a 1991 Consent Judgment (amended in 1995) that MDEQ entered against Rexair, Rexair installed a number of extraction wells and is treating groundwater using an air stripping system. Data collected in 2008 by MDNRE question whether the Rexair plume is being completely captured and treated by the LDFA remediation system. The data show TCE (40 ppb) in the deeper aquifer. An adjacent well (I-8) shows no contamination. This well is screened in the upper intermediate aquifer at less than 160 feet bgl, but there is no separating clay layer at this location. Rexair has presented data to refute this claim. The Consent Judgment requires Rexair to pump and treat groundwater from eight to nine wells purge wells located on its property and northwest of its property near the city well field for TCE removal. Rexair is not required to take any further action for groundwater contamination beyond its property boundary.

The MDNRE is currently in negotiations with Brooks and Perkins (now AAR). A TCE plume from AAR, separate and distinct from the LDFA area plume, is located far southeast of the LDFA remediation area. The Mitchell-Bentley PCE plume has migrated northward. All purge wells for the system are still operating. One monitoring well shows PCE above the TCL of five ug/L. The MDNRE currently operates the remediation system as the company went bankrupt in 2006. The PCE levels have been declining. Although the number of plumes makes the evaluation of the Kysor Site plume more complex, the amount of data from the 340 wells and the

differences in the types of VOCs present in the various releases assists in differentiating between sources.

A plume capture analysis was performed in 2007 based on data up to that time frame. The analysis sought to answer several questions regarding the efficacy of the system. The analyses suggested that the combined groundwater remedies in the vicinity of the Cadillac Industrial Park area would groundwater contaminated to above TCLs for targeted VOCs. In some areas at the edges of the groundwater capture zone, evidence supporting hydraulic capture was limited, and the period required to complete the remediation may be on the order of tens of years.

Contaminant trends suggest that throughout much of the Kysor Industrial Site, the VOC plume does not appear to be migrating and is not increasing in extent. However, three areas of uncertainty have been identified. These are areas where contaminants, potentially exceeding the TCLs, are present in groundwater, and/or where there is uncertainty in the direction and magnitude of hydraulic gradients. These are areas where future data evaluations should focus on confirming groundwater flow directions and changes in contaminant concentrations (see Figure 9).

- Area 1 is located north of Thirteenth Street and extends into the North Park and Pine View subdivisions. In this area, contaminants have been found above TCLs in several wells. Pumping of well I-8 may lead to migration of contaminants from within the Cadillac Industrial Park area northward further into the North Park and Pine View subdivisions.

- Area 2 is located northwest of Thirteenth Street and Leeson Avenue. In this area, contaminants have been identified at relatively low levels but above TCLs, particularly in shallow well F-1S. Well F-1S, which defines the extent of the PCE Shallow Target Zone northwest of Thirteenth Street and Leeson Avenue, shows no significant trend in concentration. While contaminants identified at well F1S are likely to be recovered by the combined groundwater remedies, the extent of the Composite Target Zone is not well defined in this area and hydraulic gradients do not provide strong evidence that groundwater extraction will ultimately capture contamination above TCLs from this area.

- Area 3 is located west of Leeson Avenue and the Cadillac Industrial Park. In this area, contaminants have been identified at relatively low levels but above TCLs for several wells. In particular, extraction well I-9 has shown VOC levels in excess of TCLs.

The groundwater remedy is performing as intended and the overall objectives are being met. The LDFA and Rexair groundwater systems near the Cadillac Industrial Park are recovering groundwater contaminated above TCLs and are containing the plume. Originally, the shallow LDFA wells were designed to recover shallow groundwater, but some of the shallow wells operate at less than design rates. Some shallow groundwater is recovered by upper intermediate LDFA extraction wells. The upper intermediate LDFA extraction wells generally operate at the design rates. Also, the majority of hexavalent chromium was recovered by wells designed to recover only VOCs. The Rexair groundwater remedy operates at different extraction rates than originally designed, but is consistent with revised operating plans.

The capture evaluation estimates that the combined groundwater remedies will clean up the VOC contaminated groundwater throughout most of the Site within the cleanup times projected in the ROD. It is possible that the current operation of some groundwater extraction wells, notably well I-8, increase the extent of groundwater contamination in the area of the North Park and Pine View subdivisions and may extend the time required for operating the LDFA remedy.

LDFA Groundwater Treatment System

Some statistics regarding progress of the LDFA include:

- More than 10.89 billion gallons of groundwater have been treated since 1999 (the inception of the program).
- The average gallon-per-year treatment amount over the past five years (2004-2008) was 871,414,000.
- The estimated pounds of VOCs stripped from the water per year since 1999 totaled 19,544 pounds with 7,341 pounds stripped of over the last five years (see Attachment 1).

Monitoring Results

- Concentrations of VOCs have decreased in the shallow extraction wells and in a number of the intermediate extraction wells.
- Concentrations of contaminants are above TCLs in seven shallow wells and 14 of the intermediate wells since 2005. Graphs 4-8 show concentrations for TCE in shallow wells and Graphs 9-12 for TCE and 1,1,1-TCA in intermediate extraction wells.
- Number and magnitude of exceedences have decreased since 2004.
- The LDFA influent is the combined groundwater collected by the extraction wells. Influent is pumped to the PTAS. This groundwater represents the total VOC concentration in groundwater prior to treatment and discharge to the Clam River. During the time the system has operated, the influent concentrations have shown a steady decline from about 680 ug/L total VOCs to 150 ug/l. Over the last five years, the concentration has dropped from 220 ug/L to 150 ug/L (see Graph 13 for all VOCs in groundwater influent).
- PCE and TCE are the two COCs for which multiple wells have not attained TCLs.
- None of the samples collected from private wells in the North Park subdivision have had detectable levels of any VOCs. In addition, shallow groundwater monitoring wells in the subdivision have only shown two TCL exceedences. Both exceedences were for TCE and occurred in 1997.

Forty-five wells are part of the routine groundwater monitoring program for the Kysor Site. The monitoring program for VOCs in groundwater consists of 23 shallow and 22 intermediate wells. Three intermediate wells are sampled quarterly (F-6, CMS-5W, and SLW-34), and the remaining are sampled annually (see graph 14). Well F-6 and CMS-5W are located to the west of the Site beyond the lateral extent of the plume in that direction, and SLW-34 is located in the North Park subdivision.

In 2007-2008, MDEQ conducted four deep well borings within and to the northwest of the Cadillac Industrial Park (see Figure 10 and Table 11). The borings were installed below the clay at depths ranging from about 200-300 feet bgl. Two borings (WFA-2 and WFA-3) respectively located to the west and northwest of the city well field (also screened in the deep aquifer) showed no contamination.

Another deep well field boring (WFA-1) located just north of CW-7 showed TCE levels up to 91 ug/l at 245 feet bgl, but none was detected below 257 feet. Benzene was also detected at levels ranging from 34-38 ug/L at 300 feet bgl in this boring. The chemical profile of TCE contamination in groundwater suggests the contamination is attributable to the Rexair release and further, that TCE did not migrate through the aquitard but was pulled back from north of Thirteenth Street by the pumping of the municipal wells. The vertical extent of TCE was also confirmed by the presence of TCE (40 ug/l) at 227 feet bgl in WFA-4.

Regional groundwater flow in the deep well field aquifer was previously determined to be west northwest. However, pumping influences from the city water supply wells have significantly altered localized flow towards the southeast, as apparent from the relatively lower water elevation nearest to the city well field.

Six shallow/intermediate borings were installed northwest of the Rexair facility. In general, the direction of groundwater flow is the west-northwest with several purge wells locally altering the natural direction of flow, specifically in the vicinity of the Rexair (PW-1 through PW-8) and LDFA purge wells I-3, I-7, and I-11 (see Figure 11). The vertical extent of TCE impacts in the shallow intermediate aquifer has been determined in the vicinity of the northern half of the city well field, near CW-7, to extend below 100 feet. One boring (DMW-1) directly northeast of the city well field showed TCE up to 33 ug/l at about 165 feet bgl. A second boring (DMW-2) just north of CW-7 showed TCE as high as 120 ug/l also at 165 feet bgl. The other four borings (DMW-3, DMW-4, DMW-5 and DMW-6) showed no contamination.

The chemical fingerprints of groundwater contaminants in all groundwater samples collected from the and monitoring wells, DMW-1, DMW-2 (DMW-2S and DMW-2D and DMW-3 are consistent with that of the Rexair plume, which suggest this groundwater contamination is attributed to the Rexair release.

The preliminary data assessment for monitoring well data collected since 2005 allows for some trend analysis. The following table identified discernable concentration trends for each of the COCs in the monitoring wells.

Table 5 – Contaminant Trend Summary for the Kysor/Northernaire Plume

Chemical	Trend	Wells	Exceedance Wells
1,1-TCA	Downward	Intermediate: I-4-to I-6, I-10, I-11 Shallow: S-2; S-4, S-5, SLW-25 (upper intermediate)	Intermediate: K-16D Shallow: S-2
1,1-DCE	Downward	Intermediate: I-10, I-4, I-5	Intermediate: I-1, to I-5, K-16D Shallow: S-2
cis-1,2-DCE	Downward	Intermediate: I-4 to I-6, I-10 Shallow: S-2, S-4,	Intermediate: I-1, K-16D
	Upward	I-1, I-2	
PCE	Downward	Intermediate: I-1, I-5, I-6, I-10, S-2, S-4, S-5 SLW-24 (upper intermediate), CW-17	Intermediate: I-1 to I-6, I-10, I-11, K-16D, SLW-25 (upper intermediate), DNR-14, Shallow: F-1-S, FWMW-16 F-5, SLW-9, GW-10S
	Upward	Intermediate: I-2, I-4, K-16D	
TCE	Downward	Intermediate: I-1, I-10, I-11, I-2, I-4 to I-7, Shallow: K-6-S, S1, S-2, S-4, S-5, GW-17	Intermediate: I-1, I-10, I-11, I-2 to I-7, K-16-D, F-5, FWMW-16, FWMW-9, Shallow: K-13, K-6-S, S-2 to S-4, S-6, SLW-25, GW-17

Operating Conditions and Design Estimates – Groundwater Extraction System

The objective of the OU2 ROD for groundwater at the Kysor Site is to achieve TCLs in both the monitoring wells and the extraction wells.

The ROD estimated that VOC contamination would be reduced to acceptable levels within the shallow aquifer 29 years after system startup. Given that the LDFA began operating its groundwater remedy 1996, a projected cleanup time of 29 years suggests that the VOCs would be successfully remediated by about 2025 throughout much of the area and that most remaining contamination will be recovered by 2046. Some portions of the Kysor Industrial Site, such as the area between the Rexair and LDFA remediation systems, and the area of the North Park and Pine View subdivisions, could take considerably longer to remediate due to the extent of the contaminated aquifer volume and the low hydraulic gradients in these areas. It was estimated that VOC contamination in the intermediate aquifer would be reduced to acceptable levels within the upper intermediate aquifer 64 years after system startup and that, therefore, the VOC contamination would be successfully remediated by the year 2060.

Site Inspection

The Five-Year Review site inspection of the Kysor Industrial Site was conducted on March 21, 2010 by the EPA Remedial Project Manager and city of Cadillac officials. In addition, staff from the MDNRE Cadillac area office were consulted. The purpose of the inspection was to

assess the progress of remedy implementation, ensure records and site documents were available and current, inspect treatment units to verify they were operational and have no significant problems, and view general site conditions and areas of other groundwater releases and plumes in the Industrial Park. The intent was to collect information to assess the protectiveness of the remedy and try to foresee any future remedy implementation problems and needs.

Most issues related to the remedy for the Kysor Site were identified at the time of the Site inspection or prior to it. Because of the planned decommissioning of the SVE system, it was not operating. The treatment plant appeared to be very well-maintained. A new operating control system had been installed to manage the PTAS system. No significant problems were noted (see Attachments 7-8 for site inspection and photos).

The city of Cadillac has been studying several potential locations for a new city well field. This is needed due to the age of the current system and the need to expand. The city originally had seven wells (CW-1 through CW-7). In the 1980s, the casing ruptured in CW-5 and the well had to be abandoned. CW-7, located about 350 feet north/northwest of the city well field, is no longer used due to TCE contamination. The city currently operates CW-1, CW-2, CW-3, CW-5 and CW-6. Due to the necessary isolation distance, the city cannot add new wells to the system. The selected location for the new well field is two miles southeast of the existing well field and southwest of the 55/East Division and 131 interchange. This location is downgradient of the Kysor/Northernair Sites and the Cadillac Industrial Park, as well as the groundwater contamination plumes associated with these areas. The city projects that the well field construction will be half-completed by 2012, and that the current well field will be abandoned by 2014 (see Figure 12).

VII. TECHNICAL ASSESSMENT

Question A: *Is the remedy functioning as intended by the decision documents?*

Yes. The remedy is functioning as designed and is protective in the short term. The groundwater extraction and treatment system is operating as planned. The extraction system was effectively designed and is capturing groundwater contaminated with VOCs from the Kysor Industrial Site. The air stripping system operates at 99 percent efficiency. As a result, no VOCs have been detected in treated groundwater since it began operating. All surface water discharge requirements for VOCs have been met since remediation began. Since the last Five-Year Review, over six-billion gallons of groundwater have been extracted and treated, and over 20,000 pounds of VOCs have been removed from groundwater. The SVE system has achieved the cleanup goals for contaminants in the soil, which were a continuing source of groundwater contamination. Since 1996, the SVE system has removed over 1,000 pounds of VOCs. The SVE system is being decommissioned under EPA and MDNRE approval.

The groundwater monitoring programs are adequate. The combined monitoring networks of the LDFA, Rexair and adjacent remedies provide an opportunity for wide spatial coverage of groundwater hydrogeology and water quality data for remedy evaluation. A coordinated data sharing initiative was attempted in the past, but has not been implemented to date. Another issue

of concern is the possibility that the groundwater extraction and treatment system for the Kysor Industrial Site is capturing and treating contaminated water from the Rexair site. If this is true, this may increase the time needed to achieve Kysor groundwater cleanup goals. The Kysor groundwater remedial action may continue to operate in the future in order to address Rexair contamination if there is no way to distinguish the source of the contamination, and only Rexair contamination remains. A Consent Judgment between Rexair and the State of Michigan requires Rexair to pump and treat groundwater from about eight wells on its property.

Long-term protectiveness of the remedy will be achieved when groundwater cleanup goals are achieved. In the mean time, two local ordinances exist, one in Haring Township and one in the city of Cadillac, to regulate groundwater use. In addition, LTS procedures should be reviewed to ensure that the ICs are maintained, monitored and enforced.

Question B: *Are the exposure assumptions, toxicity data, cleanup levels, and RAOs used at the time of the remedy selection still valid?*

Yes. The assumptions and information on which the OU2 ROD was based are still valid. The city well field should continue to be monitored to ensure that actual exposure to contaminated drinking water does not occur. There have been no changes in the physical conditions at the Site that would affect the protectiveness of the remedy. As mentioned, the SVE system is in the process of being decommissioned.

Changes in Standards and To-Be-Considered Requirements

There have been no changes in the site ARARs and no new standards or to be considered (TBC) requirements affecting the protectiveness of the remedy.

Changes in Exposure Pathways, Toxicity, and Other Contaminant Characteristics

There have been no changes in exposure pathways or toxicity factors for the COCs that were used in the baseline risk assessment. No changes to these assumptions appear to be needed. Furthermore, there has been no change to the risk assessment methodology that could affect the protectiveness of the remedy. The remedy is progressing as expected and it is expected that all cleanup goals will be met, as specified in the OU2 ROD.

Question C: *Has any other information come to light that could call into question the protectiveness of the remedy?*

No. Although it is not entirely clear whether the LDFA extraction well system is capturing the Rexair plume at this time, as long as treatment continues until TCLs are met and groundwater use measures are in place, the remedy will remain protective. Plume definition should continue in the deeper aquifer. The LDFA system has the capacity to capture and treat the Rexair plume; however, this may result in a longer remediation period to meet TCLs and may increase the cost of the remedy.

Technical Assessment Summary

According to the data reviewed, information gathered at the Site inspection, interviews, and a review of relevant documents, the remedy is functioning as intended by the ROD, as modified by the ESD, and is protective in the short term. No exposure pathways are complete, and none of the samples collected from private wells in the North Park subdivision had detectable levels of any VOCs. In addition, shallow groundwater monitoring wells in the subdivision have only shown two MCL exceedences. Both exceedences were for TCE and occurred in 1997. EPA should continue to explore improving the ICs in the city of Cadillac and Haring Township area. Strengthening the ICs will ensure the prevention of exposures to contaminated groundwater until the groundwater TCLs are met.

VII. ISSUES

Table 6 - Issues

Issue	Currently Affects Protectiveness (Y/N)	Affects Future Protectiveness (Y/N)
1. Haring Township and the city of Cadillac ordinances that serve to prevent drinking water exposures to contaminated groundwater should be strengthened to ensure effectiveness by 1) requiring all current and future residents to connect to their respective municipal water systems; 2) requiring current and future residents in Haring Township and the city of Cadillac to abandon their private wells, and 3) developing methods to achieve better compliance with current and future ICs.	N	Y
2. Initial IC evaluations revealed that additional steps must be taken to assess the effectiveness of ICs and to ensure long-term protectiveness. The evaluation supports follow-up actions to assure that the remedy remains protective. LTS procedures should be reviewed to ensure long-term protectiveness.	N	Y

IX. RECOMMENDATIONS AND FOLLOW-UP ACTIONS

Table 7 - Recommendations and Follow-Up Actions

Issue	Recommendations/ Follow-Up Actions	Party Responsible	Oversight Agency	Milestone Date	Affects Protectiveness? (Y/N)	
					Current	Future
1. Haring Township and the city of Cadillac ordinances that serve to prevent drinking water exposures to contaminated groundwater should be strengthened to ensure effectiveness by: 1) requiring all current and future residents to connect to their respective municipal water systems; 2) requiring current and future residents in Haring Township and the city of Cadillac to abandon their private wells; and 3) developing methods to achieve better compliance with current and future ICs.	1. EPA should encourage Haring Township and the city of Cadillac governments to tighten the existing ordinances and report the status and/or changes to the ordinances to EPA and MDNRE in order to evaluate future protectiveness.	Local Governments	EPA/ MDNRE	12/2012	N	Y
2. Initial IC evaluations revealed that additional steps must be taken to assess the effectiveness of ICs and to ensure long-term protectiveness. The evaluation supports follow-up actions to assure that the remedy remains protective. LTS procedures should be reviewed to ensure long-term protectiveness.	2. An IC Plan will be developed by EPA with input from MDNRE. The Plan will include the results of the IC evaluation, plan for additional IC activities as needed (such as mapping), and include a plan to develop more effective ordinances and LTS procedures to ensure that effective ICs are maintained, monitored and enforced. The IC Plan will also include a plan to periodically sample the potentially affected private wells in order to ensure the drinking water is safe until local ICs are improved.	EPA/ MDNRE	EPA	12/2011	N	Y
Protectiveness Statement						

X. PROTECTIVENESS STATEMENT

The remedy for the Kysor Industrial Site continues to be protective of human health and the environment in the short term. Long-term protectiveness will be achieved when groundwater cleanup goals are met and the implementation of effective ICs is monitored, maintained and enforced. In the meantime, ICs are in place in the form of two local ordinances in Haring Township and city of Cadillac to regulate groundwater use. Initial IC evaluations have revealed that these ordinances should be strengthened by: 1) requiring all current and future residents to connect to their respective municipal water systems; 2) requiring current and future residents in Haring Township and the city of Cadillac to abandon their private wells; and 3) developing methods to achieve better compliance with current and future ICs.

Long-term protectiveness requires compliance with effective groundwater use restrictions. This will be accomplished by ensuring effective ICs are in place and planning for long-term stewardship that includes maintaining, monitoring and enforcing effective ICs. In addition, the existing private wells should be periodically tested.

XI. NEXT REVIEW

The next Five-Year Review will be completed five years from the date of this review.

Tables

Table 1 – Chronology of Site Events.....	12
Table 2 – Institutional Controls Summary.....	29
Table 3 – Annual System Operations/O&M Costs.....	31
Table 4 – Actions Taken Since the Last Five-Year Review.....	32
Table 5 – Contaminant Trend Summary.....	40
Table 6 – Issues.....	43
Table 7 – Recommendations and Follow-Up Actions.....	44
Table 9 – LDFA Extraction Well Construction Summary	Attachment 9
Table 10 – LDFA Monitoring Plan	Attachment 9
Table 11 – Boring Well Placement for MDNRE Plume Investigation	Attachment 9

Figures

Figure 1 – Site Location Map
Figure 2 – Plume Map
Figure 3 – Cadillac Industrial Park Area
Figure 4 – Location of Cadillac City Well Field
Figure 5 – SVE System Layout and Former Excavation Area
Figure 6 – Diagram of Extraction Well Discharge and Piping
Figure 7 – North Park Subdivision Residential Well Users
Figure 8 – Residential Well Reconnaissance of November 2004
Figure 9 – Identified Areas of Principal Uncertainty for Contamination
Figure 10 – MDNRE Plume Investigation in Deep Well Aquifer
Figure 11 – MDNRE Plume Investigation in Shallow/Intermediate Well Aquifer
Figure 12 – City of Cadillac New Well Field Site Map and Water System Improvements

Attachments

Attachment 1 – LDFA Operating Budget and O&M Statistics
Attachment 2 – Cadillac Ordinance and Recertification
Attachment 3 – Haring Township Ordinance
Attachment 4 – Advertisement of Five-Year Review Start
Attachment 5 – List of Documents Reviewed
Attachment 6 – Graphs
Attachment 7 – Site Inspection Checklist
Attachment 8 – Inspection Documentation Photos
Attachment 9 – Additional Tables

FIGURES

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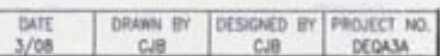
Figure 8 – Residential Well Reconnaissance of November 2004

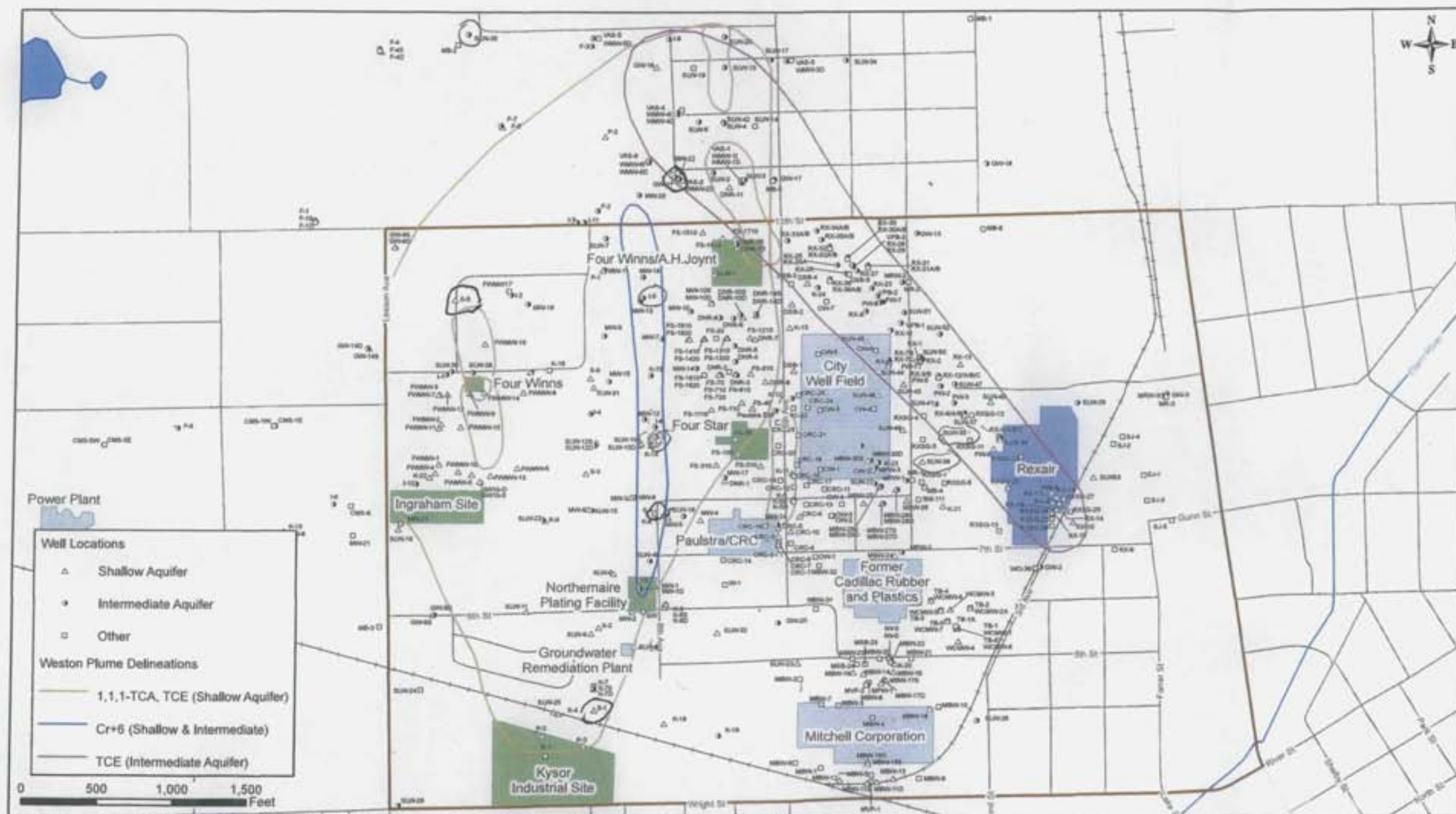
Figure 9 – Identified Areas of Principal Uncertainty for Contamination

Figure 10 – MDNRE Plume Investigation in Deep Well Aquifer

Figure 11 – MDNRE Plume Investigation in Shallow/Intermediate Well
Aquifer

Figure 12 – City of Cadillac New Well Field Site Map and Water System
Improvements





Note: Weston generalized plume depictions (Weston, 2003) are illustrated solely for the purpose of spatial reference since these have been produced previously; they form no basis of any evaluations in this report.

Note: This figure illustrates all wells for which spatial data are available. Not all wells are (a) actively pumping, or (b) included in a monitoring program. See text for details.

FIGURE 2

General Extent of Groundwater Contamination and Locations of Wells



Cadillac Area Location Map

FIGURE 3

Location of City of Cadillac Well Field



0.25 0.125 0 0.25 Miles

Plot created by Andrew Foster U.S. EPA Region 5 on 8/27/2005

FIGURE 4



LEGEND

- KB-1 ● JULY 1987 BORINGS
- KB-11 ● 1992 PILOT TEST BORINGS
- I-1 ● INDUCTION WELLS
- E-1 ● EXTRACTION WELLS
- EXTRACTION WELL RADIUS OF INFLUENCE
- (897) MAXIMUM VOCs IN BORING (µg/kg)
- (752,000) MAXIMUM VOCs ISOCONCENTRATION CONTOURS (µg/kg)
- PVC LINER
- CONCRETE/ASPHALT CAP
- ▨ FORMER EXCAVATION

NOTE: AERIAL PHOTOGRAPH PROVIDED FOR REFERENCE ONLY

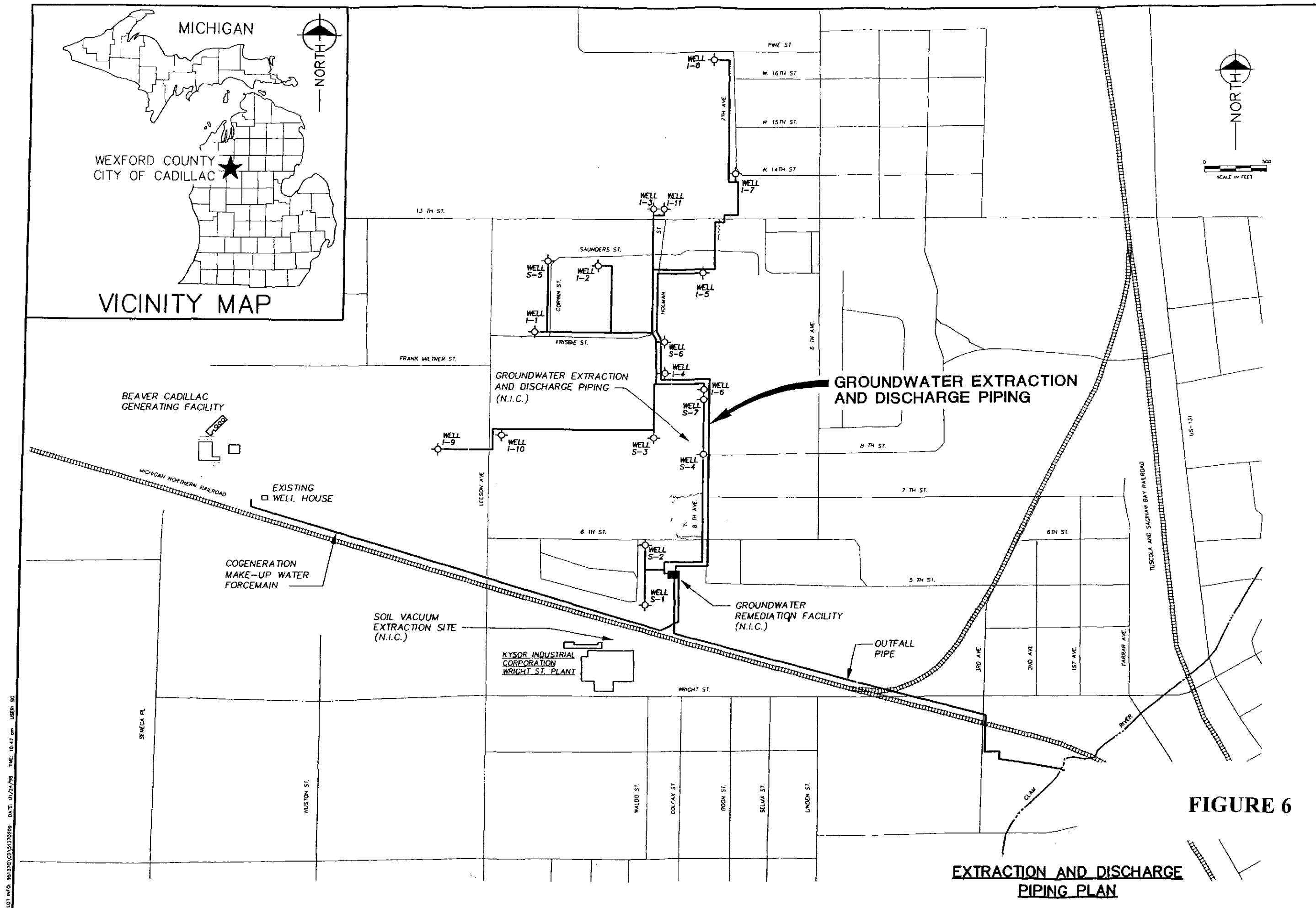


FIGURE 6

EXTRACTION AND DISCHARGE
PIPING PLAN

Plot Info: \$51370/GD/\$170288 DATE: 01/24/98 TIME: 10:47 am USER: SG

NO	DATE	Drawn By	Checked By	Designed By	DATE
					1997

REVISIONS

fishbeck, thompson, carr & huber
engineers • scientists • architects
Ada • Lansing • Kalamazoo
Michigan

fic&h
SINCE 1956
300 East State Ave., Kalamazoo, MI 49001
Phone: (269) 336-1100, Fax: (269) 336-1101

CADILLAC LDFA
CADILLAC, MICHIGAN

ANNUAL REPORT 1997

North Park Subdivision - Residential Well Uses Cadillac, MI

Superfund
U.S. Environmental Protection Agency

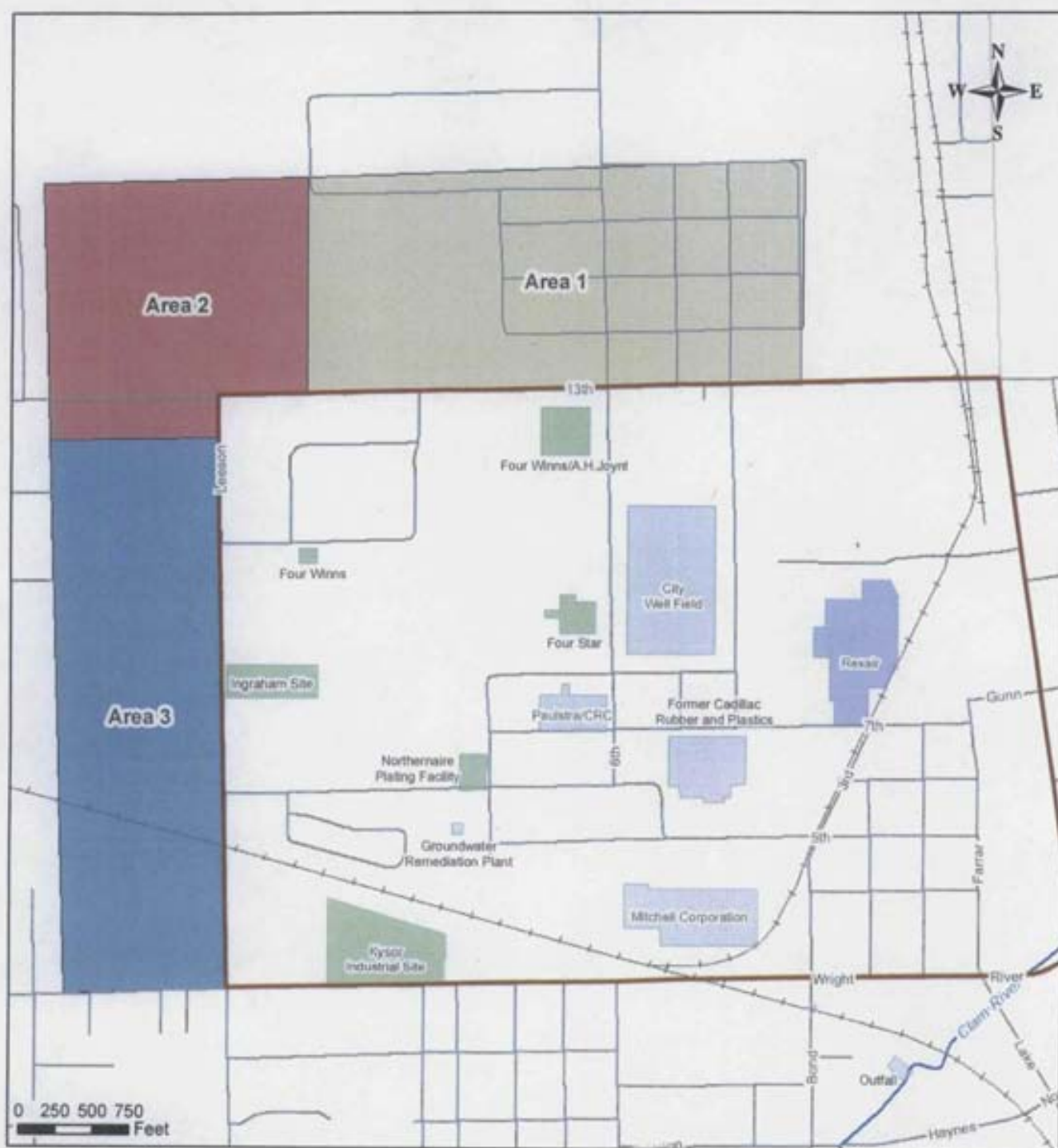


* Copyright Wexford County 2006
(*includes aerial photograph)

Created by Sarah Backhouse
U.S. EPA Region 5 on 6/27/2006

0 200 400 Feet

FIGURE 7



Areas of Principal Uncertainty

FIGURE 9



NOTE:

Groundwater Elevations in The Deep Well Field Aquifer Estimated Using Water Level Measurements Collected From 6 Monitoring Wells On January 9, 2008.

TCE Plume Estimated Using Data Collected By MSG From January 10-11, 2008.

ND = Not Detected

TCE = Concentrations in ug/L

Aerial Photography Adapted From the MDEQ, 1992.

LEGEND

- ▲ DMB-1 VAP BORING/MONITORING WELLS, SHALLOW/INTERMEDIATE AQUIFER, INSTALLED BY MANNIK & SMITH GROUP, INC.
- ▲ WFA-1 VAP BORING/MONITORING WELLS, DEEP AQUIFER, INSTALLED BY MANNIK & SMITH GROUP, INC.
- + EXISTING MONITORING WELLS, INSTALLED BY OTHER CONSULTANTS.
- + EXISTING MONITORING WELLS, INSTALLED BY OTHER CONSULTANTS AND SAMPLED DURING MANNIK & SMITH GROUP HYDROGEOLOGIC INVESTIGATION
- ⊙ BENCHMARK LOCATIONS SET OR UTILIZED DURING SURVEY
- APPROXIMATE POTENTIOMETRIC SURFACE - DEEP WELL FIELD AQUIFER, JANUARY 9, 2008, (DASHED WHERE INFERRED)
- ESTIMATED REXAIR TCE GROUNDWATER PLUME EXTENT - DEEP WELL FIELD AQUIFER JANUARY 2008.
- APPROXIMATE NORTHERN EXTENT OF WELL FIELD "CLAY"
- TCE-ND TCE-20-TRICHLOROETHENE (TCE) CONCENTRATION MEASURED GROUNDWATER ELEVATION - 1/9/08

The
Mannik & Smith
Group, Inc.
Civil Engineering, Surveying and Environmental Consulting

FIGURE 10

fic&h

engineers
scientists
architects
constructors

fishbeck, thompson,
carr & huber, inc.

Hard copy is
intended to be
8.5"x11" when
plotted. Scale(s)
indicated and
graphic quality may
not be accurate for
any other size.

City of Cadillac
Wexford County, Michigan

DWRF Project Plan / Water System Improvements

PROJECT NO.
G090025PS



LEGEND

- EXISTING TEST PRODUCTION WELL
- EXISTING OBSERVATION WELLS
- PROPOSED PRODUCTION WELLS



WELL FIELD SITE MAP

SCALE: 1" = 500'

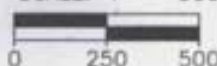


FIGURE 12

ATTACHMENTS

Attachment 1 – LDFA Operating Budget and O&M Statistics

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Attachment 3 – Haring Township Ordinance

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Attachment 9 – Additional Tables

Cadillac, Michigan

2009-2010 Annual Operating Budget
Local Development Finance Authority Operating Fund**L.D.F.A. Groundwater Treatment Statistics****Volume of groundwater pumped and treated:**

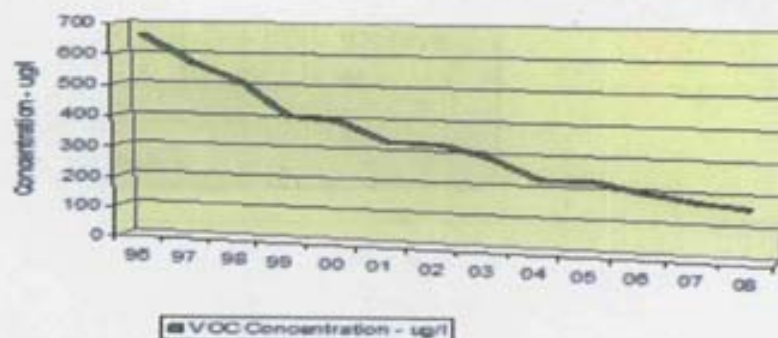
Year	Gallons	
	Per Day	Per Year
1999	2,436,500	889,330,000
2000	2,410,000	880,000,000
2001	2,384,000	870,180,000
2002	2,315,100	845,000,000
2003	2,331,500	851,000,000
2004	2,407,100	878,600,000
2005	2,339,300	853,850,000
2006	2,387,000	871,400,000
2007	2,422,000	884,190,000
2008	2,401,700	879,030,000



LDFA Water Treatment Facility

Year	Estimated pounds of volatile organics stripped from the water (per year):	Hours spent in Operation and Maintenance
1999	2,761	380
2000	2,628	435
2001	2,402	770
2002	2,322	890
2003	2,090	940
2004	1,715	970
2005	1,673	830
2006	1,453	940
2007	1,305	1,140
2008	1,195	1,014

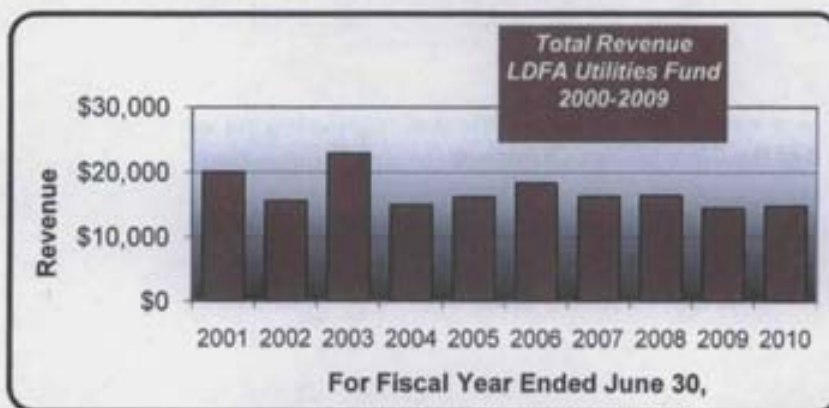
LDFA GWTP - Influent VOC Concentration



Fund Highlights

Revenue

The customer water rate is tied to the electrical rate Consumers Energy pays for wholesale power. Only one customer is served, Cadillac Renewable Energy, and they provide the electrical energy for the well operation.



System Statistics

Volume of Water Pumped	
Year	Gallons
1999	158,620,100
2000	143,838,000
2001	152,795,000
2002	162,571,000
2003	168,144,588
2004	152,780,000
2005	140,912,000
2006	148,930,000
2007	123,957,000
2008	147,034,000



LDFA well house (foreground) with Cadillac Renewable Energy in background



LDFA Well

Hours spent in operation and maintenance	
Year	Hours
1999	168
2000	154
2001	148
2002	122
2003	115
2004	170
2005	134
2006	125
2007	175
2008	151



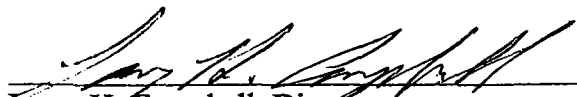
200 North Lake Street • Cadillac, Michigan 49601
231.775.0181 • fax 231.775.8755
www.cadillac-mi.net

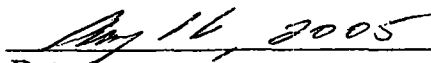
ATTACHMENT 2

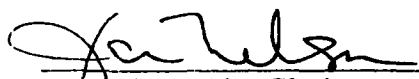
Certification of Institutional Controls

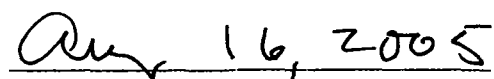
The City of Cadillac hereby certifies that Ordinance #97-10, Chapter 24, Section 2.300, p. 2-83, establishing institutional controls for the remedial action at the Kysor/Northernire site within the City of Cadillac is still in effect. Said Ordinance was adopted November 3, 1997, and is part of the Codified Ordinances of Cadillac, Michigan.

Certified By:


Larry H. Campbell, Director
Cadillac Utilities Department


Date: Aug 16, 2005


Jan Nelson, City Clerk


Date: Aug 16, 2005

AN ORDINANCE TO AMEND THE CITY CODE OF THE CITY OF CADILLAC TO ADD A NEW SECTION 2.300 TO CHAPTER 24, TO ESTABLISH INSTITUTIONAL CONTROLS FOR REMEDIAL ACTION AT THE KYSOR INDUSTRIAL CORPORATION/NORTHERNAIRE PLATING COMPANY SITE LOCATED IN THE CITY OF CADILLAC.

THE CITY OF CADILLAC ORDAINS:

Section 1.

For the purpose of protecting public health, welfare and the environment, and for the purpose of implementing the remedial action plan at the site commonly known as the Kysor Industrial Corporation/Northernair Plating Company Superfund Site, Section 2.300 is hereby added to read as follows:

Chapter 24, Section 2.300.

A. Use of the following described real estate shall be restricted by the provisions of this Subsection(a):

All land located in Township 22 North, Range 9 West, City of Cadillac, Wexford County, Michigan, described as follows:

1. The East Quarter (E 1/4) of the Northwest Quarter (NW 1/4) of Section 32.
2. The Northeast Quarter (NE 1/4) of Section 32.
3. The North Half (N 1/2) of the Southeast Quarter (SE 1/4) of Section 32.
4. The Southwest Quarter (SW 1/4) of Section 33 lying North and West of the Tuscola-Saginaw Bay Railroad.
5. The Northwest Quarter (NW 1/4) of Section 33, EXCEPT the following: South of Gunn Street and Seventh Street which is East of the Ann Arbor Railroad; the property lying East of the Pennsylvania Central Railroad; and also commencing as the Point of Beginning at the Southwest corner of Block 179 of the Improvement Board Addition; thence North to the Northwest corner of Block 188; thence East along the North line of Block 188 to the Northwest corner of Block 189; thence East along the North block line, 220 feet; thence South 71 feet; thence East 107.43 feet; thence North 71 feet; thence East 212.83 feet; thence South 16°2'30" East, 331.74 feet; thence South 3°28'30" East, 246.56 feet to the West right-of-way line of the Ann Arbor Railroad; thence Southwesterly along the West right-of-way line of the Ann Arbor Railroad to the Southeast corner of Block 177; thence West along the South line of Block 177 to the centerline of Third Avenue; thence North on the centerline of Third Avenue to the

South line of Block 179 and Block 178, if extended; thence West to the Point of Beginning of the Improvement Board Addition, City of Cadillac, Wexford County, Michigan.

(Hereafter referred to as the Kysor Industrial Corporation/Northernair Plating Company Site (the "site")).

1. No water wells used for drinking water or any other domestic use shall be installed in the Kysor Industrial Corporation/Northernair Plating Company site (the "site"). There shall be no installation or operation of any wells that may interfere with the operation or maintenance of the groundwater extraction or treatment systems set forth in paragraph 2 following, except with written consent by the United States Environmental Protection Agency.
2. There shall be no tampering with, or removal of the containment or monitoring systems that remain on the site as the result of implementation of any response action by the United States Environmental Protection Agency, or any party acting under order by the United States Environmental Protection Agency, and which is selected and/or undertaken, or ordered by, the United States Environmental Protection Agency pursuant to Section 104 and/or 106 of the Comprehensive Environmental Response, Compensation and Liability Act (CERCLA).

Section 2. This Ordinance shall take effect twenty (20) days after its passage.

TOWNSHIP BOARD
CHARTER TOWNSHIP OF HARING
WEXFORD COUNTY, MICHIGAN

Boardmember Hofstra supported by Boardmember Cerwin moved the adoption of the following ordinance:

ORDINANCE NO. 86-01

WATER ORDINANCE

The Charter Township of Haring ordains:

ARTICLE I

DEFINITIONS

1.1. Definitions. In the interpretation of this ordinance, the following definitions shall apply unless the context clearly indicates otherwise:

- (a) "Backflow" means water of questionable quality, wastes or other contaminants entering a public water supply system due to a reversal of flow.
- (b) "City" means the City of Cadillac, Wexford County, Michigan.
- (c) "Commercial user" means a person whose premises are used to offer services and/or products such as, for example, retail and wholesale stores, gasoline stations, restaurants, schools, churches, hotels, motels, nursing homes, private clubs, theaters and governmental buildings.
- (d) "County" means Wexford County, Michigan.
- (e) "Cross-connection" means a connection or arrangement of piping or appurtenances through which a backflow could occur.
- (f) "Director" means the Director of the DPW or his authorized representative.
- (g) "Domestic user" means a person whose premises are domiciles for single or multiple family use.
- (h) "DPW" means the Wexford County Department of Public Works.

- (i) "Industrial user" means a person who operates a manufacturing or process facility which is engaged in producing a product.
- (j) "Local distribution lines" means those pipes which serve only the abutting property within only one local service area.
- (k) "Premises" means each lot or parcel of land, building, premises, dwelling unit or apartment unit having any connection to the water distribution system.
- (l) "Secondary water supply" means a water supply system maintained in addition to a public water supply, including but not limited to water systems from ground or surface sources not meeting the requirements of Act No. 98 of the Public Acts of 1913, as amended, being Sections 325.201 to 325.214 of the Compiled Laws of 1948, or water from a public water supply which in any way has been treated, processed or exposed to any possible contaminant or stored in other than an approved storage facility.
- (m) "Submerged inlet" means a water pipe or extension thereto from a public water supply terminating in a tank, vessel, fixture or appliance which may contain water of questionable quality, waste or other contaminants and which is unprotected against backflow.
- (n) "System" or "Water distribution system" means the Township water supply and distribution system which may be connected to the City of Cadillac's water system as its source of water and which may distribute the water purchased from the City of Cadillac to water service customers within the Township.
- (o) "Township" means the Charter Township of Haring, Wexford County, Michigan.
- (p) "Township-City Water Agreement" means the Agreement, together with Addendum to Water Agreement, by and between the Township and the City executed in 1985 pursuant to which the City sells water to the Township on a wholesale basis to serve users within a designated area of the Township, as it may be amended from time to time.
- (q) "User" means the owner, lessee or occupant of any premises connected to or served by the System.
- (r) "Water connection" means that part of the water distribution system connecting the watermain at the curb cock with the premises served.

- (s) "Watermain" means the primary and intermediate transmission and local distribution lines of the water distribution system.

ARTICLE II

CONNECTIONS

2.1. Water Connection Required. The owner of each house, building or property used for human occupancy, employment, recreation or other purposes situated within the Township and abutting on any street, alley, or right-of-way in which there is located or may in the future be located, a watermain or local distribution line of the System shall, at his expense (unless such expense is expressly provided by other sources) within 180 days after said watermain becomes available, install suitable plumbing facilities therein and to connect such facilities directly to the water distribution system, provided said watermain is within 200 feet of the nearest property line of said premises. Provided, however, no person shall be required or permitted to connect any premises to the System if such premises will be served by water purchased from the City and service of the premises would be in breach of the Township-City Water Agreement.

2.2. Service Connections. Applications for water connections shall be made to the DPW on forms prescribed and furnished by it. Water connections, plumbing and water meters shall be installed in accordance with the rules and regulations of the DPW and upon payment of the connection fee and meter installation fee set from time to time by resolution of the Township Board. Water meters shall be installed by the DPW, unless otherwise provided by resolution of the Township Board. All meters and water connections shall be the property of the Township. Water service will not commence until payment in full for the installation has been made to the DPW. Construction of a water service line shall be done as expeditiously as possible after written notice to proceed, but the time for construction shall be at the convenience of the Township and the DPW.

2.3. Institution of or Restarting Water Service. Written notice given not less than 48 hours in advance shall be made to the DPW by the property owner and/or occupant of the premises when water service is desired. It shall be unlawful for any person to connect to or use water supplied by the Township without first giving notice as provided herein. The Township and the DPW reserve the right to require that an amount of money equal to an anticipated bill for three months of water service be placed on deposit with the DPW for the purpose of establishing or maintaining any customer's credit. No person, other than an authorized employee of the DPW, shall turn on or off any water service, except that a licensed plumber may, with the prior express approval of the DPW, turn on water service for testing his work (when it must be immediately turned off) or upon receiving a

written order from the DPW; provided, that upon written permit from the DPW, water may be turned on for construction purposes upon payment to the DPW of the charges applicable thereto.

2.4. Meters. The Township reserves to the DPW the right to determine the size and type of meter used. The Township reserves to the DPW the right to require the installation of remote meter reading equipment. The cost of that equipment and its installation shall be charged to the user at the prevailing rates and actual cost of material and labor.

2.5. Access to Meters. The DPW shall have the right to shut off the supply of water to any premises where the DPW is not able to obtain access to the meter. Any qualified employee the DPW shall, at all reasonable hours, have the right to enter the premises where such meters are installed for the purpose of reading, testing, removing or inspecting the same and no person shall hinder, obstruct, or interfere with such employee in the lawful discharge of his duties in relation to the care and maintenance of such water meter.

2.6. Curb Box. No person shall remove the cover from any curb box or place any dirt, stone or other obstruction in it or tamper with any meter or commit any act tending to obstruct the use thereof.

2.7. Injury to Facilities. No person, except an employee of the DPW in the performance of his duties, shall break, damage, destroy, uncover, deface or tamper with any structure, appurtenance or equipment which is a part of the water distribution system.

2.8. Reimbursement for Damage. Any damage which a meter may sustain resulting from carelessness of the owner, agent, or tenant or from neglect of either of them to properly secure and protect the meter, as well as any damage which may be wrought by frost, hot water, or steam backing from a boiler, shall be paid by the owner of the property to the DPW on presentation of a bill therefor; and in cases where the bill is not paid, the water may be shut off and shall not be turned on until all charges have been paid to the DPW.

2.9. Responsibility for Damages. Neither the Township nor the DPW will be responsible for any damages because of failures of or within the System, or actions by the Township or DPW to correct such failures.

2.10. Water Leakage. It shall be the responsibility of the owner to repair any leaks between the meter and the curb valve within 72 hours of notice by the DPW or an additional fee shall be charged for water loss.

2.11. Hydrant Use. No person, except an employee of the Township or the DPW in the performance of his duties, shall open

or use any fire hydrant, except in case of emergency, without first securing a written permit from the DPW and paying such charges as may be prescribed by the Township.

2.12. Unlawful Connections. No person shall make a connection on a service line between the water meter and the street mains, or install a by-pass around the meter.

2.13. Cross-Connections Control. The DPW shall eliminate and prevent all cross-connections pursuant to the plan submitted to and approved by the Department of Public Health, pursuant to the requirements of Sections 325.11401 through 325.11407 of the 1979 Michigan Administrative Code, as the same may be amended from time to time.

(a) Cross-Connections Prohibited:

- (1) A cross-connection shall not be made between the water distribution system and a secondary water supply.
- (2) A cross-connection shall not be made by submerged inlet.
- (3) A cross-connection shall not be made between the water distribution system and piping which may contain sanitary waste or a chemical contaminant.
- (4) A cross-connection shall not be made between the water distribution system and piping immersed in a tank or vessel which may contain a contaminant.

(b) Inspections. It shall be the duty of the DPW to cause inspections to be made of all properties served by the public water supply where cross connections with the public water supply is deemed possible. The frequency of inspections and reinspections based on potential health hazards involved shall be as established by the DPW.

(c) Right of Entry. The representative of the DPW shall have the right to enter at any reasonable time any property served by a connection to the water distribution system for the purpose of inspecting the piping system or systems thereof for cross-connections. On request, the owner, lessees or occupants of any property so served shall furnish to the DPW any pertinent information regarding the piping system or systems on such property. The refusal of such information or refusal of access, when reasonably requested, shall be deemed evidence of the the presence of cross connections.

2.14. Protection of Water Supply. The potable water supply made available on the premises served by the System shall be

protected from possible contamination as specified by this ordinance and by the State of Michigan's and any other applicable plumbing codes. Any water outlet which could be used for potable or domestic purposes and which is not supplied by the water distribution system must be labeled in a conspicuous manner as water unsafe for drinking.

2.15. Discontinuance of Service. In the event a water customer desires water service to be discontinued, said customer shall so request in writing not less than 48 hours prior to time of such discontinuance of service is desired. Neither the Township nor the DPW will recognize the discontinuance of water service unless the provisions of the foregoing sentence have been fully met. Under any other circumstances, discontinuance of service will be charged to the customer at the actual cost of labor, materials and equipment. The DPW may refuse or discontinue water service for any violation of any rule, regulation, or condition of service.

2.16. Shut off of Water. The DPW reserves the right at all times (and will endeavor to give due notice) to shut off the water at the watermains or local distribution lines or to require reduced use or no use of the System for the purpose of making repairs or extensions or for other purposes. All persons having equipment on their premises and depending on water from the watermains or local distribution lines are hereby cautioned against danger which might arise from emergency shutting off of water. In the event of such emergency, the Director may designate in any notice the extent of any regulation, limitation or prohibition and the date and time on which it shall take effect.

2.17. Additional Regulations. The Director or the Township Board may make and issue additional rules and regulations concerning the water distribution system, connections thereto, meter installations and maintenance, connection and meter installation fees, hydrants and watermains and the appurtenances thereto, not inconsistent herewith. Such rules and regulations shall be effective upon approval by the Township Board.

2.18. Other Laws. If any other ordinances of the County or Township or any statutes of the State of Michigan shall impose greater restrictions than herein set forth, then such ordinances or statutes shall control.

ARTICLE III

CONTROLLED USE

3.1. Limitation of Water Use. The Director, subject to veto by the Township Board, may regulate, limit or prohibit the use of water for any purpose. Such regulations shall restrict less essential water uses to the extent deemed necessary to assure an adequate supply for essential domestic and commercial needs and

for fire fighting. No such regulation, limitation or prohibition shall be effective until twenty-four (24) hours after the publication thereof in a newspaper of general circulation in the Township, except in an emergency as may be deemed by the Director. In the case of such emergency the regulation, limitation or prohibition shall be and take effect as indicated by the Director and notice shall be given on a local radio station. Any person violating such rule or regulation shall, upon conviction thereof, be punished as prescribed in this ordinance.

3.2. Water Conservation. For those users served with water purchased from the City pursuant to the Township-City Water Agreement the following requirements apply:

- (a) Watering Days. The System may only be used for outdoor sprinkling purposes ("watering purposes") on even numbered days by those residences, buildings or parcels of land having an even numbered address. The water distribution system may only be used for outdoor sprinkling purposes on odd numbered days by those residences, buildings or parcels of land having an odd numbered address.
- (b) Watering Hours. It shall be unlawful for any person, firm or corporation to use, consent to use or otherwise approve or direct the use of Township water for watering purposes at any time other than between the hours of 6:00 a.m. to 12:00 noon and 5:00 p.m. to 12:00 midnight.
- (c) Determination of Use. Whenever Township water shall be unlawfully used for outdoor watering purposes such use shall be prima facie proof that it was so used, or that such use was consented to, approved by or directed by the person or persons to whom such water was billed at the time of such use as appears from the records of the DPW.
- (d) Special Conditions. Special consideration shall be given to owners and/or occupants of parcels of land exceeding 8,000 square feet that require irrigation beyond that allowed by this section, by the DPW. Application for an Extra Use permit must be accompanied with a site plan showing the dimensions of the area, describing the system used for watering the area, stating the reason for requiring extra water use and requesting specific amounts in extra days and/or hours of watering. A fee in an amount established from time to time by resolution of the Township Board shall be charged for each permit granted. Special consideration shall also be given to owners and/or occupants of parcels where new lawns, sodding or shrubs are being established. Application for an Extra Use permit must be accompanied with a site plan showing the dimensions

of the area, the date(s) of the new planting(s) and the number of days Extra Use will be required. A maximum of sixty (60) days of daily watering within the hours permitted by this section will be allowed. A fee in an amount established from time to time by resolution of the Township Board shall be charged for each permit granted.

ARTICLE IV

WATER RATES

4.1. Basis of Charges. Except as otherwise provided in this ordinance, all water service shall be charged for on the basis of water consumed as determined by the meter installed in the premises of water customers by the DPW. No free water service shall be furnished to any person.

4.2. Water Rates and Charges.

- (a) Water rates and charges shall be levied monthly against each premises having any water connection to the water distribution system.
- (b) Said water rates and charges shall be based on the quantity of water used on or in the property, as measured by the water meter therein used, except as otherwise provided. If a meter cannot be read, an estimated charge will be made and adjustment, if necessary, will be made when the meter reading can be obtained. Rates and charges for users inside and outside the Township shall be in amounts to be established from time to time by the Township Board.
- (c) Any charges remaining unpaid shall be charged against the real property on which the service was rendered, and may be spread on the next regular Township ad valorem property tax roll after the date on which such charge shall become due and payable, and shall become a lien of the same character and effect as the lien created by State of Michigan and County taxes, until paid.

4.3. Billing and Penalties and Remedies for Nonpayment or Late Payment. The owner of the premises serviced and the occupants thereof shall be jointly and severally liable for the water service provided to said premises. Tenants or land contract purchasers requesting water services in their name shall deposit with the DPW the estimated bill for three (3) months in advance of receiving service. Deposits shall be applied to any bill, including interest, for water service more than thirty (30) days delinquent. Any deposit so applied shall be promptly refunded by the user. The deposit, in the case of tenant, shall be returned, less any balance due, when service is discontinued. In the case of

land contract purchasers, the deposit shall be returned less any balance due when the service is discontinued or the purchaser obtains a deed for the premises.

Bills for rates and charges as herein established shall be mailed to users monthly. All bills shall be payable on the 10th day of the month following the period of service and shall be payable to the DPW. If any bill shall not be paid by the 10th of the month in which it is due, a late charge of ten percent (10%) of the amount billed shall be applied to the current amount past due and collected therewith. Penalty charges will not be compounded.

Water service may be discontinued to any premises to enforce the payment of rates and charges after the user has been given the opportunity for a hearing before the Township Supervisor or his designated representative to show cause why service should not be discontinued. Services so discontinued shall not be restored until such time as all the rates, charges, and penalties are paid or satisfactory arrangements made for the payments thereof.

Delinquent bills may be collected by any method authorized by the law including, without limitation, making such amounts due a lien on the premises served.

4.4. Meter Failure. If any meter shall fail to register properly, the DPW shall estimate the consumption on the basis of former consumption and bill accordingly.

- (a) Inaccurate Meters. A user may require that the meter be tested. If the meter is found accurate, a charge as set from time to time by the Township Board will be made. If the meter is found defective, it shall be repaired or an accurate meter installed and no charge shall be made.
- (b) Accuracy Required. A meter shall be considered accurate if, when tested, it registers not to exceed two percent (2%) more to two percent (2%) less than the actual quantity of water passing through it. If a meter registers in excess of two percent (2%) more than the actual quantity of water passing through it, it shall be considered "fast" to that extent. If a meter registers in excess of two percent (2%) less than the actual quantity of water passing through it, it shall be considered "slow" to that extent.
- (c) Bill Adjustment. If a meter has been tested at the request of a user and shall have been determined to register "fast," the DPW shall credit the consumer with a sum equal to the percent "fast" multiplied by the amount of all bills incurred by said consumer, within the three (3) months prior to the test, and if a meter so tested is determined to register "slow," the DPW may collect from the consumer a sum equal to the percent

"slow" multiplied by the amount of the bills incurred by the consumer for the prior three (3) months. When the DPW, on its own initiative, makes a test of a water meter, it shall be done without cost to the consumer, other than his paying the amount due the DPW for water used by him as above provided, if the meter is found to be "slow."

ARTICLE V

WATER DISTRIBUTION LINES

5.1. Purpose of Rates and Charges. Charges for the installation of local water distribution lines of the water distribution system are hereby established for the purpose of recovering the costs of construction, reconstruction, maintenance, operation and replacement of said local water distribution lines. Such charges shall be made in accordance with the provisions hereinafter set forth.

5.2. Local Distribution Line Installation Charges:

- (a) Payment of Local Distribution Line Installation Charges. Whenever a watermain used for local distribution is constructed, except if pursuant to a special assessment project or if funded entirely by State or federal grants, the property owners whose property is served by the line and on which property the line is used by a commercial user, a domestic user or an industrial user shall pay their share of the cost of construction and installation as herein established at the time of construction. Any property owner requesting service who has not previously paid his share of the cost of constructing the watermain used for local distribution shall pay those costs as established by the then effective resolution of the Township Board setting the local water distribution installation charges.
- (b) Computation of Local Distribution Charges and Annual Review. The charge shall be known as the "front foot rate" and be determined by multiplying the rate per foot by the number of feet the property owner has fronting on the watermain used for local distribution. The front foot rate shall be as set from time to time by resolution of the Township Board. Following an annual review of the charges, the Township Board may change the charges to reflect the current cost of construction and installation of local distribution lines.
- (c) Assessment of Charges. Upon completion of construction or at the time connection is requested (if this cost has not been paid at the time of construction), the DPW shall certify to the Township Board the cost to be

charged to the property owner. The DPW shall bill the owner of the premises affected, advising him that the amount so billed is to be paid prior to connection to the local distribution line serving the property of said property owner.

- (d) Deferral of Assessments. If the Township Board so decides, such installations and connections may be made at the expense of the System when the owner of the premises signs a written agreement agreeing to pay the expense of such installation and connections upon terms agreeable to the Township Board and that the unpaid balance shall constitute a lien upon his property of the same character and subject to the same methods of collection as prescribed for special assessments.

ARTICLE VI

ENFORCEMENT

6.1. Penalties for Violation of Ordinance. Whoever violates or fails to comply with any provision of this ordinance shall be fined up to Five Hundred Dollars (\$500.00), imprisoned not more than ninety (90) days, or both. A separate offense shall be deemed committed each day during or which a violation or noncompliance occurs or continues. The Township Supervisor or the County Sheriff or his deputies may issue appearance citations for violations of this ordinance.

6.2. Civil Action. The Township or the DPW may institute any action at law or equity to compel compliance with this ordinance or to collect amounts due under this ordinance. If such action is instituted the Township or the DPW shall recover the costs and expenses incurred to bring and maintain the action including, without limitation, actual reasonable attorneys' fees.

6.3. Lien. All rates, fees and charges billed or due hereunder including those due pursuant to section 6.2 hereof shall to the extent permitted by law be liens upon the premises served from the delivery of such service which shall be enforceable in the same manner as ad valorem property tax liens.

ARTICLE VII

MISCELLANEOUS

7.1. This ordinance shall take effect ten (10) days after its publication in a newspaper of general circulation in the Charter Township of Haring.

7.2. The Township Clerk shall publish this ordinance in a newspaper of general circulation in the Charter Township of Haring, Wexford County, Michigan.

YEAS: Boardmember(s) Ziel Young, Hofstra, Corwin,
Nelson, Walker, Parker

NAYS: Boardmember(s) NONE

ABSENT: Boardmember(s) NONE

Approved this 14th day of April, 1986.

Theron Parker
Theron Parker
Township Supervisor

Andrea Hofstra
Andrea Hofstra
Township Clerk

I, Andrea Hofstra, Township Clerk of the Charter Township of Haring, certify that the foregoing ordinance was adopted by the Township Board of Haring in public session held APRIL 14th, 1986.

Dated: April 14th, 1986 Andrea Hofstra
Andrea Hofstra, Township Clerk

I, Andrea Hofstra, Township Clerk of the Charter Township of Haring, do hereby certify that this Ordinance was published in the CADILLAC EVENING NEWS on April 18th, 1986.

Dated: April 18th, 1986 Andrea Hofstra, Clerk
Andrea Hofstra, Township Clerk



Cadillac High School MME Chief Administrator Penny Phelps works on putting together packets for proctors behind the closed door of a room at the high school. The MME was administered this week on Monday, Tuesday and today.

Testing process an all-consuming task spanning several days

BY RICK CHARMOLI
CADILLAC NEWS

CADILLAC — Here is a test question for you to ponder.

If there are three days of testing every March, how long would it take to

sigh of relief for people like Penny Phelps; there is only one more day to go. Phelps is the MME Chief Administrator for Cadillac High School, and she was charged with making sure the test went on with-

ated with the test.

From that point, the school received monthly updates from the state. Eventually, the names to be placed on the school's test roster had to be submitted to ensure the right amount

school student taking the test and two new students who were registered just last week.

"Getting their information processed and on the rolls is important," Phelps said.

at school during the three days of testing.

"It is a unique situation because of the spacing issues, and things have to be quiet. We can't have students passing in the halls and no bells," Bruggema

- Acceptable photo ID — must meet all of the following criteria: Examples: driver's license, passport, school ID; Current (not expired); Includes your first and last names; Issued by school, employer or government agency; Current photo clearly recognizable
- A calculator for mathematics assessments only (optional)
- If you use a calculator, be sure to bring a calculator that you are comfortable using, along with fresh batteries. Check the Prohibited Calculator List at www.act.org to make sure that your calculator is permitted for use on the MME mathematics tests.
- A watch (if you wish to pace yourself). Alarms must be turned off.

WHAT NOT TO BRING

cell phone, paper, iPod



EPA Begins Review of the Kysor Industrial Corp. and Northernair Superfund sites Cadillac, Michigan

U.S. Environmental Protection Agency is conducting a five-year review of the Kysor Industrial Corp. and Northernair Plating Superfund sites. The sites are located in southern quarter of an industrial park on the northwest side of Cadillac. Kysor Industrial is adjacent to Wright Street and Northernair Plating is on Sixth Street. The Superfund law requires regular checkups of sites that have been cleaned up — with waste managed on-site — to make sure the cleanup continues to protect people and the environment. This is the third five-year review of the Kysor site and the fourth of the Northernair site.

EPA will check to see if institutional controls are protecting the neighboring subdivision, including any private wells. In addition, EPA will review the adequacy of current local ordinances in place to restrict ground water (underground water) use, stop new well installation. The review will also look into any activities that would compromise the initial cleanup. A reevaluation of air emission data will also be done to determine the effectiveness of the cleanup.

More information is available at the Cadillac-Westford Library, 411 S. Lake St., Cadillac. The review should be completed by mid July 2010.

The five-year review is an opportunity for you to tell EPA about site conditions and any concerns you have. Contact:

Sheila Sullivan
Remedial Project Manager
312-886-5251
sullivan.sheila@epa.gov

Robert Paulson
Community Involvement Coordinator
312-886-0272
paulson.robert@epa.gov

You may also call Region 5 toll-free at 800-621-8431, 9:30 a.m. to 5:30 p.m., weekdays

EPA Region 5 (SI-7J)
77 W. Jackson Blvd.
Chicago, IL 60604

Special Discounts On Cash Parts Sales

Filters: Save 10% on 1-24 New Holland/FleetGuard filters or
Save 20% when you purchase 25 or more filters

Oil: 55 gallon 15W40 engine oil \$550.00
5 gallon hydraulic oil or 15W40 engine oil \$58.00
55 gallon hydraulic oil \$535.00



Ellens Equipment

104 W. Maple, McBain, MI 49657 231-8
Mon.-Fri. 8 AM-5 PM, Sat. 8 AM-12

Documents Used to Prepare Five-Year Review Report

Charter Township of Haring, Wexford Co., 1986. Michigan Ordinance no. 86-01 Zoning Ordinance (Ord. No. 86-01, Sect. 2.1) 1986.

City of Cadillac, 1997. Ordinance No. 97-10, ch. 25, sect. 2.300), Title II, Chpt. 24, Water Supply and Wastewater Systems, recertification in August 2005.

City of Cadillac, March 25, 2010. Presentation to the LDFA Board.

Cadillac Local Development Finance Authority Remediation Project: Summary, via fax on January 18, 2005.

Fishbeck, Thompson, Carr & Huber, April 1995. Final Operation and Maintenance Manual, Volume I, Northernnaire/Kysor Sites Remediation.

Fishbeck, Thompson, Carr & Huber, April 1995. Final Performance Monitoring Plan for Northernnaire/Kysor Sites Remediation.

Fishbeck, Thompson, Carr & Huber Ingraham Property, January 1997. Remedial Action Report.

GeoTrans, Inc., August 17, 2009. Soil Vapor Extraction System Closure report, prepared for the City of Cadillac.

Longshore Environmental Services, 2000, 2002, and 2003. Annual Reports to City of Cadillac.

Mannik & Smith Group, Inc. April 2009. Technical Memorandum Supplemental Hydrogeologic Investigation, Rexair Inc., Cadillac, Michigan. Prepared for the Michigan Department of Natural Resources and Environment.

Michigan Department of Natural resources, 1996. Final Discharge Monitoring Permit, Discharge Requirements to Clam River.

Roy F. Weston Inc., February 2002. Additional Hydrogeologic Investigation Report, Northwest of the Rexair, Inc. Site, Cadillac Industrial Park.

S.S. Papadopoulos and Associates, Inc., April 2007. Hydraulic Capture Zone Evaluation, Kysor Northernnaire Superfund Site, Cadillac, Michigan, prepared for U.S. EPA.

TetraTech, Ann Arbor, MI, February 8, 2006. Kysor Preliminary Soil Investigation, prepared for LDFA.

U.S. EPA, September 29, 1989. Record of Decision, Northernnaire/Kysor sites, OU2.

U.S. EPA, March 3, 1994. Explanation of Significant Differences. Northernnaire/Kysor Sites, OU2.

U.S. EPA, January 30, 1995. Unilateral Administrative Order for Remedial Action, RA-V-W-95-C-282.

U.S. EPA, September 23, 1996. Preliminary Site Close-Out Report, Kysor Industrial Corp.

U.S. EPA, July 26, 2000. First Five-Year Review Report, Kysor Industrial Corp. Site, Cadillac, Michigan.

U.S. EPA, September 2005. Remedial Action Work Plan for the Northernnaire/ Kysor Sites, Cadillac, Michigan.

U.S. EPA, September 2005. Second Five-Year Review Report, Kysor Industrial Corp. Site, Cadillac, Michigan.

Various memoranda and correspondence between U.S. EPA, MDEQ, MDNRE, City of Cadillac, Haring Township, and others.

ATTACHMENT 6 – GRAPHS

Graph 1 -- SVE System Extraction Well VOC Concentration Measurements

Graph 2 -- Extraction Well S-1 Groundwater VOC Trends

Graph 3 -- Well SLW-25 Groundwater VOC Trends

Graph 4 -- Extraction Well S-2 TCE Concentrations

Graph 5 -- Extraction Well S-3 TCE Concentrations

Graph 6 -- Extraction Well S-4 TCE Concentrations

Graph 7 -- Extraction Well S-5 TCE Concentrations

Graph 8 -- Extraction Well S-6 TCE Concentrations

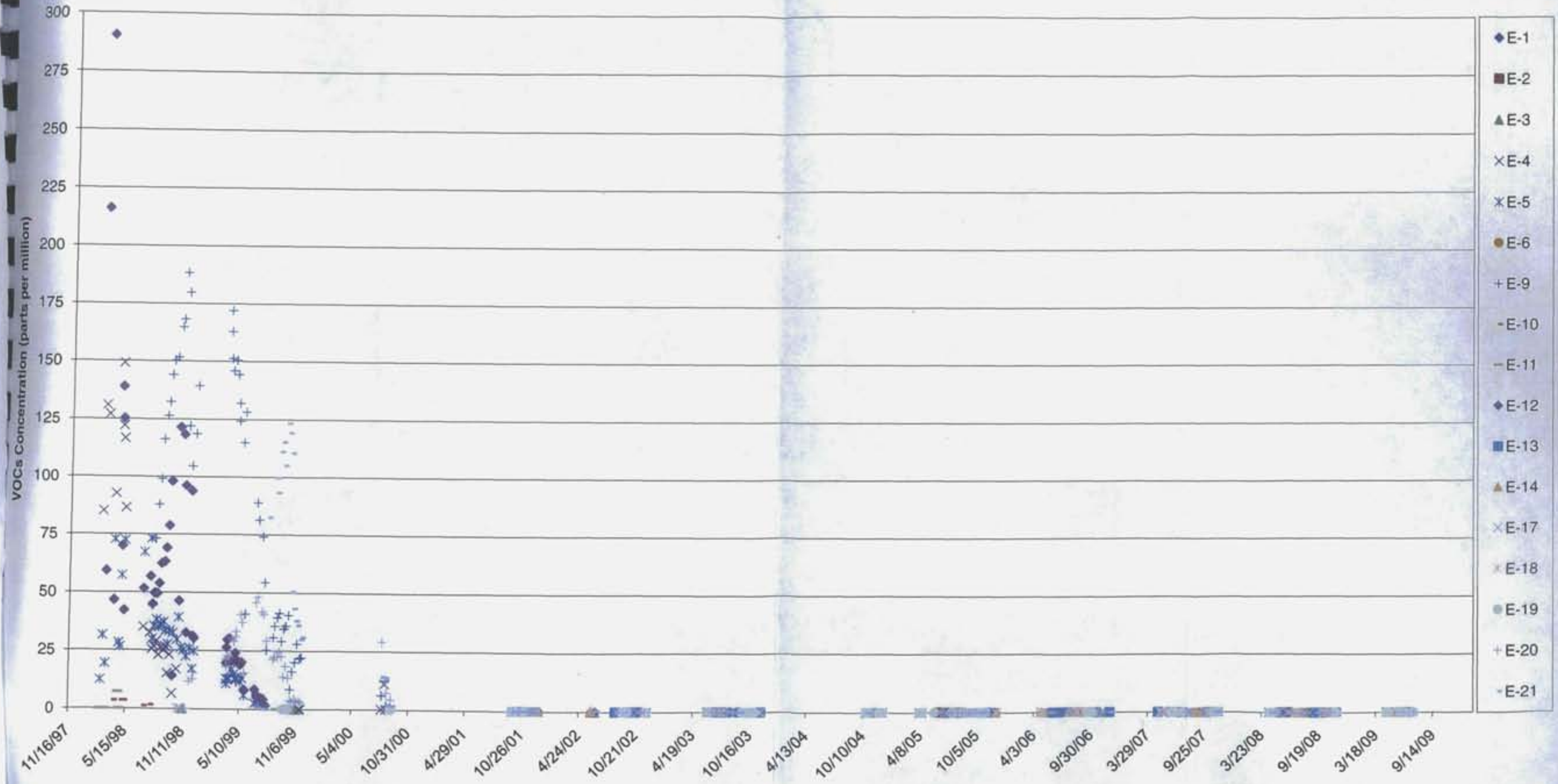
Graphs 9-10 -- 1,1,1- TCA Concentrations in Intermediate Purge Wells

Graphs 11-12 -- TCE Concentrations in Intermediate Purge Wells

Graph 13 -- VOC Concentrations in LDFA Treatment Plant Influent

Graph 14 -- Quarterly TCE Concentrations in LDFA Fringe Wells

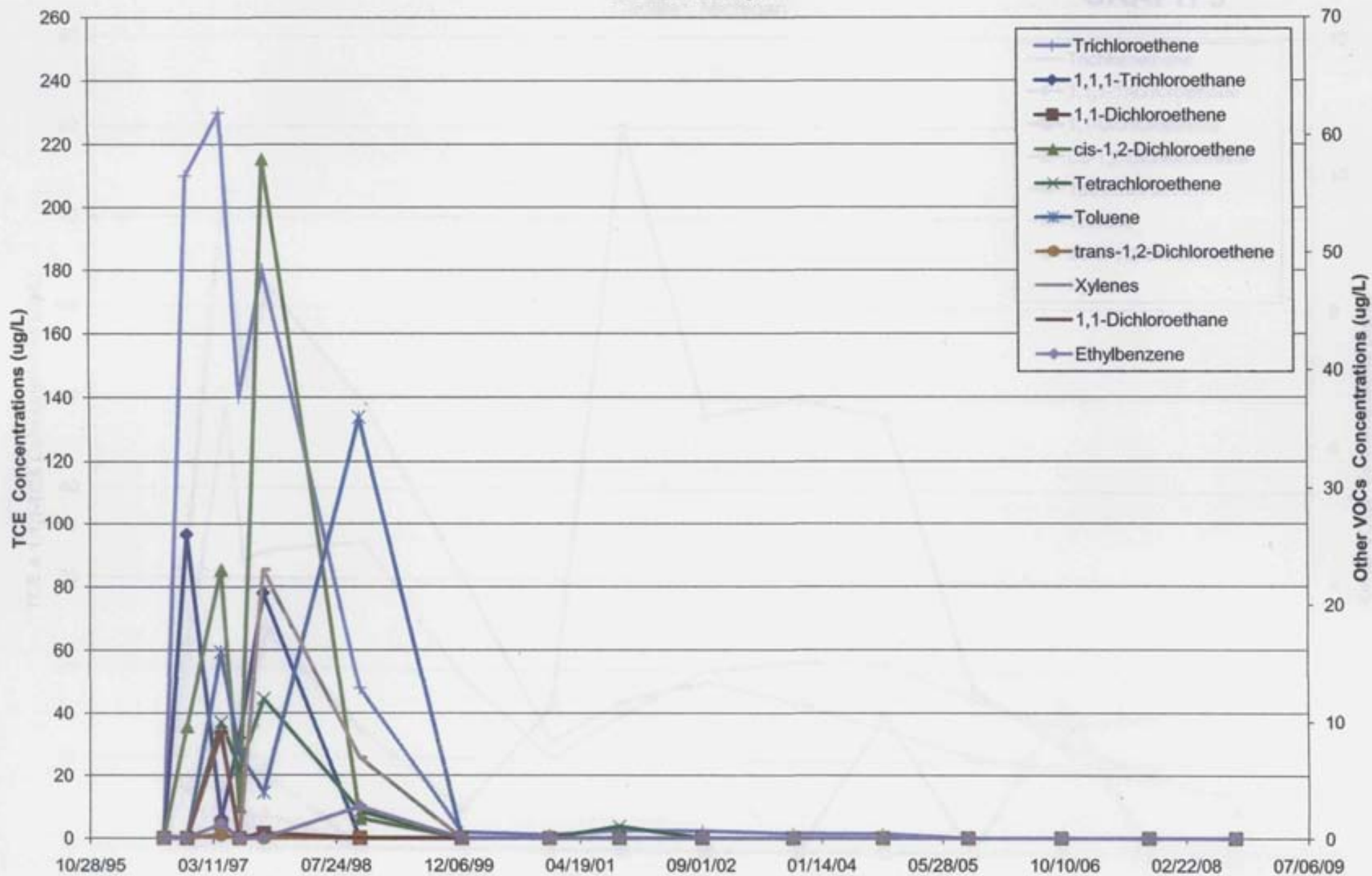
GRAPH 1
Extraction Well VOCs Concentration Measurements
 Kysor Site
 Cadillac, Michigan



S-1 Groundwater VOCs Trends

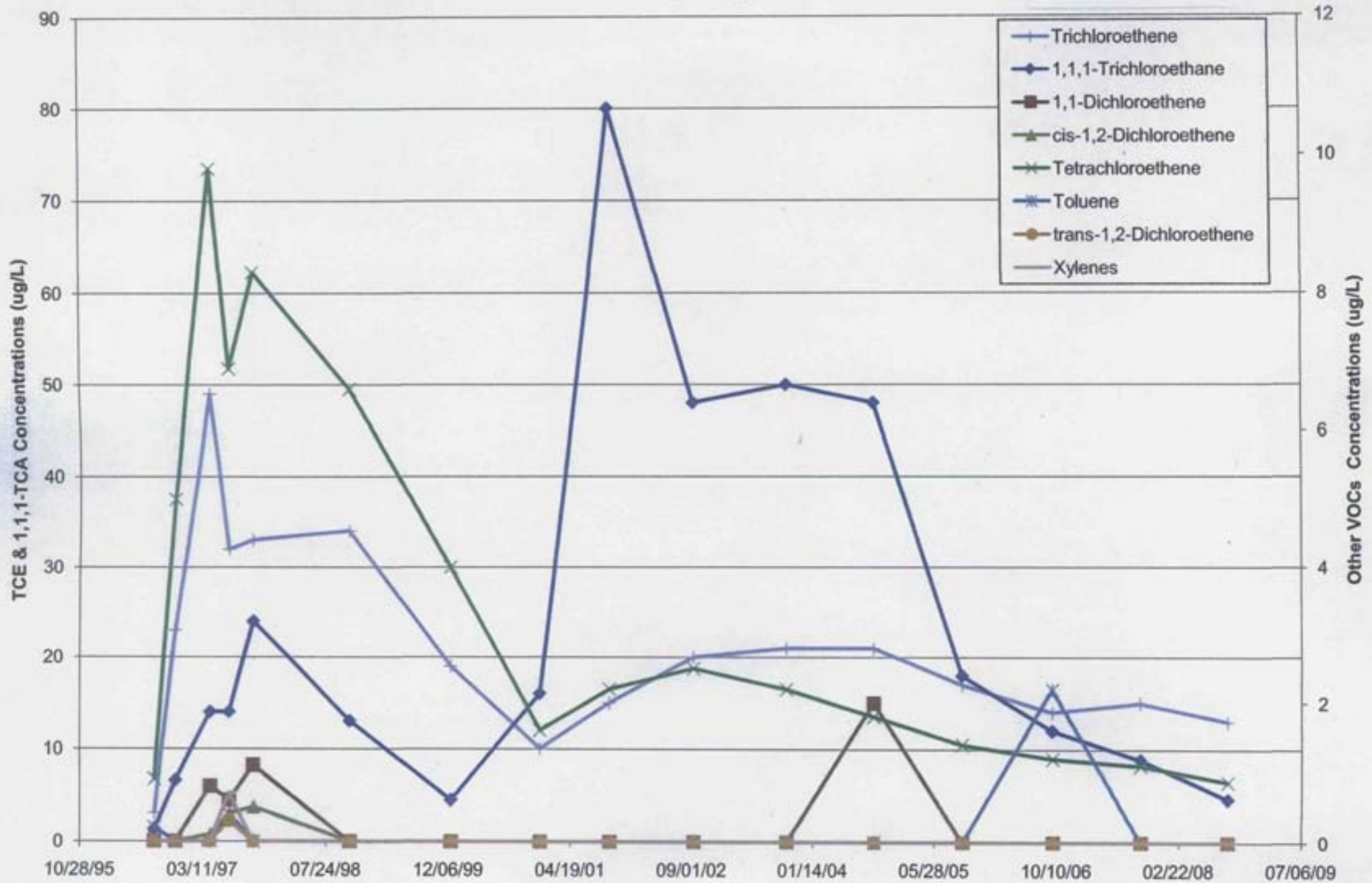
Kysor Site
Cadillac, Michigan

GRAPH 2



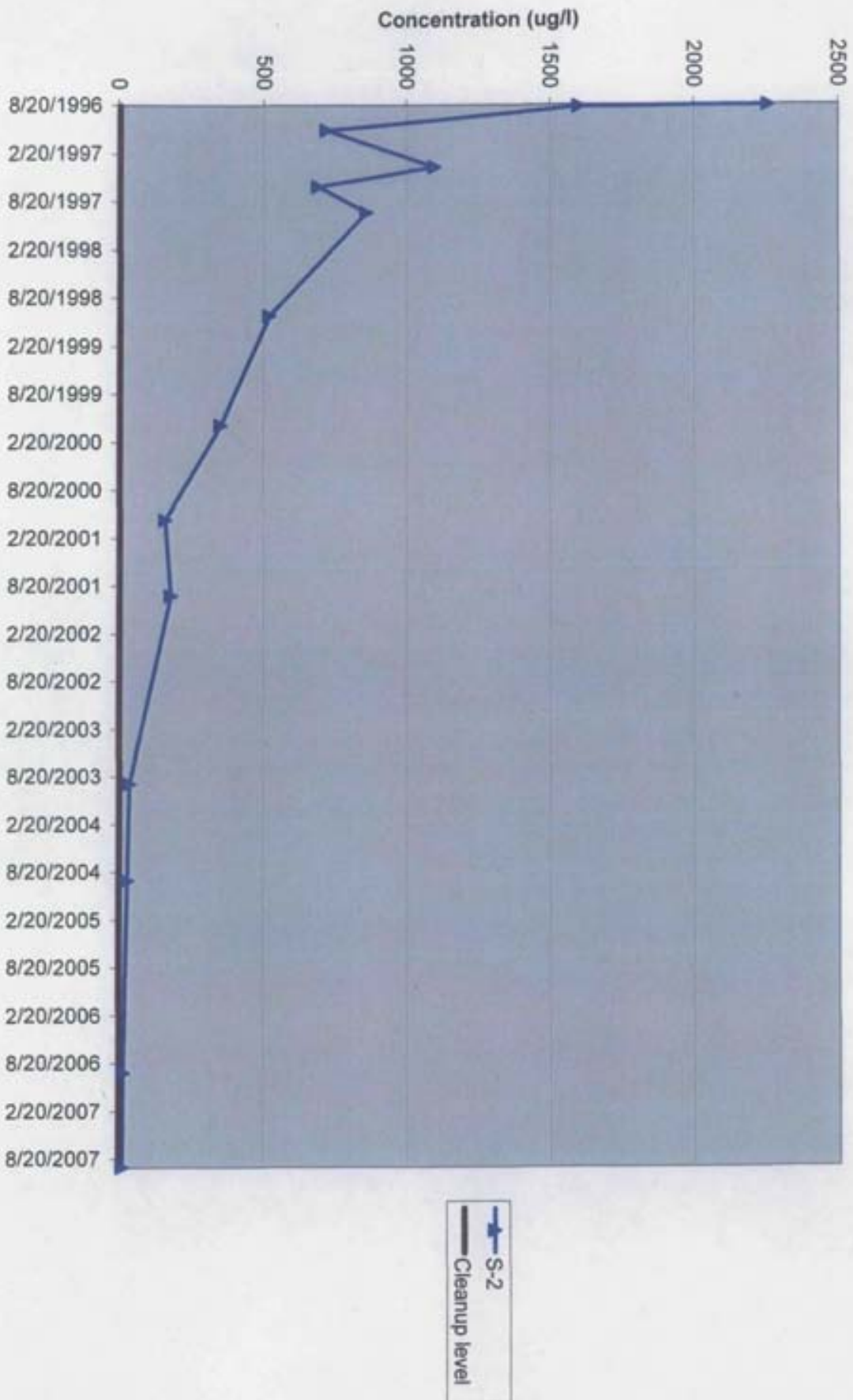
SLW-25 Groundwater VOCs Trends
Kysor Site
Cadillac, Michigan

GRAPH 3



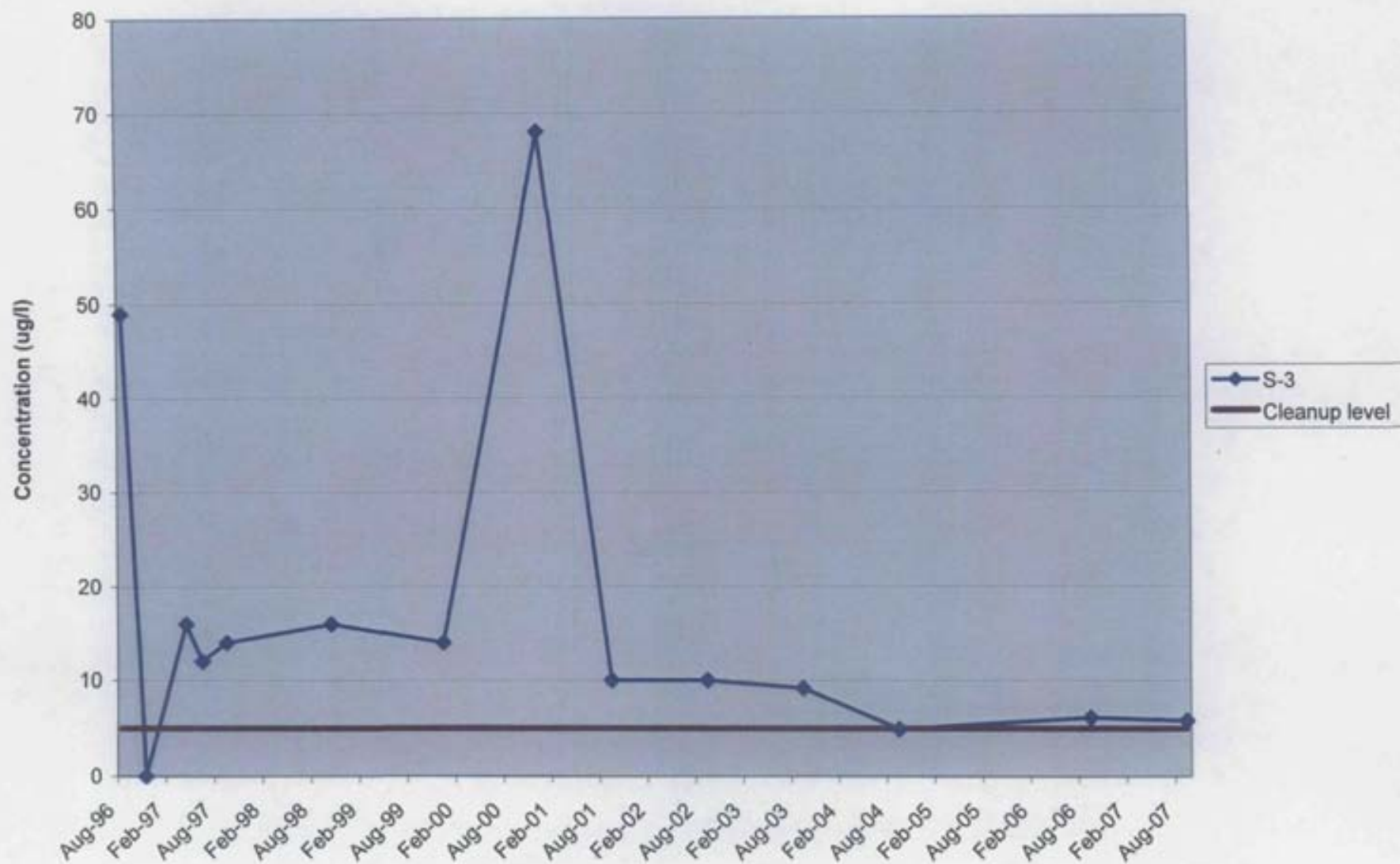
GRAPH 4

TCE in S-2



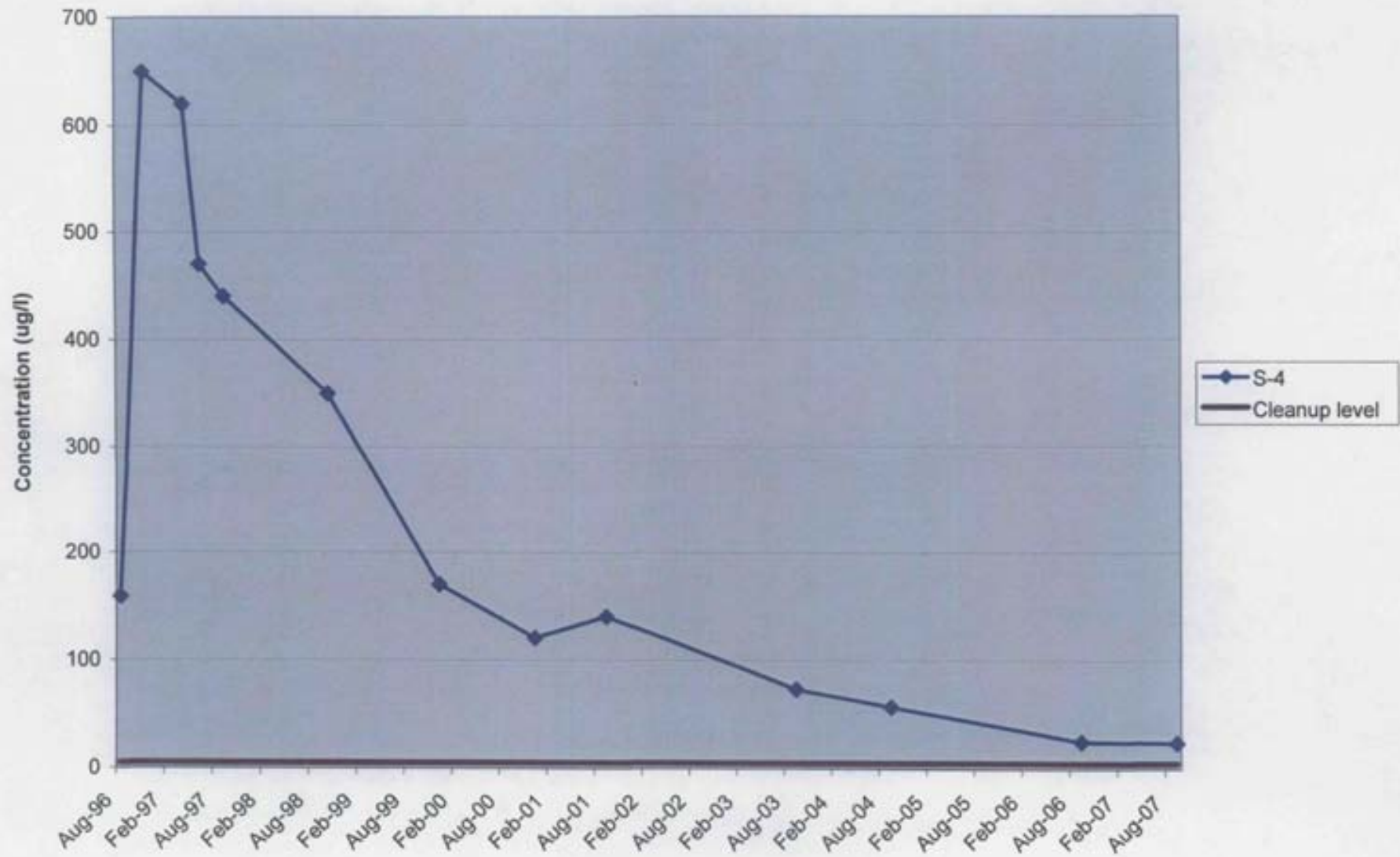
GRAPH 5

TCE in Extraction Well S-3



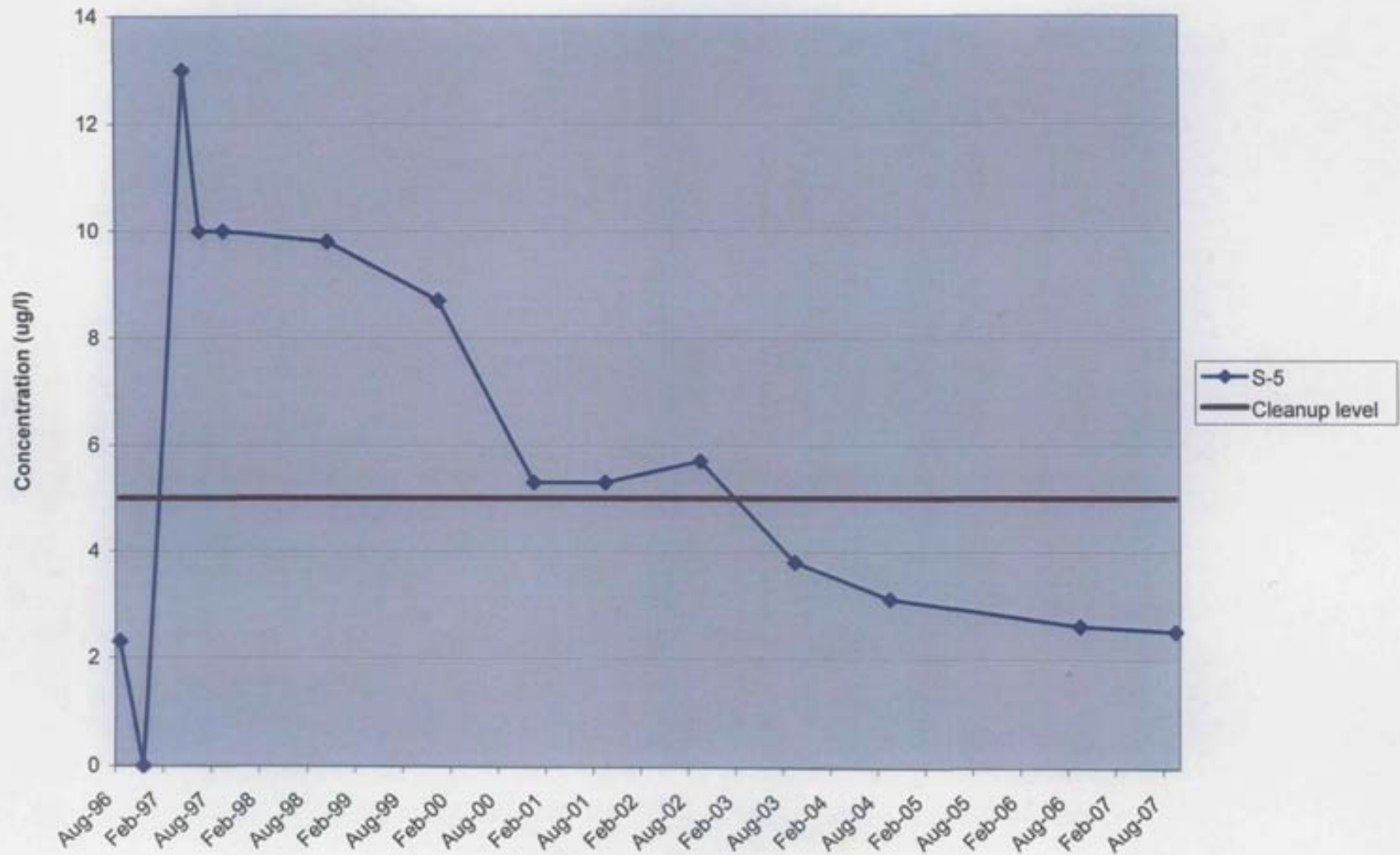
GRAPH 6

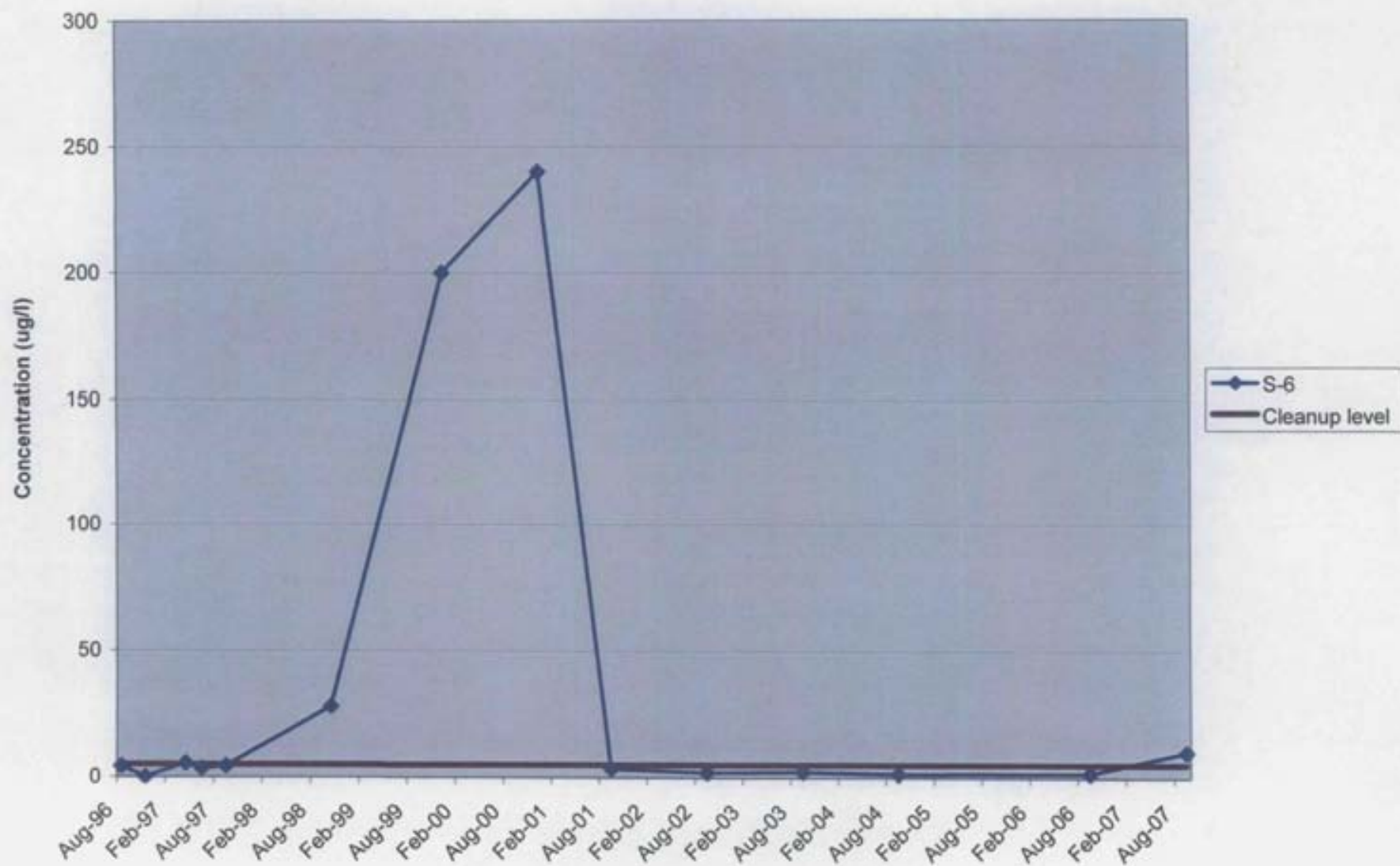
TCE in Extraction Well S-4



GRAPH 7

TCE in Extraction Well S-5

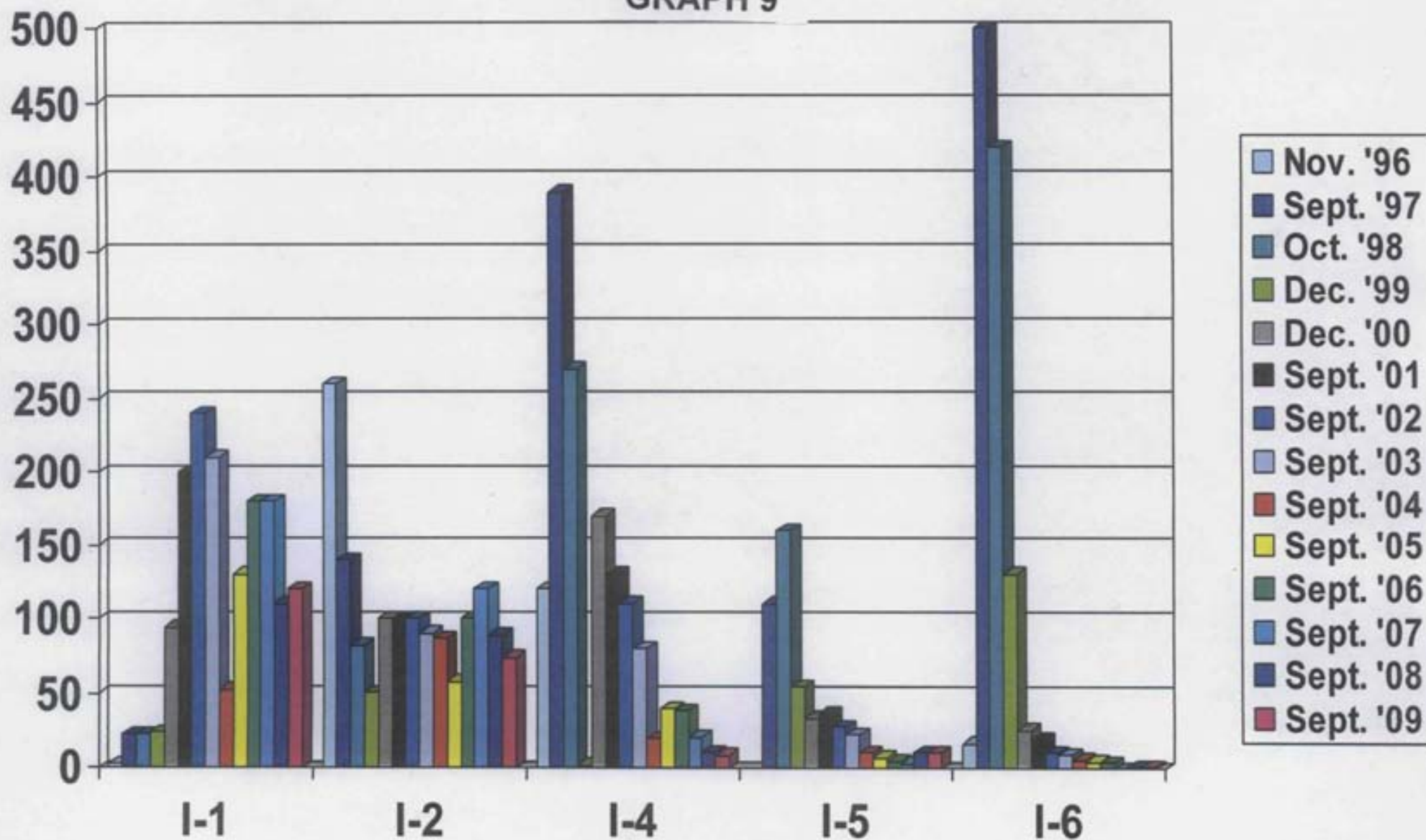


GRAPH 8**TCE in Extraction Well S-6**

Intermediate Purge Wells

1,1,1 – Trichloroethane – ug/l

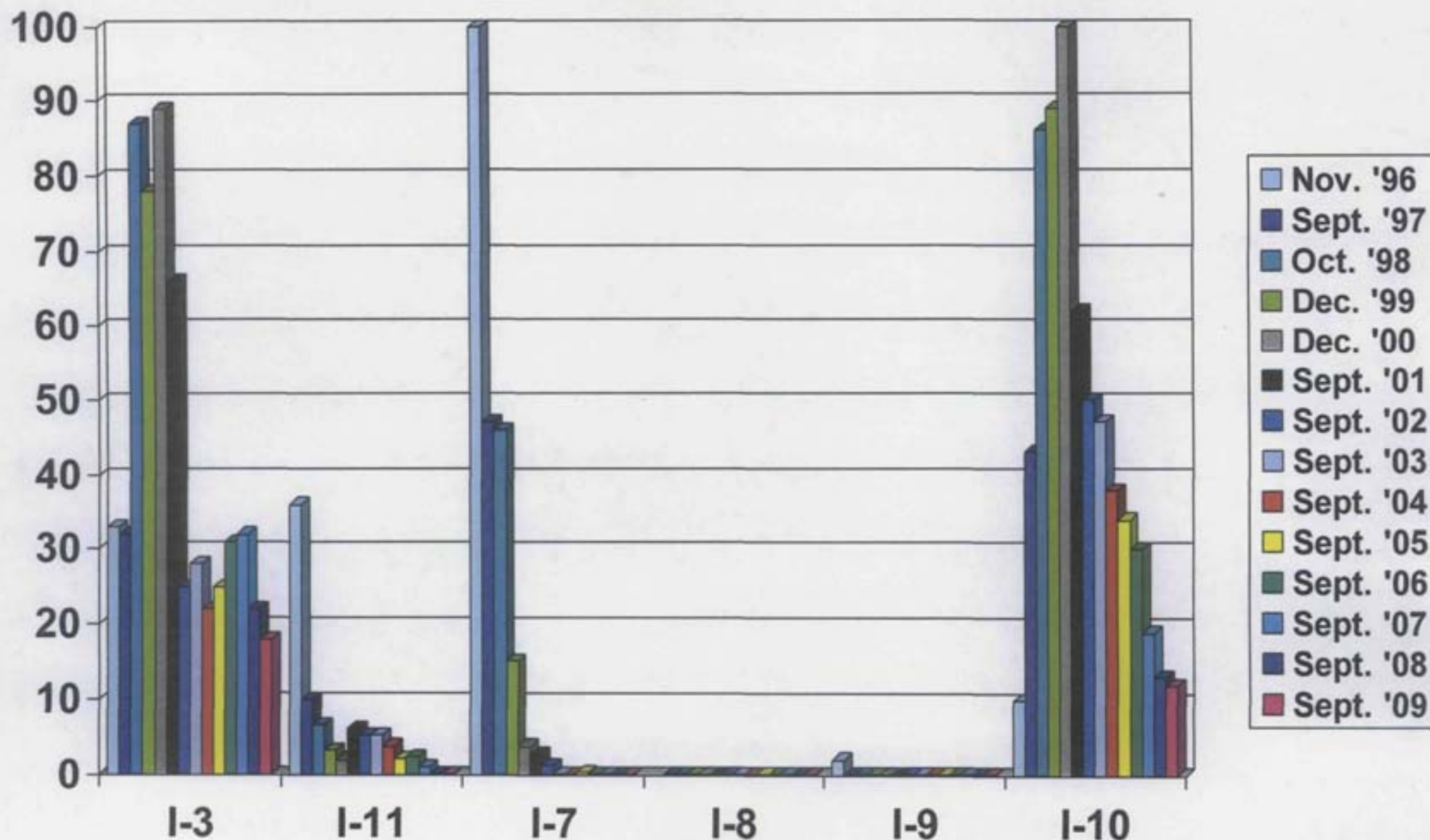
GRAPH 9



Intermediate Purge Wells

1,1,1 - Trichloroethane – ug/l

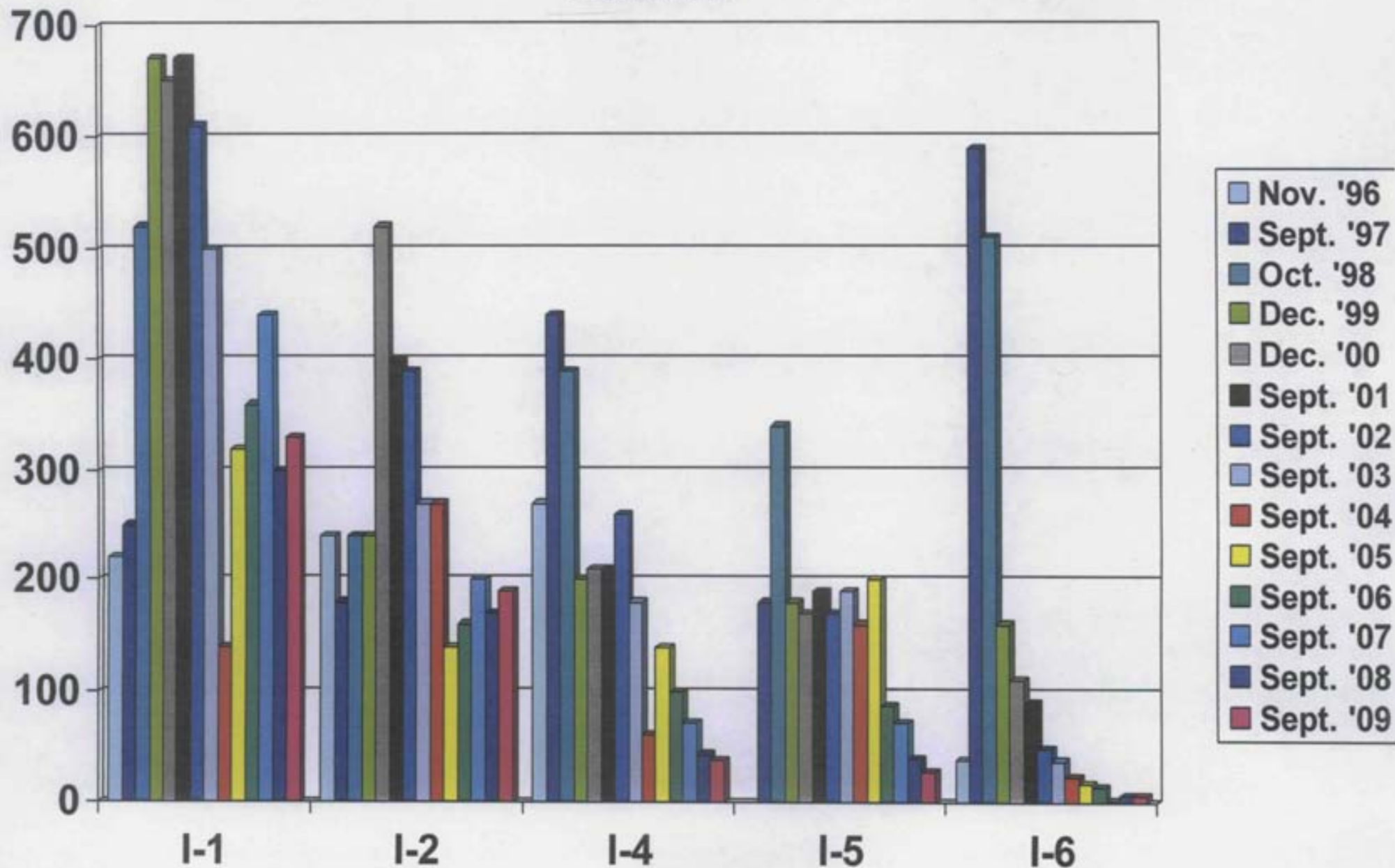
GRAPH 10



Intermediate Purge Wells

Trichloroethene – ug/l

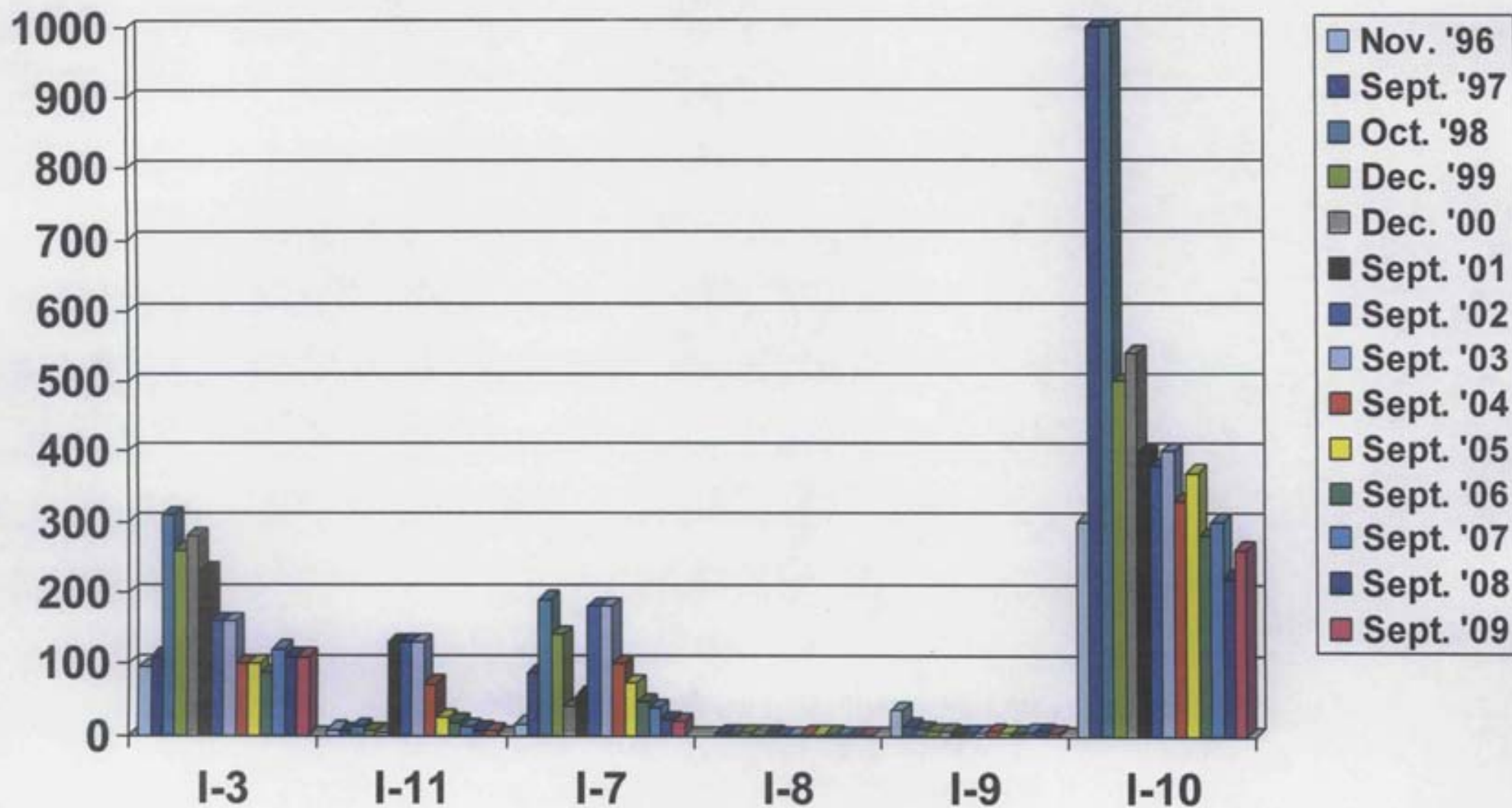
GRAPH 11



Intermediate Purge Wells

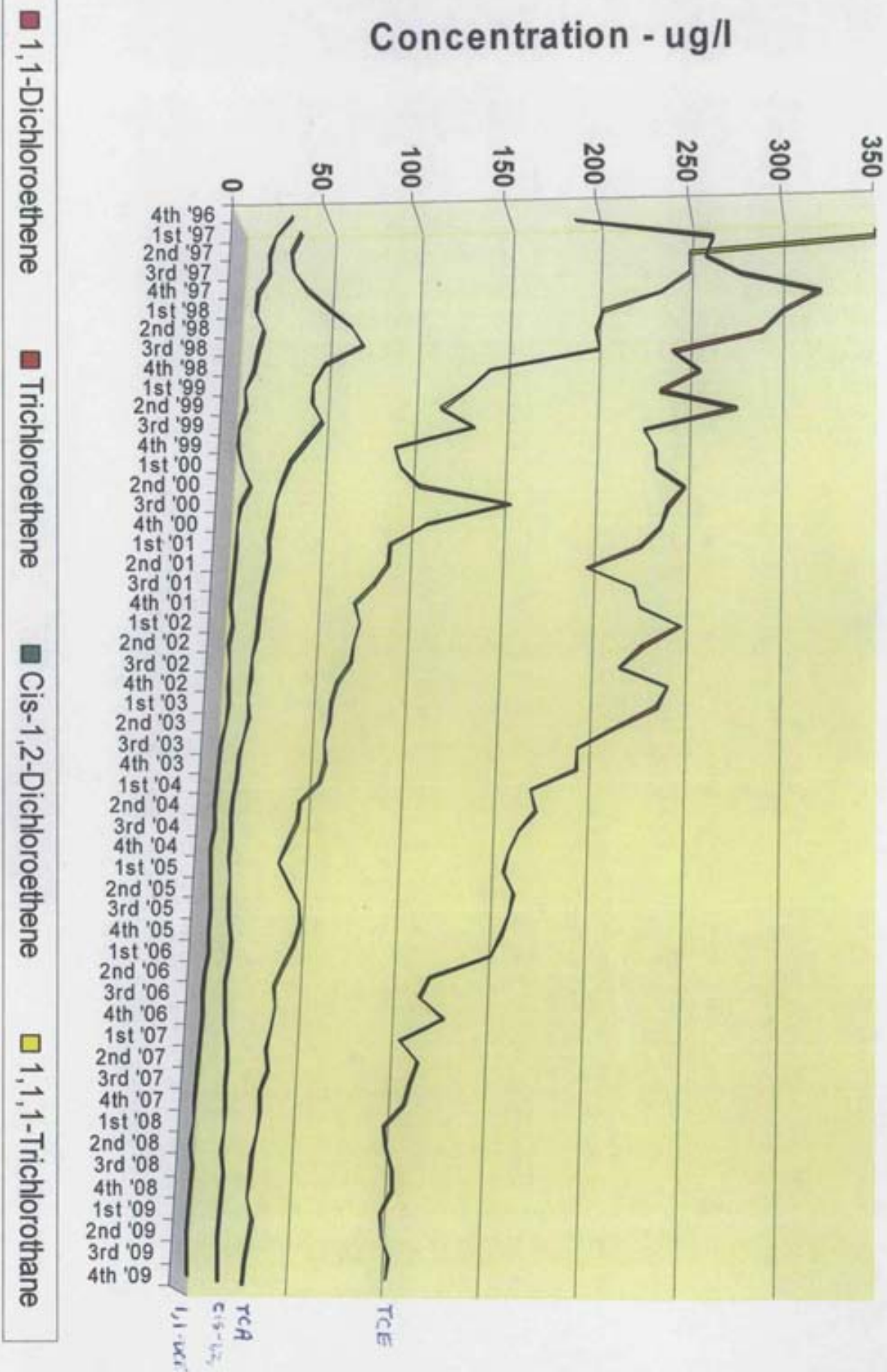
Trichloroethene – ug/l

GRAPH 12



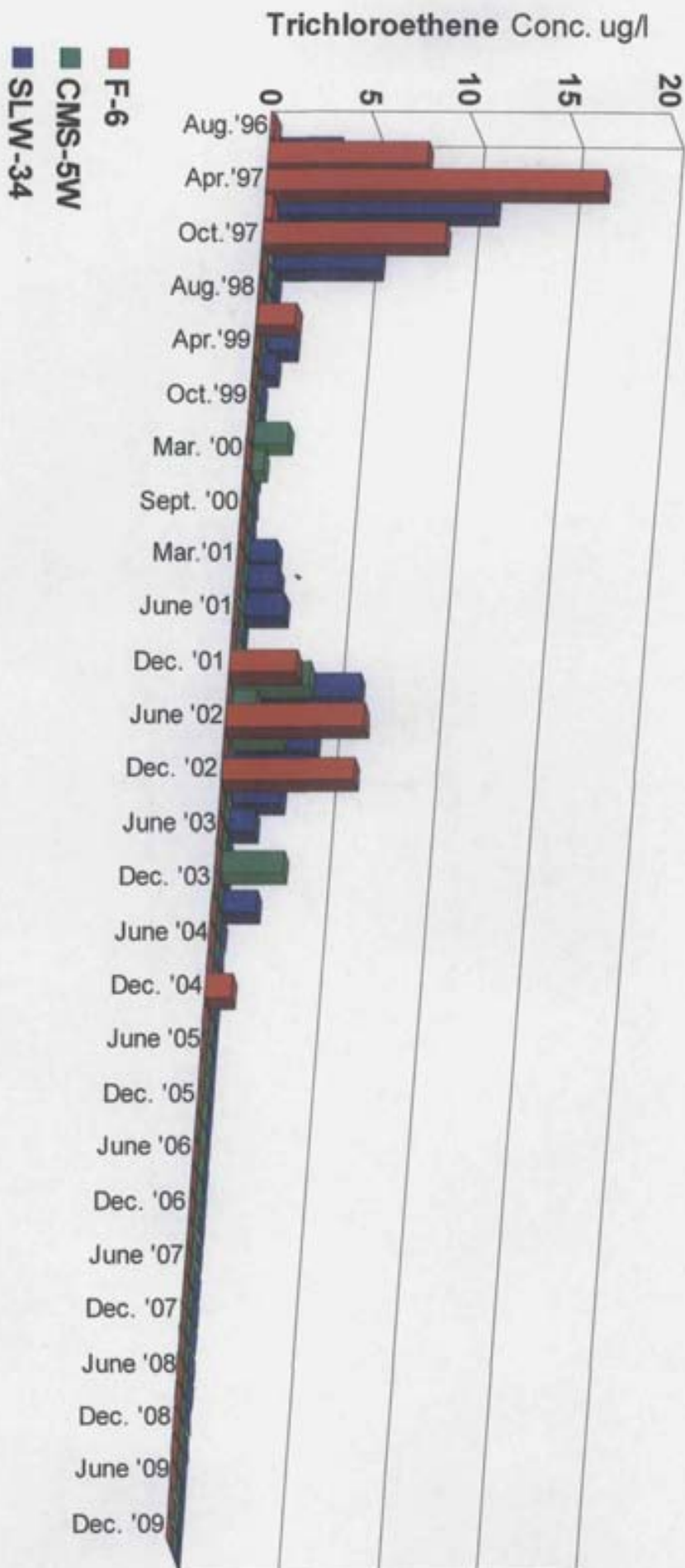
GRAPH 13

LDFA - Groundwater Treatment Plant Influent



GRAPH 14

Cadillac LDFA Quarterly Fringe Organic Monitoring



Five-Year Review Site Inspection Checklist (Template)

I. SITE INFORMATION					
Site name: <u>Northernair/Kysor</u>			Date of inspection: <u>May 21, 2010</u>		
Location and Region: <u>Cadillac, MI, Reg-5</u>			EPA ID: <u>MTD043681840(Kysor) MID02088</u> 3609 (Nor)		
Agency, office, or company leading the five-year review: <u>U.S.EPA</u>			Weather/temperature: <u>Sunny, warm ~60°F</u>		
Remedy Includes: (Check all that apply)					
☐ Landfill cover/containment		☐ Monitored natural attenuation			
☒ Access controls		☐ Groundwater containment			
☒ Institutional controls		☐ Vertical barrier walls			
☒ Groundwater pump and treatment					
☐ Surface water collection and treatment					
☒ Other <u>SVE</u>					
Attachments: ☐ Inspection team roster attached ☐ Site map attached					
II. INTERVIEWS (Check all that apply)					
1. O&M site manager <u>Jeff Dietlin</u>		<u>Cadillac Utilities Director</u> <u>5/21/2010</u>			
Name		Title		Date	
Interviewed ☒ at site ☒ at office ☒ by phone Phone no. _____					
Problems, suggestions; ☐ Report attached <u>No Problems</u>					
2. O&M staff _____					
Name		Title		Date	
Interviewed ☐ at site ☐ at office ☐ by phone Phone no. _____					
Problems, suggestions; ☐ Report attached _____					

3. **Local regulatory authorities and response agencies** (i.e., State and Tribal offices, emergency response office, police department, office of public health or environmental health, zoning office, recorder of deeds, or other city and county offices, etc.) Fill in all that apply.

Agency Wexford County, Dept. of Pub. Works
 Contact John DiVozzo Director 6/17/10
 Name Title Date Phone no.
 Problems; suggestions; ☐ Report attached _____

Agency MDNRE
 Contact Nick Swiger _____ 5/24/10
 Name Title Date Phone no.
 Problems; suggestions; ☐ Report attached _____

Agency MDNRE
 Contact Steven Cunningham Cadillac District Enf. Coord. 5/21/10
 Name Title Date Phone no.
 Problems; suggestions; ☐ Report attached _____

Agency _____
 Contact _____
 Name Title Date Phone no.
 Problems; suggestions; ☐ Report attached _____

4. **Other interviews** (optional) ☐ Report attached.

III. ON-SITE DOCUMENTS & RECORDS VERIFIED (Check all that apply)

- | | | | | |
|----|---|---|--|------------------------------|
| 1. | O&M Documents | | | |
| | ☒ O&M manual office/plant | ☒ Readily available | ☒ Up to date | ☐ N/A |
| | ☒ As-built drawings | ☒ Readily available | ☒ Up to date | ☐ N/A |
| | ☒ Maintenance logs - LDEFA | ☒ Readily available | ☒ Up to date | ☐ N/A |
| | Remarks <u>Daily Bench logs kept at LDEFA</u> | | | |
| 2. | Site-Specific Health and Safety Plan | ☒ Readily available | ☒ Up to date | ☐ N/A |
| | ☐ Contingency plan/emergency response plan | ☒ Readily available | ☒ Up to date | ☐ N/A |
| | Remarks <u>A Risk Management Program plan kept at Cadillac Utilities office</u> | | | |

3.	O&M and OSHA Training Records	☒ Readily available	☒ Up to date	☐ N/A
Remarks <u>In-house OSHA training records kept at PTX plant</u> <u>Eight-hour Incident Response training</u>				
4.	Permits and Service Agreements			
	☒ Air discharge permit	☒ Readily available	☒ Up to date	☐ N/A
	☒ Effluent discharge	☒ Readily available	☒ Up to date	☐ N/A
	☒ Waste disposal, POTW	☒ Readily available	☒ Up to date	☐ N/A
	☐ Other permits _____	☐ Readily available	☐ Up to date	☐ N/A
Remarks _____				
5.	Gas Generation Records	☐ Readily available	☐ Up to date	☐ N/A
Remarks _____				
6.	Settlement Monument Records	☐ Readily available	☐ Up to date	☐ N/A
Remarks _____				
7.	Groundwater Monitoring Records	☒ Readily available	☒ Up to date	☐ N/A
Remarks <u>Records kept at Cadillac utilities office</u>				
8.	Leachate Extraction Records	☐ Readily available	☐ Up to date	☐ N/A
Remarks _____				
9.	Discharge Compliance Records			
	☐ Air	☐ Readily available	☐ Up to date	☐ N/A
	☒ Water (effluent)	☒ Readily available	☒ Up to date	☐ N/A
Remarks _____				
10.	Daily Access/Security Logs	☒ Readily available	☐ Up to date	☐ N/A
Remarks <u>The treatment buildings are kept locked.</u> <u>Fences are locked</u>				
IV. O&M COSTS				
1.	O&M Organization			
	☐ State in-house	☐ Contractor for State		
	☐ PRP in-house	☐ Contractor for PRP		
	☐ Federal Facility in-house	☐ Contractor for Federal Facility		
	☒ Other <u>L DFA & City of Cadillac, see report for costs</u>			

2.	O&M Cost Records	☒ Readily available ☒ Up to date ☐ Funding mechanism/agreement in place Original O&M cost estimate _____ ☒ Breakdown attached		
Total annual cost by year for review period if available				
	From _____ Date	To _____ Date	_____ Total cost	☐ Breakdown attached
	From _____ Date	To _____ Date	_____ Total cost	☐ Breakdown attached
	From _____ Date	To _____ Date	_____ Total cost	☐ Breakdown attached
	From _____ Date	To _____ Date	_____ Total cost	☐ Breakdown attached
	From _____ Date	To _____ Date	_____ Total cost	☐ Breakdown attached

3.	Unanticipated or Unusually High O&M Costs During Review Period Describe costs and reasons: <u>New Telemetry system installed for Monitoring Well system</u>	_____ _____ _____ _____		
----	---	----------------------------------	--	--

V. ACCESS AND INSTITUTIONAL CONTROLS ☐ Applicable ☐ N/A				
A. Fencing				
1.	Fencing damaged ☐ Location shown on site map ☐ Gates secured ☐ N/A Remarks: <u>The SVE system is fenced, All wells and electrical systems on Monitoring wells are locked</u>			
B. Other Access Restrictions				
1.	Signs and other security measures ☐ Location shown on site map ☐ N/A Remarks: _____			
C. Institutional Controls (ICs)				

1.	Implementation and enforcement		☐ Yes ☒ No ☐ N/A
	Site conditions imply ICs not properly implemented		☒ Yes ☐ No ☐ N/A
	Site conditions imply ICs not being fully enforced		
	Type of monitoring (e.g., self-reporting, drive by) _____		
	Frequency <u>Periodically</u>		
	Responsible party/agency <u>City of Cadillac, Herring Township, Wexford Co. DPA</u>		
	Contact <u>Jeff Dietlin (Cadillac)</u>	<u>Utilities Mgr</u>	<u>5/21/10</u>
	Name <u>John DiVozzo</u>	Title <u>Wexford Co</u>	Date _____ Phone no. _____
	Reporting is up-to-date		☒ Yes ☐ No ☐ N/A
	Reports are verified by the lead agency		☒ Yes ☐ No ☐ N/A
	Specific requirements in deed or decision documents have been met		☒ Yes ☐ No ☐ N/A
	Violations have been reported		☐ Yes ☐ No ☐ N/A
	Other problems or suggestions: ☐ Report attached		<u>Somewhat</u>
2.	Adequacy		☐ ICs are adequate ☐ ICs are inadequate ☐ N/A
	Remarks <u>ICs are somewhat adequate. Township and City are upgrading ordinances to require connection to public water</u>		
D. General			
1.	Vandalism/trespassing		☐ Location shown on site map ☒ No vandalism evident
	Remarks _____		
2.	Land use changes on site		☐ N/A
	Remarks <u>None</u>		
3.	Land use changes off site		☐ N/A
	Remarks <u>None in general</u>		
VI. GENERAL SITE CONDITIONS			
A. Roads		☐ Applicable ☐ N/A	
1.	Roads damaged		☐ Location shown on site map ☒ Roads adequate ☐ N/A
	Remarks _____		
B. Other Site Conditions			
Remarks _____			
VII. LANDFILL COVERS			
A. Landfill Surface		☐ Applicable ☐ N/A	

1.	Settlement (Low spots) Areal extent _____ Remarks _____	☐ Location shown on site map Depth _____	☐ Settlement not evident
2.	Cracks Lengths _____ Widths _____ Depths _____ Remarks _____	☐ Location shown on site map	☐ Cracking not evident
3.	Erosion Areal extent _____ Remarks _____	☐ Location shown on site map Depth _____	☐ Erosion not evident
4.	Holes Areal extent _____ Remarks _____	☐ Location shown on site map Depth _____	☐ Holes not evident
5.	Vegetative Cover ☐ Trees/Shrubs (indicate size and locations on a diagram) Remarks _____	☐ Grass ☐ Cover properly established	☐ No signs of stress
6.	Alternative Cover (armored rock, concrete, etc.) Remarks _____	☐ N/A	
7.	Bulges Areal extent _____ Remarks _____	☐ Location shown on site map Height _____	☐ Bulges not evident
8.	Wet Areas/Water Damage ☐ Wet areas ☐ Ponding ☐ Seeps ☐ Soft subgrade Remarks _____	☐ Wet areas/water damage not evident ☐ Location shown on site map ☐ Location shown on site map ☐ Location shown on site map ☐ Location shown on site map	Areal extent _____ Areal extent _____ Areal extent _____ Areal extent _____
9.	Slope Instability Areal extent _____ Remarks _____	☐ Slides ☐ Location shown on site map	☐ No evidence of slope instability
B. Benches ☐ Applicable ☐ N/A (Horizontally constructed mounds of earth placed across a steep landfill side slope to interrupt the slope in order to slow down the velocity of surface runoff and intercept and convey the runoff to a lined channel.)			
1.	Flows Bypass Bench Remarks _____	☐ Location shown on site map	☐ N/A or okay

2.	Bench Breached	☐ Location shown on site map	☐ N/A or okay
Remarks _____			
3.	Bench Overtopped	☐ Location shown on site map	☐ N/A or okay
Remarks _____			
C. Letdown Channels ☐ Applicable ☐ N/A (Channel lined with erosion control mats, riprap, grout bags, or gabions that descend down the steep side slope of the cover and will allow the runoff water collected by the benches to move off of the landfill cover without creating erosion gullies.)			
1.	Settlement	☐ Location shown on site map	☐ No evidence of settlement
Areal extent _____ Depth _____			
Remarks _____			
2.	Material Degradation	☐ Location shown on site map	☐ No evidence of degradation
Material type _____ Areal extent _____			
Remarks _____			
3.	Erosion	☐ Location shown on site map	☐ No evidence of erosion
Areal extent _____ Depth _____			
Remarks _____			
4.	Undercutting	☐ Location shown on site map	☐ No evidence of undercutting
Areal extent _____ Depth _____			
Remarks _____			
5.	Obstructions	Type _____	☐ No obstructions
☐ Location shown on site map		Areal extent _____	
Size _____			
Remarks _____			
6.	Excessive Vegetative Growth	Type _____	
☐ No evidence of excessive growth			
☐ Vegetation in channels does not obstruct flow			
☐ Location shown on site map		Areal extent _____	
Remarks _____			
D. Cover Penetrations ☐ Applicable ☐ N/A			
1.	Gas Vents	☐ Active	☐ Passive
☐ Properly secured/locked		☐ Functioning	☐ Routinely sampled
☐ Evidence of leakage at penetration		☐ Good condition	
☐ N/A		☐ Needs Maintenance	
Remarks _____			
2.	Gas Monitoring Probes	☐ Properly secured/locked	☐ Functioning
☐ Evidence of leakage at penetration		☐ Routinely sampled	☐ Good condition
		☐ Needs Maintenance	☐ N/A
Remarks _____			

3.	Monitoring Wells (within surface area of landfill) ☐ Properly secured/locked ☐ Functioning ☐ Routinely sampled ☐ Good condition ☐ Evidence of leakage at penetration ☐ Needs Maintenance ☐ N/A Remarks _____
4.	Leachate Extraction Wells ☐ Properly secured/locked ☐ Functioning ☐ Routinely sampled ☐ Good condition ☐ Evidence of leakage at penetration ☐ Needs Maintenance ☐ N/A Remarks _____
5.	Settlement Monuments ☐ Located ☐ Routinely surveyed ☐ N/A Remarks _____
E. Gas Collection and Treatment ☐ Applicable ☐ N/A	
1.	Gas Treatment Facilities ☐ Flaring ☐ Thermal destruction ☐ Collection for reuse ☐ Good condition ☐ Needs Maintenance Remarks _____
2.	Gas Collection Wells, Manifolds and Piping ☐ Good condition ☐ Needs Maintenance Remarks _____
3.	Gas Monitoring Facilities (e.g., gas monitoring of adjacent homes or buildings) ☐ Good condition ☐ Needs Maintenance ☐ N/A Remarks _____
F. Cover Drainage Layer ☐ Applicable ☐ N/A	
1.	Outlet Pipes Inspected ☐ Functioning ☐ N/A Remarks _____
2.	Outlet Rock Inspected ☐ Functioning ☐ N/A Remarks _____
G. Detention/Sedimentation Ponds ☐ Applicable ☐ N/A	
1.	Siltation Areal extent _____ Depth _____ ☐ N/A ☐ Siltation not evident Remarks _____
2.	Erosion Areal extent _____ Depth _____ ☐ Erosion not evident Remarks _____
3.	Outlet Works ☐ Functioning ☐ N/A Remarks _____
4.	Dam ☐ Functioning ☐ N/A Remarks _____

H. Retaining Walls		☐ Applicable	☐ N/A
1.	Deformations	☐ Location shown on site map	☐ Deformation not evident
	Horizontal displacement _____	Vertical displacement _____	
	Rotational displacement _____		
	Remarks _____		
2.	Degradation	☐ Location shown on site map	☐ Degradation not evident
	Remarks _____		
I. Perimeter Ditches/Off-Site Discharge		☐ Applicable	☐ N/A
1.	Siltation	☐ Location shown on site map	☐ Siltation not evident
	Areal extent _____	Depth _____	
	Remarks _____		
2.	Vegetative Growth	☐ Location shown on site map	☐ N/A
	☐ Vegetation does not impede flow		
	Areal extent _____	Type _____	
	Remarks _____		
3.	Erosion	☐ Location shown on site map	☐ Erosion not evident
	Areal extent _____	Depth _____	
	Remarks _____		
4.	Discharge Structure	☐ Functioning	☐ N/A
	Remarks _____		
VIII. VERTICAL BARRIER WALLS		☐ Applicable	☐ N/A
1.	Settlement	☐ Location shown on site map	☐ Settlement not evident
	Areal extent _____	Depth _____	
	Remarks _____		
2.	Performance Monitoring	Type of monitoring _____	
	☐ Performance not monitored		
	Frequency _____	☐ Evidence of breaching	
	Head differential _____		
	Remarks _____		
IX. GROUNDWATER/SURFACE WATER REMEDIES		☒ Applicable	☐ N/A
A. Groundwater Extraction Wells, Pumps, and Pipelines		☒ Applicable	☐ N/A
1.	Pumps, Wellhead Plumbing, and Electrical		
	☐ Good condition	☒ All required wells properly operating	
		☐ Needs Maintenance	
		☐ N/A	
	Remarks <u>Shallow well S-7 is dry and has not been operating</u>		
	<u>Well S-3 platform was crushed on one edge. Electrical panels and</u>		
	<u>telemetry in good shape</u>		
2.	Extraction System Pipelines, Valves, Valve Boxes, and Other Appurtenances		
	☒ Good condition	☐ Needs Maintenance	
	Remarks _____		

3.	Spare Parts and Equipment		
	☒ Readily available	☐ Good condition	☐ Requires upgrade ☐ Needs to be provided
	Remarks _____		
B. Surface Water Collection Structures, Pumps, and Pipelines ☐ Applicable ☐ N/A			
1.	Collection Structures, Pumps, and Electrical		
	☒ Good condition	☐ Needs Maintenance	
	Remarks _____		
2.	Surface Water Collection System Pipelines, Valves, Valve Boxes, and Other Appurtenances		
	☒ Good condition	☐ Needs Maintenance	
	Remarks _____		
3.	Spare Parts and Equipment		
	☒ Readily available	☐ Good condition	☐ Requires upgrade ☐ Needs to be provided
	Remarks _____		
C. Treatment System ☐ Applicable ☐ N/A			
1.	Treatment Train (Check components that apply)		
	☒ Metals removal	☐ Oil/water separation	☐ Bioremediation
	☒ Air stripping	☐ Carbon adsorbers	
	☐ Filters _____		
	☐ Additive (e.g., chelation agent, flocculent) _____		
	☐ Others _____		
	☒ Good condition	☐ Needs Maintenance	
	☒ Sampling ports properly marked and functional		
	☒ Sampling/maintenance log displayed and up to date		
	☒ Equipment properly identified		
	☐ Quantity of groundwater treated annually <u>~2.7 MGD</u>		
	☐ Quantity of surface water treated annually _____		
	Remarks _____		
2.	Electrical Enclosures and Panels (properly rated and functional)		
	☐ N/A	☒ Good condition	☐ Needs Maintenance
	Remarks _____		
3.	Tanks, Vaults, Storage Vessels		
	☐ N/A	☒ Good condition	☐ Proper secondary containment ☐ Needs Maintenance
	Remarks _____		
4.	Discharge Structure and Appurtenances		
	☐ N/A	☒ Good condition	☐ Needs Maintenance
	Remarks _____		

5.	Treatment Building(s) ☐ N/A ☒ Good condition (esp. roof and doorways) ☐ Needs repair ☒ Chemicals and equipment properly stored Remarks _____
6.	Monitoring Wells (pump and treatment remedy) ☒ Properly secured/locked ☐ Functioning ☒ Routinely sampled ☒ Good condition ☒ All required wells located ☐ Needs Maintenance ☐ N/A Remarks _____
D. Monitoring Data	
1.	Monitoring Data ☒ Is routinely submitted on time ☒ Is of acceptable quality
2.	Monitoring data suggests: ☒ Groundwater plume is effectively contained ☒ Contaminant concentrations are declining
D. Monitored Natural Attenuation	
1.	Monitoring Wells (natural attenuation remedy) ☐ Properly secured/locked ☐ Functioning ☐ Routinely sampled ☐ Good condition ☐ All required wells located ☐ Needs Maintenance ☒ N/A Remarks _____
X. OTHER REMEDIES	
If there are remedies applied at the site which are not covered above, attach an inspection sheet describing the physical nature and condition of any facility associated with the remedy. An example would be soil vapor extraction.	
XI. OVERALL OBSERVATIONS	
A.	Implementation of the Remedy Describe issues and observations relating to whether the remedy is effective and functioning as designed. Begin with a brief statement of what the remedy is to accomplish (i.e., to contain contaminant plume, minimize infiltration and gas emission, etc.). <i>Remedy is to contain plume and remove source of contaminant from soil.</i> <i>SVE system effective and is being decommissioned.</i> <i>Groundwater extraction and treatment system is well-maintained and effectively capturing the groundwater plume.</i> <i>The city is in process of planning a new well field due to age of the existing system.</i>
B.	Adequacy of O&M

Describe issues and observations related to the implementation and scope of O&M procedures. In particular, discuss their relationship to the current and long-term protectiveness of the remedy.

O & M practices are effective

*The remedy is protective but needs stricter ordinances
regarding private well use*

C. Early Indicators of Potential Remedy Problems

Describe issues and observations such as unexpected changes in the cost or scope of O&M or a high frequency of unscheduled repairs, that suggest that the protectiveness of the remedy may be compromised in the future.

None -

D. Opportunities for Optimization

Describe possible opportunities for optimization in monitoring tasks or the operation of the remedy.

ATTACHMENT 8

Site Inspection Documentation Photos

Five-Year Review Site Inspection – Kysor and Northernaire Superfund Sites,
May 21, 2010



Figure 1 – Electrical panel for Local Development Finance Authority (LDFA) extraction well I-7, located about 1,200 feet northwest of the edge of the Cadillac city well field. The antenna on top right side of panel is radio telemetry control system.



Figure 2 – View of shallow aquifer well GW-11 located about 500 feet northwest of the Four Winns/A.H. Joynt facility. The well is used for water level and VOC monitoring.

Five-Year Review Site Inspection – Kysor and Northernaire Superfund Sites,
May 21, 2010



Figure 3 – View of the Four Winns facility as seen from the intersection of Seventh Avenue and Fourteenth Street, where well I-7 is located.



Figure 4 – View of electrical panel for LDFA intermediate depth extraction well I-8, located at the intersection of Seventh Avenue and Grahek Street in the North Park subdivision area. The well is about 2,250 feet northwest of, and the farthest distance from, the Cadillac well field.

Five-Year Review Site Inspection – Kysor and Northernaire Superfund Sites,
May 21, 2010



Figure 5 – View of well SLW-35. This shallow-aquifer well is located about 1,025 feet north of Thirteenth Street and northwest of the North Park subdivision area.



Figure 6 – View of electrical panel for LDFA intermediate extraction well I-5, located north of the Michigan Rubber facility and 1,000 feet northwest of the Four Star facility. Well I-5 is the dedicated chromium extraction well.

Five-Year Review Site Inspection – Kysor and Northernaire Superfund Sites,
May 21, 2010



Figure 7 – A view of the back of the Four Winns Frisbie facility, which manufactures recreational boats. The electrical panel for the LDFA shallow-aquifer extraction well S-6 is visible at the far right background of the photo.



Figure 8 - A closer view of the back side of the Four Winns Frisbie facility.

Five-Year Review Site Inspection – Kysor and Northernaire Superfund Sites,
May 21, 2010



Figure 9 – A view of LDFA shallow-aquifer extraction well S-3, located about 1,250 feet west of the south west corner of the Cadillac city well field. The well cover in the middle of the photograph is cracked around the metal cover. The electrical panel is at the top right corner of the photo.



Figure 10 – A view of the south side of the LDFA groundwater treatment plant. The two air stripping towers are visible in the center mid-ground of the photo. The plant is in front of the towers.

Five-Year Review Site Inspection – Kysor and Northernaire Superfund Sites,
May 21, 2010



Figure 11 –View from atop the air-stripping towers looking south toward the Kysor plant (now owned by Borg-Warner) just south of the railroad tracks. The Cadillac city Department of Public Works yard is immediately north (below) the railroad tracks. The soil vapor extraction (SVE) system (not visible in photo) is located between the Kysor plant and the railroad tracks.

Five-Year Review Site Inspection – Kysor and Northernnaire Superfund Sites,
May 21, 2010



Figure 12 – View of one of the two GAC treatment tanks inside the LDFA treatment plant. The chromium stream from dedicated extraction well I-5 and the VOCs that may also be present in the extracted groundwater from I-5 are processed in these tanks. Most of VOC stream does not enter these vessels.

Five-Year Review Site Inspection – Kysor and Northernaire Superfund Sites,
May 21, 2010



Figure 13 – View inside the treatment plant of the joining of the dedicated chromium well (I-5) extraction stream with the VOC extraction wells. The thin vertical pipe carries the combined streams to the GAC filter tanks before entering the air strippers.

Five-Year Review Site Inspection – Kysor and Northernaire Superfund Sites,
May 21, 2010



Figure 14 – View from atop the air stripping towers looking north. The Northernaire facility is in the mid-ground. The Paulstra/CRC facility (long low red building) is to the right and immediately northeast of Northernaire.



Figure 15 – View from atop the air stripping facility looking east. The Mitchell Corporation is in the background.

Five-Year Review Site Inspection – Kysor and Northernaire Superfund Sites,
May 21, 2010



Figure 16 – View from atop the air stripping tower looking west. The Co-Gen power plant is barely visible in the far back center of the photo.



Figure 17 – View of the SVE system, now decommissioned after meeting soil cleanup goals. The system is located directly north of the Kysor plant and south of the railroad tracks. The foreground area is capped with a PVC liner. The concrete-asphalt pad is visible behind the blue GAC tanks.

ATTACHMENT 9 – ADDITIONAL TABLES

Table 8 – LDFA Extraction Well Construction Summary

Table 9 – LDFA Monitoring Plan

Table 10 – Boring Placement and VAP Rationale for MDNRE Plume Investigation

TABLE 8

LDFA Extraction Well Construction Summary			
Extraction Well No.	Date Completed	Well Depth (feet bgl)	Screened Interval (feet bgl)
S-1	7/7/95	92	77 - 92
S-2	7/8/95	71.5	56.5 - 71.5
S-3	8/20/95	67	52 - 67
S-4	7/10/95	50	35 - 50
S-5	7/9/95	67	57 - 67
S-6	7/24/95	53	43 - 53
S-7	8/5/95	46	36 - 46
I-1	7/10/95	187	162 - 187
I-2	7/19/95	185	160 - 185
I-3	9/29/95	221	201 - 221
I-4	8/20/95	177	152 - 177
I-5	9/8/95	150	125 - 150
I-6	8/2/95	179	154 - 179
I-7	10/5/95	164	139 - 164
I-8	8/30/95	158	133 - 158
I-9	7/25/95	190	165 - 190
I-10	8/15/95	191	166 - 191
I-11	10/12/95	163	133 - 163

TABLE 9



S.S. PAFADOPULOS & ASSOCIATES, INC.

LDFA Monitoring Plan

Well	Easting ¹	Northing ¹	Aquifer ²	Compliance Wells ³	VOC Monitoring ³	Chromium Monitoring since March 2006 ⁴	Water Level Monitoring ⁵
S-1	2053830.4	16081779.7	S	X	A	X	
S-2	2053856.4	16082337.5	S	X	A	X	
S-3	2053763.0	16083387.3	S	X	A	X	
S-4	2054228.0	16083121.6	S	X	A	X	
S-5	2052889.8	16084571.7	S	X	A	X	
S-6	2053787.1	16084036.9	S	X	A	X	
S-7	2054218.2	16083613.7	S	X		X	
DNR-10S	2054800.5	16084453.6	S	X	A		
FS-110	2054796.2	16083874.3	S	X	A		
FS-310	2054632.6	16083448.5	S				
DNR-1	2054690.2	16083338.1	S				
F-5	2053184.7	16085742.6	S		A		X
GW-10S	2053066.9	16083334.0	S	X	A		X
GW-11	2054337.5	16085384.6	S		A		X
GW-14S	2052312.9	16084237.8	S		A		X
GW-40S	2052479.3	16084929.0	S				
GW-9S	2052760.4	16082424.0	S	X	A		X
K-10	2051760.3	16082990.2	S				
K-1	2055129.8	16083390.0	S		A		X
K-2	2055106.1	16084383.4	S		A		X
K-4	2054198.5	16083580.6	S				
K-7	2053660.2	16080928.6	S				
K-22	2056132.9	16083160.3	S				
K-25	2052732.9	16083361.1	S				X
K-45	2054305.6	16082497.1	S				X
MW-1	2054133.9	16082616.0	S				
MW-2	2054078.8	16082446.3	S				
MW-3	2054067.1	16083225.3	S				
MW-5	2054319.4	16083085.2	S			X	X
MW-5	2053781.3	16083144.2	S				
P-1	2053864.0	16084771.5	S				
P-2	2053868.3	16085677.3	S				
SLW-1	2054608.9	16084745.6	S	X			
SLW-10S	2054132.2	16083590.4	S				
SLW-11	2053377.9	16082449.0	S				
SLW-12S	2053829.1	16083600.1	S				
SLW-16	2052526.6	16083004.9	S				
SLW-21S	2053498.2	16080917.9	S				
SLW-23	2055174.7	16082098.0	S		A		
SLW-25	2053563.5	16081767.8	S	X	A		X
SLW-31	2053808.2	16083974.3	S	X	A		X
SLW-32	2054646.0	16082304.1	S				
SLW-33	2056120.7	16083636.3	S				
SLW-37	2056269.8	16083780.7	S				
SLW-39	2056505.4	16083616.7	S				
SLW-40	2056442.9	16083872.6	S				
SLW-43	2055865.8	16083993.9	S				
SLW-44	2055761.4	16084121.9	S				
SLW-45	2055619.9	16084289.2	S				
SLW-46	2055864.5	16083685.5	S				
SLW-48	2055701.6	16083921.8	S				
SLW-49	2053954.7	16082704.8	S				
SLW-51	2054090.2	16082209.5	S				
SLW-54	2053806.3	16082296.8	S	X	A		X
PWMV-16	2053089.8	16084273.1	S	X	A		X
PWMV-9	2052993.9	16083935.2	S	X	A		X
F-1S	2051945.1	16085109.4	S		A		X
I-1	2052837.5	16084063.2	UI	X	A	X	
I-2	2053259.1	16084604.5	UI	X	A	X	
I-3	2053686.7	16085096.6	LI	X	A	X	
I-4	2053790.2	16083794.1	UI	X	A	X	



LDFA Monitoring Plan

Well	Easting ¹	Northing ¹	Aquifer ²	Compliance Wells ³	VOC Monitoring ³	Chromium Monitoring since March 2006 ⁴	Water Level Monitoring ⁵
I-1	2054136.8	16084584.8	UI	X	A	X	
I-6	2054223.1	16083705.6	UI	X	A	X	
I-7	2054340.4	16085405.0	UI	X	A	X	
I-8	2054293.1	16086333.5	UI	X	A	X	
I-9	2052093.6	16083174.1	UI	X	A	X	
I-10	2052643.2	16083315.1	UI	X	A	X	
I-11	2053734.2	16085096.6	UI	X	A	X	
DNR-14D	2054888.0	16084477.7	UI	X	A		X
F-10	2051939.6	16085106.1	UI		A		
F-1	2053772.5	16086286.1	UI		A		X
F-4	2051036.7	16083691.0	UI		Q	X	
F-7	2053177.1	16085752.5	LI		A		X
F-2	2053825.0	16085175.0	UI	X		X	X
GW-10D	2053072.2	16083321.3	UI	X			
GW-13	2055950.4	16085020.8	LI				
GW-14D	2052304.9	16084239.1	UI		A		X
GW-17	2054998.0	16085388.7	LI	X	A		X
GW-18	2056405.4	16085491.7	LI				
GW-19	2056362.9	16080979.2	UI				
GW-2	2056782.2	16082745.3	UI				
GW-20	2055046.4	16082370.3	UI				
GW-3	2057615.9	16083922.1	LI				
GW-6D	2052481.7	16084922.4	UI				
GW-7	2054652.0	16087281.7	UI				
GW-8	2051762.7	16082982.3	UI				
GW-9D	2052770.3	16082424.4	UI	X			
CMS-3A	2050546.2	16083572.3	UI		Q	X	X
K-12	2055125.8	16083901.4	UI			X	X
K-13	2054191.2	16084056.3	UI				
K-16C	2053397.0	16084079.9	UI	X	A		X
K-21	2055704.7	16083460.6	UI				
K-24	2055291.3	16084651.9	UI				
K-30	2054304.0	16082495.5	UI				
K-8	2054233.7	16083078.9	UI				
MRW-3	2057586.5	16083908.6	UI				
MW-10	2054455.8	16084494.1	UI			X	X
MW-1	2053875.5	16084774.8	UI			X	X
MW-2	2054151.2	16083757.5	UI				
MW-3	2054116.3	16084561.5	UI			X	X
MW-4	2054495.8	16084105.5	UI				
MW-5	2053911.6	16084014.0	UI				
MW-6	2053370.1	16084542.7	UI				
MW-7	2054690.8	16083359.9	UI			X	X
MW-8	2054134.9	16084725.6	UI	X		X	X
MW-20	2054101.4	16085280.0	UI			X	X
MW-7	2054262.3	16084305.6	UI			X	X
MW-8	2054095.2	16083225.0	UI	X	A		X
MW-9	2053877.8	16084325.6	UI				
SLW-10D	2054125.6	16083576.0	UI				
SLW-12D	2053822.6	16083577.3	UI				
SLW-13	2054660.3	16086146.2	UI				
SLW-15	2053805.5	16083134.4	UI				
SLW-17	2054965.8	16086196.2	UI				
SLW-18	2054420.5	16083095.0	UI				
SLW-2	2054594.0	16085434.2	UI				
SLW-20	2054661.5	16086350.7	UI				
SLW-22	2053483.4	16083070.4	UI				
SLW-26	2056366.5	16081707.5	UI				
SLW-27	2055681.4	16083316.3	UI				
SLW-29	2057019.1	16083863.8	UI				



LDFA Monitoring Plan

Well	Easting ¹	Northing ¹	Aquifer ²	Compliance Wells ³	VOC Monitoring ³	Chromium Monitoring since March 2006 ⁴	Water Level Monitoring ⁵
SLW-3	2054800.3	16085386.6	UI				
SLW-34	2055462.8	16086189.8	UI		Q	X	
SLA-35	2052957.3	16086363.7	UI		A		X
SLW-4	2054656.9	16085775.1	UI				X
SLA-41	2056083.0	16083859.4	UI				
SLA-42	2056206.7	16083993.9	UI				
SLA-49	2054198.2	16082789.9	UI				
SLW-5	2054495.0	16085778.7	UI				
SLA-50	2055975.8	16084181.0	UI				
SLA-51	2055869.0	16084482.8	UI				
SLA-52	2056106.3	16084335.2	UI				
SLW-7	2053881.6	16084988.0	UI				

Notes

1 Coordinates sources:

a) December 2005 GPS City of Cadillac Survey in NAD 83, STATE Plane "City_Wells NAVD88.xls".

b) Extrapolated from "Dragun Northing_Easting Wells.xls," using a 2-point transformation to place in the NAD83 UTM z16N coordinate system. Extrapolated coordinates were corrected to match December 2006 survey by offset of UTM meters X =11.18 and UTM meters Y offset=-0.16 on estimates from "Dragun Northing_Easting Wells.xls" that were surveyed by Dragun in 2000/2002, and by offset of UTM meters X=10.79 and UTM meters Y offset=4.74 for estimate from "Dragun Northing_Easting Wells.xls" not surveyed by Dragun in 2000/2002.

c) Extrapolated from EC Jordan, 1987, Remedial Investigation Report using a 2-point transformation to place in the NAD83 UTM z16N coordinate system.

J: LDFA.xls

2 S = Shallow; UI = Upper Intermediate; LI = Lower Intermediate

3 Source = Tetra Tech MPS, 2005, 2004 Annual Performance Monitoring Report for LDFA Northern Kysor Superfund Site, July. Q= Quarterly, A=Annual

4 Source = Chromium data collected on a quarterly basis since March 2006 in preparation for closure (Larry Campbell, LDFA, City of Cadillac, personal communication)

5 Based on data available in 2004

TABLE 10

BORING PLACEMENT AND VAP SAMPLING RATIONALE

Rexair, Inc. Site (Site ID: 83000102)

Cadillac, Wexford County, Michigan

VAP Boring ID	Drilling Method	Total Boring Depth (ft)	VAP Boring/Sampling Rationale	Vertical Aquifer Profiling (VAP)	Number of VAP Samples Collected (Including Duplicates)	Monitoring Well ID Screened Interval (ft)
Shallow/Intermediate Aquifer - VAP and TCE Plume Delineation Monitoring Wells (DMW)						
DMW-1	Lead Screened Auger	230 +/-	Assess deep TCE data gap within the intermediate aquifer near the City well field above the wellfield clay.	150-237 ft	10	DMW-1 (160-165)
DMW-2	Lead Screened Auger	230 +/-		100-225 ft	14	DMW-2I (160-165) DWM-2D (220-225)
DMW-3	Lead Screened Auger	230 +/-	Assess vertical and horizontal extent of TCE impacts at the down-gradient edge of the TCE plume beyond the LDFA purge well capture zone near F-2.	40-230 ft	22	DMW-3 (200-205)
DMW-4	Lead Screened Auger	230 +/-	Investigate historical TCE impacts to the north beyond the LDFA radius of influence and down-gradient of SLW-20.	40-215 ft	20	None
DMW-5	Lead Screened Auger	230 +/-		50-215 ft	19	None
DMW-6	Lead Screened Auger	230 +/-		40-225 ft	20	None
Deep Well Field Aquifer - VAP and Hydrogeologic Monitoring Wells (WFA)						
WFA-1 WFA-1R	Sonic	330 +/-	Assess vertical extent of TCE impacts within the deep well field aquifer near CW-7(screened from 260 and 330 feet bgs).	240-327 ft	12	WFA-1RS (244.5-249.5) WFA-1RD (276 -281)
WFA-2	Sonic	280 +/-	Assess potential TCE impacts within the deep wellfield aquifer and triangulate groundwater flow.	207-287 ft	8	WFA-2 (262-267)
WFA-3	Sonic	280 +/-		207-287 ft	8	WFA-3 (262-267)
WFA-4	Sonic	280 +/-		217-307 ft	12	WFA-4S (235-240) WFA-4D (257-262)